Class 10

Physics

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1. Light - Reflection

LEVEL-I

1. Answer: D

Explanation: Light can travel through a vacuum, which is why sunlight reaches the Earth through space. Therefore, the given statement is incorrect.

2. Answer: C

Explanation: When light falls on water, only a part of it is reflected while the rest enters the water. Hence, the reflected image appears less bright than the actual object.

3. Answer: A

Explanation: A concave mirror produces a parallel beam of light when a point source is placed at its focus. This principle is used in headlights and searchlights.

4. Answer: A

Explanation: A convex mirror always forms a virtual, erect, and diminished image. It never forms a real or inverted image under any condition.

5. Answer: D

Explanation: The focal length of a spherical mirror depends only on its radius of curvature. It remains the same for all colours of light.

6. Answer: C

Explanation: For every spherical mirror, whether concave or convex, the focal length is half the radius of curvature. Thus, $f = \frac{R}{2}$.

7. Answer: D) Zero

Explanation:

When a real object is placed at the centre of curvature (2f) of a concave mirror, its real image is also formed at the centre of curvature (2f). Since the object and image coincide at the same position, the distance between them is zero, which is the minimum possible.

8. Answer: B

Explanation: For a convex mirror, the focal length is negative. Using $P = \frac{1}{f}$with $f = -0.20$m gives the power as −5 D.

9. Answer: C) $\frac{(n+1)}{n} f$

Explanation:

For a concave mirror, the magnification is $m = \frac{v}{u} = n$, so $v = nu$. Using the mirror formula,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} = \frac{1}{u} + \frac{1}{nu} = \frac{n+1}{nu}$$

which gives

$$u = \frac{(n+1)}{n} f.$$

Hence, the distance of the object from the mirror is

$$\boxed{\frac{(n+1)}{n} f},$$

10. Answer: A

Explanation: Since the Sun is extremely far away, its image is formed at the focus of the concave mirror. Here, $f = \frac{R}{2} = 3$ m. The image diameter is $D_i = f \times \theta =$

$3 \times \frac{8,64,100}{9,29,00,000} \approx 0.0279\,\text{m} = 27.9\,\text{mm} \approx 28\,\text{mm}$. Note: This calculation gives the diameter of the image. If the question asks for the radius, the correct value is approximately 14 mm.

11. Answer: C

Explanation: A dentist uses a concave mirror to obtain an enlarged, erect image of the tooth. The calculated radius of curvature is 1.6 cm.

12. Answer: D

Explanation: A four-times enlarged erect image can only be formed by a concave mirror with the object placed between the pole and the focus. The focal length is 24 cm.

13. Answer: B

Explanation: The rays converge behind the convex mirror, forming a virtual object at $u = +10$ cm. Using the mirror formula with $f = +15$ cm, we get $v = -30$ cm, indicating a real image. The magnification is $m = -\frac{v}{u} = +3$, so the image is erect and three times the size of the object.

14. Answer: D

Explanation: For a real image on a wall, the mirror must be concave ($f = -1$ m). If the object distance is $-u$, the image distance is $-(u + 1.5)$. Using $\frac{1}{-u} + \frac{1}{-(u+1.5)} = \frac{1}{-1}$, solving the quadratic equation gives $u = 1.5$ m.

15. Answer: D

Explanation: A concave mirror can produce an image twice the size of the object in two positions. The first position is when the object lies between the pole and the focus.

16. Answer: C

Explanation: Similar to Q13, the virtual object is at $u = +20$ cm with $f = +30$ cm. The mirror formula yields $\frac{1}{v} = \frac{1}{30} - \frac{1}{20} = -\frac{1}{60} \Rightarrow v = -60$ cm (Real image).

Magnification $m = -\frac{-60}{20} = +3$ (Erect image).

17. Answer: B

Explanation: Using the given magnification and the mirror formula, the focal length of the concave mirror is found to be 15 cm.

18. Answer: C

Explanation: Applying the mirror formula for a convex mirror gives the image distance as $\frac{20}{3}$cm behind the mirror. The image is virtual and erect.

19. Answer: A

Explanation: Given concave mirror with $R = 15$ cm $\Rightarrow f = -7.5$ cm and $u = -10$ cm.

$\frac{1}{v} = \frac{1}{-7.5} - \frac{1}{-10} = -\frac{2}{15} + \frac{1}{10} = -\frac{1}{30} \Rightarrow v = -30$ cm

The negative sign confirms it is real and inverted, and $|m| = \left|-\frac{v}{u}\right| = 3$.

20. Answer: D

Explanation: Given $u = -20$ cm and a real image with $m = -4.5 = -\frac{9}{2}$. Using $m = \frac{f}{f-u}$:

$-\frac{9}{2} = \frac{f}{f+20} \Rightarrow -9f - 180 = 2f \Rightarrow f = -\frac{180}{11}$ cm

21. Answer: C

Explanation: A convex mirror always forms a virtual, erect, and diminished image for a real object. No other mirror always produces this type of image.

22. Answer: C

Explanation: The calculated image height is about 0.56 cm, which is approximately 0.55 cm after rounding. Therefore, option C is the closest answer.

23. Answer: A

Explanation: For an inverted image that is twice the size of the object, the object should be placed at 75 cm from the mirror.

24. Answer: A

Explanation: At an object distance of $4f$, the magnification is 1/3. Therefore, a 6 cm object forms an image of 2 cm.

25. Answer: D

Explanation: Convex mirror has $f = +30$ cm and $u = -30$ cm. Using the mirror formula:

$\frac{1}{v} = \frac{1}{30} - \frac{1}{-30} = \frac{2}{30} = \frac{1}{15} \Rightarrow v = +15$ cm

A positive value signifies the image forms 15 cm behind the mirror.

LEVEL-II

1. Answer: A

Explanation: The object is placed at $2f$(40 cm for a mirror with $f = 20$cm), i.e., at the centre of curvature. A concave mirror forms a real, inverted, and same-sized image at the centre of curvature.

2. Answer: D

Explanation: A convex mirror always forms a virtual, erect, and diminished image for a real object. Hence, the statement "Image is real" is incorrect.

3. **Answer: A**

Explanation: For a convex mirror, the image is virtual and erect, so $m = +\frac{1}{2}$. Using the magnification relation,

$m = \frac{f}{f-u}$

$\frac{1}{2} = \frac{10}{10-u} \Rightarrow u = -10$ cm

Thus, the object is 10 cm in front of the mirror.

4. Answer: B

Explanation: The focal length of a spherical mirror depends only on its radius of curvature $\left(f = \frac{R}{2}\right)$. It is independent of the surrounding medium, so immersing the mirror in water does not change its focal length.

5. **Answer: A**

Explanation: Since the image is formed on a wall, it is real, so a concave mirror is used. With magnification $|m| = 16$, $v = 16u$. Given the object-image separation is 120 cm,

$16u - u = 120$

$15u = 120 \Rightarrow u = 8$ cm

Hence, the flower is 8 cm from the mirror.

6. Answer: C

Explanation: A concave mirror forms an image twice the object's size in two cases: one real and inverted and the other virtual and erect. In both cases, the distance between the object and image positions is 30 cm.

7. Answer: B

Explanation: The image is erect and reduced from 6 m to 2 m, so the magnification is $+\frac{1}{3}$. Only a convex mirror produces an erect, diminished image of a real object.

8. Answer: C

Explanation: Parabolic reflectors eliminate spherical aberration. When the light source is placed at the focus, the reflected rays become parallel, producing a bright, well-focused beam used in automobile headlights.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

9. Correct Options: A, B, C, D

Explanation:

A convex mirror forms a virtual image for a real object, but it can also form a real image when the incident rays are converging (i.e., for a virtual object). A concave mirror forms a magnified virtual image when the object is placed between the Focus (F) and the Pole (P), and it forms an inverted real image when the object is placed beyond the Focus (F).

10. Correct Options: A, B

Explanation:

For any position of a real object, both a plane mirror and a convex mirror always produce virtual and erect images. A concave mirror can produce either a real, inverted image or a virtual, erect image depending on the object's position relative to the focus.

11. Correct Options: A, B, C *(The question asks for statements that are not correct.)*

Explanation:

The focal length of a mirror depends only on its geometry ($f = R/2$). It does not depend on the wavelength of light or the surrounding medium, making statements A and B incorrect. Spherical mirrors suffer from spherical aberration because marginal and paraxial rays do not meet at the same point, making statement C incorrect. Mirrors do not produce chromatic aberration since reflection is independent of wavelength, so statement D is correct.

12. Correct Options: A, B

Explanation:

A concave mirror can form both real and virtual images depending on the object's position. When the object is beyond the focus, the image is real and inverted. When the object is between the pole and the focus, the image is virtual and erect. Therefore, the image is not always inverted or always erect.

13. Correct Options: A, B

Explanation:

A concave mirror can produce a magnified image that is either real (inverted, $m = -2$) or virtual (erect, $m = +2$). Using the magnification formula $m = \frac{f}{f-u}$, the object distance is $\frac{f}{2}$for the virtual image and $\frac{3f}{2}$for the real image. Hence, both object distances are possible.

14. Correct Options: A, C, D

Explanation:

The Pole, Radius of Curvature, and Principal Axis are fixed geometric properties of a spherical mirror and do not change with the incident rays. However, due to spherical aberration, marginal rays do not converge at **the** paraxial focus, so the

effective focus shifts for marginal rays. Therefore, only the paraxial focus remains fixed.

15. Correct Option: A

Using the mirror formula:

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}+\frac{1}{-30}=\frac{1}{-20}$$

$$\frac{1}{v}=\frac{1}{30}-\frac{1}{20}=-\frac{1}{60}$$

$v = -60$ cm

The negative sign indicates that the image is formed 60 cm in front of the mirror.

16. Correct Option: B

Using the magnification formula:

$$m=-\frac{v}{u}$$

$$m=-\frac{-60}{-30}=-2$$

The magnitude of magnification is 2.

17. Correct Option: A

If the object is moved 10 cm towards the mirror:

$u = -30 + 10 = -20$ cm

Since the object is now at the focal point ($f = -20$ cm), the reflected rays become parallel and the image is formed at infinity.

Passage – II (Convex Mirror)

18. Correct Option: B

Using the mirror formula:

$$\frac{1}{v}+\frac{1}{-20}=\frac{1}{10}$$

$$\frac{1}{v}=\frac{1}{10}+\frac{1}{20}=\frac{3}{20}$$

$$v=\frac{20}{3}\text{ cm}$$

The positive sign indicates that the image is formed behind the mirror.

19. Correct Option: A

Using the magnification formula:

$$m=-\frac{v}{u}$$

$$m=-\frac{20/3}{-20}=\frac{1}{3}$$

20. Correct Option: C

Image height:

$$h_i = m \times h_o$$

$$h_i=\frac{1}{3}\times 0.5=\frac{1}{6}\text{ cm}\approx 0.167\text{ cm}$$

Assertion and Reason Type

21. Correct Option: A

Assertion (A): True. A concave mirror forms a virtual image when the object is placed between the pole and the focus.

Reason (R): True. In this position, the reflected rays diverge and appear to meet behind the mirror, producing a virtual image.

Since the reason correctly explains the assertion:

22. Correct Option: A

Assertion (A): True. For a real image formed by a spherical mirror, both $u$and $v$are negative, making

$m = -\frac{v}{u}$

negative.

Reason (R): True. Real images formed by spherical mirrors are inverted, so their magnification is negative.

Since the reason correctly explains the assertion:

Matrix / Matching Type

23.

- a → Q: For a convex mirror, when $u = -f$,
  $v = \frac{f}{2}, m = +\frac{1}{2}$
- b → S: An object at the centre of curvature of a concave mirror forms an inverted image of the same size at the centre of curvature ($m = -1$).
- c → P: An object placed at the focus of a concave mirror forms its image at infinity.
- d → T: For a convex mirror, when $u = -2f$,
  $v = \frac{2f}{3}, m = +\frac{1}{3}$

24.

- a → Q, T: $m = +\frac{1}{4}$represents a diminished, erect image formed by a convex mirror. The image is virtual.
- b → P, S: $m = -1$represents a real, inverted image of the same size formed by a concave mirror.
- c → P, T: $m = +2$represents a magnified, erect, virtual image formed by a concave mirror.
- d → R, T: $m = +1$represents an erect image of the same size formed by a plane mirror. The image is virtual.

Integer Type

25. Answer: 2

Given:

- Concave mirror
- $u = -5$cm
- $f = -10$cm

Using the mirror formula:

$\frac{1}{v} + \frac{1}{-5} = \frac{1}{-10}$

$\frac{1}{v} = \frac{1}{5} - \frac{1}{10} = \frac{1}{10}$

$v = +10$ cm

Magnification:

$m = -\frac{v}{u}$

$m = -\frac{10}{-5} = 2$

26 & 27. Answer: 10 (Both are same question)

Given:

- Convex mirror

- $f = +15\text{cm}$
- $u = -30\text{cm}$

Using the mirror formula:

$\frac{1}{v} + \frac{1}{-30} = \frac{1}{15}$

$\frac{1}{v} = \frac{1}{15} + \frac{1}{30} = \frac{1}{10}$

$v = +10$ cm

28. Answer: 20

For a spherical mirror:

$f = \frac{R}{2}$

Given:

$R = 40$ cm

$f = \frac{40}{2} = 20$ cm

PREVIOUS YEAR QUESTIONS

1. Correct option: B)

Find Initial Position: Using the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$with $u = -19\,m$and $f = +1\,m$, the initial image position is

$v_1 = \frac{19}{20} = 0.95\,m.$

Find Final Position: After 1 second, the object moves 4 m closer ($u = -15\,m$), which results in a new image position of

$v_2 = \frac{15}{16} = 0.9375\,m.$

Calculate Speed: The average speed is the distance moved by the image divided by time:

$|\,0.95 - 0.9375\,|/1\,s = 0.0125\,m/s$(or $\frac{1}{80}\,m/s$).

2. Correct option: A)

Initially, the man is at distance y; after moving 5 m away, his new distance from the mirror is $y + 5$.

The total distance between the man and his image is twice his distance from the mirror, so $2(y + 5) = 20$.

Dividing by 2 gives $y + 5 = 10$, which results in the initial distance $y = 5$ *metres*.

3. Correct option: B)

Case 1: With $m = -6$and object distance $u = -x$, the formula

$-6 = \frac{-F}{-F - (-x)}$

simplifies to the relation $7F = 6x$.

Case 2: When moved to $u = -(x + 3)$with $m = -4$, the formula

$-4 = \frac{-F}{-F + x + 3}$

simplifies to the equation $5F - 12 = 4x$.

Final Step: Substituting $x = \frac{7F}{6}$from the first equation into the second gives

$5F - 12 = \frac{14F}{3}$,

which solves to $F = 36$ cm.

4. Correct option: C)

When a plane mirror rotates by an angle θ, the reflected ray rotates by double that angle (2θ). Since the mirror rotates by 15°, the reflected ray shifts by an angle of 2 × 15° = 30°. The displacement d on a screen at distance L is found using the trigonometric formula d = L tan (2θ). Substituting L = 2.0 m and 2θ = 30°, the displacement is d = 2 × tan (30°) ≈ 1.154 m.

5. Correct option: C)

Statement 1 is true, as convex mirrors always form upright *(positive m)* virtual images, while concave mirrors form both upright virtual and inverted *(negative m)* real images.

Statement 2 is false, because magnification is defined as the ratio of image height to object height *($m = h_i / h_o$)*, and its sign is an absolute physical property indicating orientation, not a choice of convention.

6. Correct option: C)

For a convex mirror, the magnification is

$m = \frac{1}{2N+1}$

and the focal length is $f = \frac{R}{2}$.

Using the relationship $m = \frac{f-v}{f}$,

we substitute the values to get $\frac{1}{2N+1} = \frac{\frac{R}{2}-v}{\frac{R}{2}}$,

which simplifies to

$1 - \frac{2v}{R} = \frac{1}{2N+1}$.

Solving for $v$gives

$\frac{2v}{R} = 1 - \frac{1}{2N+1} = \frac{2N}{2N+1}$,

resulting in the image distance

$v = \frac{NR}{2N+1}$.

7. Correct option: D)

For a short linear object lying along the principal axis, the **longitudinal magnification** is calculated as

$m_L = \left(\frac{f}{f-u}\right)^2$.

Substituting the given values ($f = -15$ cm and $u = -26.5$cm), the magnification is approximately

$\left(\frac{-15}{-15+26.5}\right)^2 = \left(\frac{-15}{11.5}\right)^2 \approx 1.7$.

Multiplying the object length by this factor (3 cm × 1.7), the size of the image is approximately **5.1 cm**.

8. The correct answer is B.

Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.

Statement 1 is True: We can identify a mirror by the size of the erect image: a Plane mirror shows the same size, a Convex mirror shows a diminished image, and a Concave mirror shows a magnified image.

Statement 2 is True: For a concave mirror, when a real object is placed at the Center of Curvature (C), the real image is formed at the exact same location, making the distance between them zero.

Relationship: Statement 2 describes a specific property of real images in concave mirrors, which does not explain the logic used in Statement 1 to identify mirrors using erect (virtual) images.

2. Light – Refraction

TOPIC-1: Lenses

LEVEL – I

1. Answer: A

Explanation: The standard formula for refraction at a spherical surface separating two media of refractive indices $\mu_1$and $\mu_2$is

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}.$$

2. Answer: D

Explanation: Lateral (linear) magnification is the ratio of the height of the image to the height of the object:

$$m = \frac{h_i}{h_o}.$$

3. Answer: D

Explanation: Each curved surface of a lens forms part of a sphere. The centre of that sphere is called the **centre of curvature**.

4. Answer: C

Explanation: Parallel rays incident on a convex lens converge at the **principal focus (F)** after refraction.

5. Answer: B

Explanation: A ray passing through the optical centre of a thin lens passes **undeviated**, without changing its direction.

6. Answer: B

Explanation: A convex lens forms a virtual, erect, and magnified image only when the object is placed **between the optical centre and the principal focus**.

7. Answer: B

Explanation: The thin lens formula is

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}.$$

8. Answer: D

Explanation: The ray enters the first surface without deviation, so $\mu_1 = \mu$. At the second surface it bends away from the normal, indicating $\mu > \mu_2$. Therefore, $\mu_1 = \mu > \mu_2$.

9. Answer: None of the given options is correct.

For a real image formed by a convex lens, the image is inverted, meaning it is rotated by 180° (both top-bottom and left-right are reversed).

Original arrangement:

1 2
3 4

After a 180°inversion:

4 3
2 1

So the correct image should be:

- Top-left = 4
- Top-right = 3
- Bottom-left = 2
- Bottom-right = 1

10. Answer: A

Explanation: Using

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R},$$

with $\mu_1 = 1.5$, $\mu_2 = 1$, $u = -3$ cm, $R = -5$ cm,

$$\frac{1}{v} + 0.5 = 0.1$$

giving

$v = -2.5$ cm.

Hence, the image is **2.5 cm inside the glass**.

11. Answer: B

Explanation: The combined power is

$P = P_1 + P_2 - dP_1P_2$.

For zero power,

$d = f_1 + f_2$.

12. Answer: C

Explanation: Since the image is diminished and real,

$m = -\frac{1}{n}$.

Using the magnification relation,

$u = -(n+1)f$.

Hence, the object is placed at a distance of $(n+1)f$.

13. Answer: A

Explanation: For a real image,

$m = -m_{\text{linear}}$.

Using the lens formula,

$u = -\left(\frac{m+1}{m}\right)f$.

Thus, the object distance is

$\left(\frac{m+1}{m}\right)f$.

14. Answer: D

Explanation: Each material has a different refractive index and focal length, so each section forms its own image. Therefore, **3 images** are formed.

15. Answer: A

Explanation: Using

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R},$$

with $\mu_1 = 1$, $\mu_2 = 1.5$, $u = -15$ cm, $R = 30$ cm,

$v = -30$ cm.

The image is formed **30 cm to the left of the surface**.

16. Answer: B

Explanation: The uncovered portions of the lens still produce the complete image at the same position, but **its brightness decreases** because less light passes through.

17. Answer: C

Explanation: The object height is the geometric mean of the two image heights:

$$h_o = \sqrt{8 \times 2} = 4 \text{ cm}.$$

18. Answer: D

Explanation: The effective power is

$P = 2P_L + P_M$.

Since

$P_L = \frac{1}{20}$ D, $P_M = 0$,

$P = \frac{1}{10}$ D,

giving an effective focal length of

$$f = 10 \text{ cm}.$$

19. Answer: C

Explanation: Using

$$P_w = P_a\left(\frac{\frac{\mu_g}{\mu_w} - 1}{\mu_g - 1}\right),$$

$$P_w = 4\left(\frac{1.2 - 1}{0.6}\right) = \frac{4}{3} \approx 1.3 \text{ D}.$$

20. Answer: A

Explanation: The upper and lower transparent portions still focus light at the same point, so **only one image** is formed, although its brightness is reduced.

21. Answer: (B) Move to the right

Moving lens B to the right means it intercepts the converging rays from lens A at a later stage. As the virtual object distance for lens B decreases, applying the lens formula shows that the final image shifts further to the right.

22. Answer: (B) Convex, 50cm

The net power of the lens combination is:

$P = P_1 + P_2 = +6\,\text{D} - 4\,\text{D} = +2\,\text{D}$

Since the net power is positive, the combination behaves as a convex lens with focal length:

$f = \frac{1}{P} = \frac{1}{2}\,\text{m} = 50\,\text{cm}$

23. Answer: A) Parallel to the principal axis

A parallel beam first converges at the focal point of the convex lens (1.0 m behind it). The concave lens is placed 0.75 m away, so this focal point is 0.25 m behind the concave lens, exactly at its virtual focus. Hence, the emerging rays become parallel to the principal axis.

24. Answer: C) 109.1cm

A silvered plano-convex lens behaves as an effective concave mirror with power:

$P = 2P_L + P_M$

Since the plane surface is silvered, $P_M = 0$, so:

$P = \frac{2}{f_L} = -\frac{1}{F}$

Using the mirror formula:

$$\frac{1}{-120}+\frac{1}{-100}=\frac{1}{F}$$
$$F=-\frac{600}{11}\text{ cm}$$
Therefore,
$$f_L=-2F=\frac{1200}{11}\approx 109.1\text{ cm}$$

25. Answer: A) 6cm from S_1

For both images to coincide, one must be real and the other virtual. Using the lens formula for sources at distances $x$and $24-x$cm and equating the image positions gives:
$$x^2-24x+108=0$$
The valid solution is:
$$x=6\text{ cm}$$
Thus, the source should be placed **6 cm from** S_1.

26. Answer: D) 2f

Cutting a lens horizontally does not change the radii of curvature, so the focal length remains f. Cutting it vertically halves the aperture without changing curvature, effectively doubling the focal length to $2f$. Hence, a quarter piece (Part A), obtained by both cuts, has focal length:
$2f$

27. Answer: D) 20 cm

For the final image position to remain independent of the separation b, the rays emerging from the first lens must be parallel. This occurs when the object is placed at the focal point of the plano-convex lens.
Using the lens maker's formula:
$$\frac{1}{f_1}=(1.5-1)\left(\frac{1}{10}-\frac{1}{\infty}\right)=\frac{1}{20}$$

Therefore,
$$a=f_1=20\text{ cm}$$

LEVEL – II

1. Answer: B) $\frac{\mu R}{\mu-1}$

Using the refraction formula at a spherical surface:
$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$
For light passing from a medium (μ)to air with $v=\infty$:
$$\frac{1}{\infty}-\frac{\mu}{-d}=\frac{1-\mu}{-R}$$
$$\frac{\mu}{d}=\frac{\mu-1}{R}$$
Hence,
$$d=\frac{\mu R}{\mu-1}$$

2. Answer: B) $n_2>n_1$

A convex lens converges parallel rays in air. If it causes them to diverge, the surrounding medium must have a refractive index greater than that of the lens material:

$n_{\text{medium}} > n_{\text{lens}}$

3. Answer: B) $\frac{f_1 f_2}{f_1+f_2}$

For two thin lenses in contact, the equivalent power is:

$P = P_1 + P_2$

or

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

Thus, the equivalent focal length is:

$$F = \frac{f_1 f_2}{f_1 + f_2}$$

4. Answer: A) 3.5D (closest approximation to 3.33 D)

The power of a refracting spherical surface is:

$$P = \frac{\mu_2 - \mu_1}{R}$$

Substituting the given values:

$$P = \frac{\frac{4}{3} - 1}{0.1} = \frac{1}{0.3} = 3.33\ \text{D}$$

This is closest to **3.5 D**.

5. Answer: B) 15.0 cm

Using the lens maker's formula:

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Substituting the values:

$$\frac{1}{10} = (1.5 - 1)\left(\frac{1}{7.5} - \frac{1}{R_2}\right)$$

$$\frac{1}{5} = \frac{1}{7.5} - \frac{1}{R_2}$$

Solving gives:

$|R_2| = 15.0\ \text{cm}$

6. Answer: A) 1 D

The powers of the individual lenses are:

$$P_1 = \frac{1}{0.5} = +2\ \text{D}$$

$$P_2 = \frac{1}{-1} = -1\ \text{D}$$

Hence, the net power is:

$P = P_1 + P_2 = +2 - 1 = +1\ \text{D}$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

7. Answer: A, C

The general formula for refraction at a spherical surface is:

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

For the First Principal Focus (f_1), the image is formed at infinity ($v = \infty$).

$$\frac{\mu_2}{\infty} - \frac{\mu_1}{f_1} = \frac{\mu_2 - \mu_1}{R}$$

$$f_1 = \frac{-\mu_1 R}{\mu_2 - \mu_1}$$

Hence, Option A is correct.

For the Second Principal Focus (f_2), the object is at infinity ($u = -\infty$).

$$\frac{\mu_2}{f_2} - \frac{\mu_1}{-\infty} = \frac{\mu_2 - \mu_1}{R}$$

$$f_2 = \frac{\mu_2 R}{\mu_2 - \mu_1}$$

Hence, Option C is correct.

8. Answer: A, C

For a plane refracting surface, the radius of curvature is infinite ($R = \infty$).

Substituting this into the refraction formula:

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{\infty} = 0$$

Therefore,

$$\frac{\mu_2}{v} = \frac{\mu_1}{u}$$

Rearranging,

$$\frac{v}{u} = \frac{\mu_2}{\mu_1}$$

Hence, Option A is correct.

Cross-multiplying,

$$v\mu_1 = u\mu_2$$

Hence, Option C is also correct.

9. Answer: A, D

Using the spherical refraction formula for light traveling from air ($\mu_1 = 1$) into a medium ($\mu_2 = 1.5$) through a convex surface of radius $R = +20$cm, with the object at infinity ($u = -\infty$):

$$\frac{1.5}{v} - \frac{1}{-\infty} = \frac{1.5 - 1}{20}$$

$$\frac{1.5}{v} = \frac{0.5}{20} = \frac{1}{40}$$

$$v = +60 \text{ cm}$$

Since the image distance is positive, the refracted rays actually converge and meet inside the second medium. Therefore, Option A and Option D are correct.

10. Answer: B, C, D

When light passes from vacuum into another medium:

- The frequency remains unchanged because it depends only on the source.

  $f = f_0$

  Hence, Option D is correct.
- The speed decreases according to

  $$v = \frac{v_0}{\mu}$$

  Hence, Option C is correct.
- Since $v = f\lambda$

  and frequency remains constant,

  $$\lambda = \frac{\lambda_0}{\mu}$$

Hence, Option B is correct.

11. Answer: A, C

Using the Lens Maker's Formula, the ratio of the focal length in water (f') to that in air (f) is:

$$\frac{f'}{f} = \frac{\mu_g - 1}{\left(\frac{\mu_g}{\mu_w}\right) - 1} = \frac{\frac{3}{2} - 1}{\left(\frac{3/2}{4/3}\right) - 1} = \frac{\frac{1}{2}}{\frac{9}{8} - 1} = \frac{1/2}{1/8} = 4$$

Hence, the new focal length is:

$f' = 4f$

For Option C (Nature of the New Image):

Initially, the image formed in air is real and magnified, which means the object is placed between the focal point and twice the focal length of the lens, i.e., $f < u < 2f$.

When the lens is immersed in water, its focal length becomes $f' = 4f$. Since the object distance $u$is less than $2f$, it is also less than the new focal length ($u < 4f$).

A convex lens always forms a virtual, erect, and magnified image when the object is placed within its focal length. Therefore, after immersing the lens in water, the new image becomes virtual and magnified.

12. Answer: C) R/2

Tracing the rays through the curved surface and then the plane surface shows that the final image depends on the sign of ($x-R$). For the image to remain virtual, the emerging rays must diverge, which requires:

$x < R$

Among the given choices, only:

$x = \frac{R}{2}$

satisfies this condition.

13. Answer: D) Inverted and same size

When:

$x = 2R$

the magnification at the curved surface is:

$m_1 = -1$

and at the plane surface:

$m_2 = +1$

Therefore, the total magnification is:

$m = m_1 m_2 = (-1)(+1) = -1$

Hence, the final image is **inverted and of the same size as the object**.

14. Correct Option: A

Analysis: The object is at $x = R$, so its distance from the center of curvature $C$of the hemisphere is $2R$. For grazing incidence at the first spherical surface, the angle of incidence is $i = 90°$.

Using Snell's law at the curved surface:

$$1 \times \sin 90° = 2 \times \sin r$$

$$\sin r = \frac{1}{2} \Rightarrow r = 30°$$

From the geometry of the hemisphere, the refracted ray reaches the plane surface at an angle of 30°with the normal.

Applying Snell's law at the plane surface:

$2 \times \sin 30° = 1 \times \sin e$

$2 \times \frac{1}{2} = \sin e = 1$

$e = 90°$

15. Correct Option: D

Analysis: Region A represents an object placed beyond the principal focus of a convex lens. A real object beyond the focus of a converging lens always forms a real, inverted image on the opposite side of the lens beyond its focus, which corresponds to region D.

16. Correct Option: C

Analysis: The incident rays are converging, so the object is virtual ($u > 0$). Using the lens formula,

$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$

Since both f and u are positive,

$\frac{1}{v} > \frac{1}{f}$

which gives

$0 < v < f$.

Hence, the image is real and forms between the lens and the right principal focus, i.e., in region C.

17. Analysis: Since the two lenses are in contact,

$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Substituting,

$\frac{1}{F} = \frac{1}{f} - \frac{1}{3f} = \frac{2}{3f}$

Therefore,

$F = \frac{3f}{2} = 1.5f$.

However, the answer choices are given as image regions (A, B, C, D) instead of focal length values. Hence, the question is structurally invalid.

18. Correct Option: A

Analysis: For a convex lens in air, the refractive index of the lens is greater than that of air, and

$\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

is positive. Therefore, by the lens maker's formula, the focal length is always positive. The reason correctly explains the assertion.

19. Correct Option: D

Analysis: The assertion is false because a concave lens does not always form a virtual, erect image. If the incident rays are sufficiently converging (virtual object), it can form a real image. The reason correctly describes the behaviour of a concave lens for a real object.

20 Match the Column

- **a → Q:** A convex lens placed in a denser medium behaves like a diverging lens because its focal length changes sign.
- **b → Q:** A concave lens in air naturally behaves as a diverging lens.
- **c → R:** A silvered plano-convex lens acts as a concave mirror with positive power.
- **d → S:** A plano-concave lens silvered on its plane surface behaves like a convex mirror with negative power.

21 Match the Column

- Converging lens in air → Positive focal length.
- Diverging lens in air → Negative focal length.
- Lens immersed in a denser medium → Nature reverses (converging becomes diverging and vice versa).
- Mirror immersed in a liquid → Focal length remains unchanged because it depends only on the radius of curvature.

22. **Analysis:** Using the lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Given,

$u = -30$ cm, $v = +15$ cm

$$\frac{1}{f} = \frac{1}{15} - \left(-\frac{1}{30}\right) = \frac{1}{15} + \frac{1}{30} = \frac{3}{30} = \frac{1}{10}$$

Therefore,

$f = 10$ cm.

23. **Analysis:** A convex lens forms an image at infinity only when the object is placed at its principal focus. Hence, the focal length is equal to the object distance.

$f = 20$ cm

24. **Analysis:** This is refraction at a single spherical surface. Using the formula,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Given,

$u = -\infty, \mu_1 = 1, \mu_2 = 1.5, R = 10$ cm

Since $\frac{1}{u} = 0$,

$$\frac{1.5}{v} = \frac{1.5 - 1}{10} = \frac{0.5}{10}$$

$$v = \frac{1.5 \times 10}{0.5} = 30 \text{ cm}$$

Therefore, the image is formed **30 cm** inside the second medium.

3. Electrostatics

LEVEL-I

1. **Answer :** C)

A static electric field is a conservative field.

$\oint \vec{E} \cdot d\vec{l} = 0$

Therefore, any diagram showing electric field lines forming closed continuous loops represents a non-static electric field.

2. **Answer :** B) 1:1

Let the two point charges be x and y.

The total charge is Q.

$x + y = Q$

$y = Q - x$

The electrostatic force between the charges is given by Coulomb's law.

$F = \frac{k \cdot x \cdot (Q - x)}{R^2}$

To find the condition for maximum force, we differentiate F with respect to x and set it to zero.

$\frac{dF}{dx} = \frac{k}{R^2} \cdot \frac{d}{dx}(xQ - x^2) = 0$

$Q - 2x = 0$

$x = \frac{Q}{2}$

Therefore, the second charge y is calculated as follows.

$y = Q - \frac{Q}{2} = \frac{Q}{2}$

The ratio of the charges $x:y$ is evaluated.

$x:y = \frac{Q}{2}:\frac{Q}{2} = 1:1$

3. **Answer :** B) $-\frac{Q}{4}$

Let the distance between the two $+Q$ charges be r.

The distance from either $+Q$ charge to the center charge q is $\frac{r}{2}$.

For the entire system to be in equilibrium, the net force on any individual charge must be zero.

Let us evaluate the net force on one of the outer $+Q$ charges.

$F_{net} = \frac{k \cdot Q \cdot Q}{r^2} + \frac{k \cdot Q \cdot q}{\left(\frac{r}{2}\right)^2} = 0$

$\frac{k \cdot Q^2}{r^2} + \frac{4 \cdot k \cdot Q \cdot q}{r^2} = 0$

We can divide the entire equation by the common factor $\frac{k \cdot Q}{r^2}$.

$Q + 4q = 0$

$4q = -Q$

$q = -\frac{Q}{4}$

4. **Answer :** B) $1:-\sqrt{3}$

Let the side length of the equilateral triangle be a.

The distance from any corner to the centroid (where q is placed) is $r = \frac{a}{\sqrt{3}}$.

For the system to be in equilibrium, the net force on any corner charge Q must be zero.
The force on a corner charge Q due to the other two Q charges is the vector sum of two forces at a 60° angle.

$$F_{Q_resultant} = \sqrt{F^2 + F^2 + 2F^2\cos(60°)} = \sqrt{3}F = \sqrt{3}\frac{k \cdot Q^2}{a^2}$$

This resultant force must be perfectly balanced by the attractive force from the center charge q.

$$F_q = \frac{k \cdot Q \cdot q}{\left(\frac{a}{\sqrt{3}}\right)^2} = \frac{3 \cdot k \cdot Q \cdot q}{a^2}$$

Equating the sum of forces to zero.

$$\sqrt{3}\frac{k \cdot Q^2}{a^2} + \frac{3 \cdot k \cdot Q \cdot q}{a^2} = 0$$

$$\sqrt{3} \cdot Q + 3 \cdot q = 0$$

$$3 \cdot q = -\sqrt{3} \cdot Q$$

$$\frac{q}{Q} = -\frac{\sqrt{3}}{3} = -\frac{1}{\sqrt{3}}$$

The ratio of q to Q is $1{:}-\sqrt{3}$.

5. **Answer :** D) $Q/8$

When an uncharged identical conducting sphere touches a charged sphere, the total charge is shared equally.
Initially, the first sphere has charge Q.
After touching the first identical uncharged sphere, the charge is divided by 2.

$$Q_1 = \frac{Q + 0}{2} = \frac{Q}{2}$$

After touching the second identical uncharged sphere, the new charge is divided by 2 again.

$$Q_2 = \frac{\frac{Q}{2} + 0}{2} = \frac{Q}{4}$$

After touching the third identical uncharged sphere, the charge is halved once more.

$$Q_3 = \frac{\frac{Q}{4} + 0}{2} = \frac{Q}{8}$$

6. **Answer :** A) 2L

Let the point where the net electric field is zero be at a distance x from the origin ($x = 0$).
Since the charges $+8q$ and $-2q$ have opposite signs, the neutral point must lie outside the charges, closer to the smaller charge in magnitude (which is $-2q$ at $x = L$).
Let the point be at $x = L + d$.
The distance to $+8q$ is $L + d$, and the distance to $-2q$ is d.
We equate the magnitudes of the electric fields produced by both charges at this point.

$$\frac{k \cdot (8q)}{(L + d)^2} = \frac{k \cdot (2q)}{d^2}$$

Dividing both sides by $2kq$.

$$\frac{4}{(L+d)^2} = \frac{1}{d^2}$$

Taking the square root of both sides.

$$\frac{2}{L+d} = \frac{1}{d}$$

Cross-multiplying to solve for d.

$2d = L + d$

$d = L$

The location on the x-axis is $x = L + d$.

$x = L + L = 2L$

7. **Answer :** B) $\frac{Q}{4}\left(1 + 2\sqrt{2}\right)$

Let the side length of the square be a.

The distance from any corner to the center of the square is $r = \frac{a}{\sqrt{2}}$.

For the system to be in equilibrium, the net force on any corner charge $-Q$ must be zero.

The force on one $-Q$ charge due to the other three $-Q$ charges consists of two adjacent forces and one diagonal force.

$$F_{adjacent} = \sqrt{\left(\frac{k \cdot Q^2}{a^2}\right)^2 + \left(\frac{k \cdot Q^2}{a^2}\right)^2} = \sqrt{2}\frac{k \cdot Q^2}{a^2}$$

$$F_{diagonal} = \frac{k \cdot Q^2}{\left(\sqrt{2}a\right)^2} = \frac{k \cdot Q^2}{2a^2}$$

The total repulsive force from the other $-Q$ charges is their sum.

$$F_{repulsion} = \frac{k \cdot Q^2}{a^2}\left(\sqrt{2} + \frac{1}{2}\right)$$

This must be balanced by the attractive force from the center charge q.

$$F_q = \frac{k \cdot (-Q) \cdot q}{\left(\frac{a}{\sqrt{2}}\right)^2} = \frac{2 \cdot k \cdot (-Q) \cdot q}{a^2}$$

Setting the net force to zero.

$$\frac{k \cdot Q^2}{a^2}\left(\sqrt{2} + 0.5\right) - \frac{2 \cdot k \cdot Q \cdot q}{a^2} = 0$$

$$Q\left(\frac{2\sqrt{2}+1}{2}\right) = 2q$$

$$q = \frac{Q}{4}\left(1 + 2\sqrt{2}\right)$$

8. **Answer :** $-2\sqrt{2}$

Let the side length of the square be a.

The force on one charge Q is due to the other charge Q (at the opposite diagonal) and the two q charges (adjacent).

The diagonal distance between the Q charges is $\sqrt{2}a$.

The repulsive force between the two Q charges is along the diagonal.

$$F_{QQ} = \frac{k \cdot Q^2}{\left(\sqrt{2}a\right)^2} = \frac{k \cdot Q^2}{2a^2}$$

The forces due to the two q charges act along the sides of the square. Their resultant must act along the diagonal to cancel F_{QQ}.

$$F_{Qq_resultant} = \sqrt{\left(\frac{k \cdot Q \cdot q}{a^2}\right)^2 + \left(\frac{k \cdot Q \cdot q}{a^2}\right)^2} = \sqrt{2}\frac{k \cdot Q \cdot q}{a^2}$$

For the net electrical force on Q to be zero, the sum of these forces must be zero.

$$\frac{k \cdot Q^2}{2a^2} + \sqrt{2}\frac{k \cdot Q \cdot q}{a^2} = 0$$

Dividing by $\frac{k \cdot Q}{a^2}$.

$$\frac{Q}{2} + \sqrt{2} \cdot q = 0$$

$$\frac{Q}{2} = -\sqrt{2} \cdot q$$

$$\frac{Q}{q} = -2\sqrt{2}$$

9. **Answer :** C) -,+,+,-,+,-

Let E_0 be the electric field at the center O due to a single positive charge $+q$ at distance r.

If only one positive charge is at R, the field is E_0 directed towards U.

We need the net field to be $2E_0$ towards U.

Opposite vertices in a hexagon are (P,S), (Q,T), and (R,U).

To maximize the field along the R-U axis towards U, we should place $+q$ at R and $-q$ at U. This gives $2E_0$ towards U.

The remaining four positions (P, Q, S, T) must have charges that cancel each other's fields out at the center.

We have two $+q$ and two $-q$ left. We must place identical charges at opposite vertices so their fields cancel.

Thus, P and S must have the same charge, and Q and T must have the same charge.

However, we only have 3 positive and 3 negative charges total. Since R is + and U is −, we have two + and two − left.

We can place + at P and + at Q. But wait, opposite vertices must cancel. So if P is +, S must be +. If Q is −, T must be −.

This gives the configuration: P(+), Q(-), R(+), S(+), T(-), U(-).

Alternatively, we can place remaining + charges at Q and T and – charges at P and S

This gives the configuration: P(-), Q(+), R(+), S(-), T(+), U(-).

10. **Answer :** A) $-\frac{1}{2}$

Generally, the ratio of charges $\frac{q_1}{q_2}$ is equal to the ratio of the number of electric field lines originating or terminating at the respective charges.

$$\frac{q_1}{q_2} = \frac{N_1}{N_2}$$

From the field-line diagram:

Electric field lines originate from positive charges and terminate on negative charges.

Here, the arrows are directed from charge 1 toward charge 2.

Therefore, $q_1$is positive.

$q_2$is negative.

Now count the field lines:

4 field lines originate from charge 1.

8 field lines terminate on charge 2.
Of these, 4 come from charge 1 and 4 come from infinity, indicating that the magnitude of $q_2$is twice that of q_1.
Since the number of field lines is proportional to the magnitude of the charge,
$\frac{|\,q_1\,|}{|\,q_2\,|} = \frac{4}{8} = \frac{1}{2}$.
Including the signs,
$\boxed{\frac{q_1}{q_2} = -\frac{1}{2}}$.

11. **Answer :** C) 10^4 volt
The work done by an external agent to bring a test charge from infinity to a point P is related to the electric potential at P.
$W = q \cdot V_P$
We are given $W = 100$ J and $q = 10^{-2}$ C.
$100 = 10^{-2} \cdot V_P$
$V_P = \frac{100}{10^{-2}} = 100 \cdot 10^2 = 10^4$ V

12. **Answer :** B) -ve
Work done is defined as the change in potential energy.
$W = \Delta U = q' \cdot \Delta V$
Since the external work done by the charge q is negative, the electrostatic force between q and q' must be repulsive.
Because q' is a negative charge, for the force to be repulsive, q must also be a negative charge.
The electric potential V due to a negative point charge at any distance r is always negative.
$V = \frac{k \cdot (-q)}{r} < 0$

13. **Answer :** A) +ve
The electrostatic force between a positively charged fixed nucleus and a negatively charged electron is attractive.
To move the electron slowly away from the nucleus, an external agent must apply a force in the direction of the displacement (outwards) to counteract the inward attractive electrostatic force.
Since the external force and the displacement are in the same direction, the work done by the external agent is positive.
$W_{ext} = \int \vec{F}_{ext} \cdot d\vec{r} > 0$

14. **Answer :** B) -ve
The electric potential is a scalar quantity, and the net potential at a point is the algebraic sum of the potentials due to individual charges.
Let the initial potential at point P due to q_1 be V_1.
$V_1 = +10$ V
When charge q_2 is brought around point P, the new net potential V_{net} becomes 4 V.
Let the potential due to q_2 at P be V_2.
$V_{net} = V_1 + V_2$
$4\text{ V} = 10\text{ V} + V_2$
$V_2 = 4\text{ V} - 10\text{ V} = -6$ V

Since the potential V_2 due to charge q_2 is negative, the charge q_2 itself must be negative.

15. **Answer :** B) $3.2 \times 10^{-16} J$

The work done by the electric field is equal to the negative change in electrostatic potential energy.

$W_{electric} = -\Delta U = -q \cdot \Delta V$

The charge of an electron is $q = -1.6 \times 10^{-19}$ C.

The change in potential is the final potential minus the initial potential.

$\Delta V = V_{final} - V_{initial} = 1500 \text{ V} - (-500 \text{ V}) = 2000 \text{ V}$

Now, calculate the work done by the electric field.

$W_{electric} = -(-1.6 \times 10^{-19} \text{ C}) \cdot (2000 \text{ V})$

$W_{electric} = 1.6 \times 10^{-19} \cdot 2000 = 3.2 \times 10^{-16} \text{ J}$

16. **Answer :** D) $\left(2 - \sqrt{2}\right)\frac{KQ^2}{l}$

The work done by an external agent to shift a charge q_0 from point C to point D is equal to the change in potential energy.

$W_{ext} = q_0 \cdot (V_D - V_C)$

Here, $q_0 = -Q$. Let the side of the square be l. We calculate the potential at points C and D due to the other two fixed charges: $+Q$ at B and $-Q$ at A.

Potential at point C (V_C): The distance from B to C is l, and from A to C is $\sqrt{2}l$.

$$V_C = \frac{K \cdot Q}{l} + \frac{K \cdot (-Q)}{\sqrt{2}l} = \frac{KQ}{l}\left(1 - \frac{1}{\sqrt{2}}\right)$$

Potential at point D (V_D): The distance from B to D is $\sqrt{2}l$, and from A to D is l.

$$V_D = \frac{K \cdot Q}{\sqrt{2}l} + \frac{K \cdot (-Q)}{l} = \frac{KQ}{l}\left(\frac{1}{\sqrt{2}} - 1\right)$$

Now, find the difference $V_D - V_C$.

$$V_D - V_C = \frac{KQ}{l}\left(\frac{1}{\sqrt{2}} - 1\right) - \frac{KQ}{l}\left(1 - \frac{1}{\sqrt{2}}\right)$$

$$V_D - V_C = \frac{KQ}{l}\left(\frac{2}{\sqrt{2}} - 2\right) = \frac{KQ}{l}\left(\sqrt{2} - 2\right)$$

Calculate the work done.

$$W_{ext} = -Q \cdot \frac{KQ}{l}\left(\sqrt{2} - 2\right) = \frac{KQ^2}{l}\left(2 - \sqrt{2}\right)$$

17. **Answer :** C) $-\frac{KQq}{a}\left(1 - \frac{1}{\sqrt{5}}\right)$

The work done by the electrical force is the negative of the change in potential energy.

$W_{elec} = -\Delta U = -Q \cdot \left(V_{final} - V_{initial}\right) = Q \cdot (V_O - V_P)$

We need to find the potential at the origin $O(0,0)$ and at point $P(a,a)$ due to the fixed charges $-q$ at $(-a,0)$ and $+q$ at $(a,0)$.

Potential at the origin O:

$$V_O = \frac{K \cdot (-q)}{a} + \frac{K \cdot (+q)}{a} = 0$$

Potential at point $P(a,a)$: The distance from $(-a,0)$ to (a,a) is $\sqrt{\left(a - (-a)\right)^2 + (a-0)^2} = \sqrt{4a^2 + a^2} = \sqrt{5}a$. The distance from $(a,0)$ to (a,a) is $\sqrt{(a-a)^2 + (a-0)^2} = a$.

$$V_P = \frac{K \cdot (-q)}{\sqrt{5}a} + \frac{K \cdot (+q)}{a} = \frac{Kq}{a}\left(1 - \frac{1}{\sqrt{5}}\right)$$

Now, calculate the work done by the electrical force.

$$W_{elec} = Q \cdot \left(0 - \frac{Kq}{a}\left(1 - \frac{1}{\sqrt{5}}\right)\right) = -\frac{KQq}{a}\left(1 - \frac{1}{\sqrt{5}}\right)$$

18. **Answer :** B) decreases

The electrostatic force between a positively charged nucleus and a negatively charged electron is attractive.

When the electron is brought nearer to the nucleus, the displacement is in the direction of the conservative attractive force.

The work done by the internal conservative electrostatic force is positive.

$W_{internal} = -\Delta U > 0$

Therefore, the change in potential energy ΔU is negative, meaning the potential energy of the system decreases.

19. **Answer :** B) zero

The potential at any point due to a point charge q at the origin is given by:

$$V = \frac{k \cdot q}{r}$$

We calculate the distance of point A $(\sqrt{2}, \sqrt{2})$ from the origin (0,0).

$$r_A = \sqrt{\left(\sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2} = \sqrt{2+2} = \sqrt{4} = 2 \text{ m}$$

We calculate the distance of point B (2,0) from the origin.

$r_B = \sqrt{(2)^2 + (0)^2} = \sqrt{4} = 2$ m

Since $r_A = r_B$, both points lie on the same equipotential surface created by the charge at the origin.

$$V_A = V_B = \frac{k \cdot q}{2}$$

The potential difference between points A and B is:

$V_A - V_B = 0$

20. **Answer :** A) 0.1 V

The potential difference (ΔV) between two points is defined as the work done (W) per unit charge (q) in moving the charge between those points.

$$\Delta V = \frac{W}{q}$$

Given values are $W = 2$ J and $q = 20$ C. The distance of 2 cm is extra information not needed for finding the potential difference directly.

$$\Delta V = \frac{2 \text{ J}}{20 \text{ C}}$$

$\Delta V = 0.1$ V

21. **Answer :** B) $-\frac{2q}{2+\sqrt{2}}$

Let the right-angled isosceles triangle have sides of length a forming the right angle. The diagram implies charges $+q$ and $+q$ are separated by distance a, and the charge Q is at a distance a from one $+q$ charge and forms the hypotenuse distance $\sqrt{2}a$ with the other $+q$ charge.

The net electrostatic potential energy U of the three-charge system is the sum of the potential energies of all charge pairs.

$U = U_{q,q} + U_{Q,q_adjacent} + U_{Q,q_opposite} = 0$

$$\frac{k \cdot q \cdot q}{a} + \frac{k \cdot Q \cdot q}{a} + \frac{k \cdot Q \cdot q}{\sqrt{2}a} = 0$$

We can divide the entire equation by $\frac{k \cdot q}{a}$.

$$q + Q + \frac{Q}{\sqrt{2}} = 0$$

Factor out Q.

$$q + Q\left(1 + \frac{1}{\sqrt{2}}\right) = 0$$

$$Q\left(\frac{\sqrt{2}+1}{\sqrt{2}}\right) = -q$$

Isolate Q and rationalize by multiplying numerator and denominator by $\sqrt{2}$.

$$Q = \frac{-\sqrt{2}q}{\sqrt{2}+1} = \frac{-2q}{2+\sqrt{2}}$$

22. **Answer :** B) $\frac{q}{8\epsilon_o}$

According to Gauss's Law, the total electric flux emerging from a complete closed spherical surface enclosing a charge q is $\frac{q}{\epsilon_o}$.

The diagram shows a surface S bounded by the first octant of a 3D coordinate system (x, y, z all positive).

This curved surface represents exactly $\frac{1}{8}$th of a complete sphere centered at the origin.

By symmetry, the flux passing uniformly outward through this $\frac{1}{8}$th sector is:

$$\Phi = \frac{1}{8} \cdot \Phi_{total} = \frac{q}{8\epsilon_o}$$

23. **Answer :** D) -1:3

According to Gauss's Law, the electric flux Φ_E through any closed surface is proportional to the net charge enclosed by that surface.

$$\Phi = \frac{Q_{enclosed}}{\epsilon_o}$$

For surface S_1, the enclosed charges are Q and $-4Q/3$.

$$\Phi_{S1} = \frac{Q - \frac{4Q}{3}}{\epsilon_o} = \frac{-\frac{Q}{3}}{\epsilon_o} = -\frac{Q}{3\epsilon_o}$$

For surface S_2, the only enclosed charge is Q.

$$\Phi_{S2} = \frac{Q}{\epsilon_o}$$

The ratio of the flux passing through S_1 to that passing through S_2 is:

$$\frac{\Phi_{S1}}{\Phi_{S2}} = \frac{-\frac{Q}{3\epsilon_o}}{\frac{Q}{\epsilon_o}} = -\frac{1}{3}$$

This corresponds to a ratio of $-1:3$.

24. **Answer :** A) 0

According to Gauss's law, the net electric flux passing through a closed surface depends only on the total net charge enclosed within that surface.

$$\Phi_{net} = \frac{Q_{enclosed}}{\epsilon_o}$$

The problem text specifies that the Gaussian surface S encloses two charges: $q_1 = q$ and $q_2 = -q$. (Charge $q_3 = q$ is situated outside the surface S).

Calculate the net enclosed charge.

$Q_{enclosed} = q_1 + q_2 = q + (-q) = 0$

Therefore, the net flux passing through the surface S is:

$\Phi_{net} = \frac{0}{\epsilon_o} = 0$

25. **Answer :** D) due to all

While Gauss's Law states that the *net electric flux* through a closed surface depends only on the charges enclosed within it, the *electric field* $\vec{E}$ at any specific point on the surface itself is determined by the principle of superposition.

$\vec{E}_{net} = \vec{E}_{q1} + \vec{E}_{-q1} + \vec{E}_{q2}$

The electric field at any point on the dotted Gaussian boundary is the vector sum of the electric fields produced by every single charge present in space, both inside and outside the surface.

Therefore, the electric field on the surface is due to all charges.

26. **Answer :** A) q/ϵ_o

According to Gauss's Law, the total net electric flux emerging from any closed surface is equal to the total enclosed charge divided by the permittivity of free space (ϵ_o).

$\Phi_{total} = \frac{Q_{enclosed}}{\epsilon_o}$

A cube is a closed surface. It completely encloses the point charge $+q$ placed at its midpoint.

The dimensions or side length L of the cube do not affect the total flux emerging from the entirely closed surface.

$\Phi = \frac{q}{\epsilon_o}$

The flux through a single face would be $\frac{q}{6\epsilon_o}$, but the question asks for the flux emerging from the "cube" (the whole closed structure).

27. **Answer :** B) equal to the total flux through S due to charges q_3 and q_4

Gauss's Law states that the net electric flux through any closed surface depends exclusively on the net charge physically enclosed within that boundary.

$\oint \vec{E}_{net} \cdot d\vec{A} = \frac{Q_{enclosed}}{\epsilon_o}$

The diagram shows that charges q_3 and q_4 are inside the closed surface S, while charges q_1 and q_2 are outside.

The total flux is simply the sum of the charges inside divided by ϵ_o.

$\Phi_{total} = \frac{q_3 + q_4}{\epsilon_o}$

This is mathematically identical to calculating the total flux if only charges q_3 and q_4 were present, making statement B correct.

28. **Answer :** C) The flux of electric field through a closed surface due to all the charges (inside or outside) is equal to the flux due to the charges enclosed by the surface

Let's evaluate the options. Gauss's Law states $\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enclosed}}{\epsilon_0}$. The electric field $\vec{E}$ in the integral is the resultant field due to ALL charges (both inside and outside). However, when the integral is evaluated over a closed surface, the contribution to the net flux from all external charges exactly cancels out (flux in equals flux out). Therefore, the total flux calculated using the net electric field of all charges equates identically to the flux contributed solely by the internal enclosed charges.

This makes statement C the correct conceptual representation of Gauss's Law.

29. **Answer :** C) $\frac{q}{6\epsilon_o}$

Consider the given square of side L as one face of a perfect cube of side L.

The charge $+q$ is located exactly at a distance of $\frac{L}{2}$ above the center of this square.

This geometry places the charge $+q$ precisely at the geometrical center of the imaginary cube.

According to Gauss's Law, the total flux emerging from the entire closed cube is:

$\Phi_{total} = \frac{q}{\epsilon_o}$

Because the charge is symmetrically placed at the center, the flux is distributed equally among all six identical square faces of the cube.

$\Phi_{one_face} = \frac{\Phi_{total}}{6} = \frac{q}{6\epsilon_o}$

LEVEL-II

1. **Answer :** B) 11.25 N

Let the initial distance between the charges be $r_1 = 0.06$ m.

The initial force $F_1 = 5$ N.

$F_1 = \frac{k \cdot q_1 \cdot q_2}{r_1^2}$

Each charge is moved towards the other by 0.01 m. The total reduction in distance is 0.01 m + 0.01 m = 0.02 m.

The new distance r_2 is:

$r_2 = 0.06 - 0.02 = 0.04$ m

Since the charges remain the same, the force is inversely proportional to the square of the distance.

$\frac{F_2}{F_1} = \left(\frac{r_1}{r_2}\right)^2$

$F_2 = 5 \cdot \left(\frac{0.06}{0.04}\right)^2$

$F_2 = 5 \cdot \left(\frac{3}{2}\right)^2 = 5 \cdot \frac{9}{4}$

$F_2 = \frac{45}{4} = 11.25$ N

2. **Answer :** A) -10 N

Let the initial charges be $q_1 = +3$ muC and $q_2 = +8$ muC.

The initial repulsive force is $F_1 = 40$ N.

$F_1 = \frac{k \cdot (3) \cdot (8)}{r^2} = \frac{24k}{r^2} = 40$

A charge of -5 muC is added to each. The new charges are:

$q_1' = +3 - 5 = -2$ muC

$q_2' = +8 - 5 = +3$ muC

The new force F_2 is calculated using the new charges. A negative sign will indicate an attractive force.

$F_2 = \frac{k \cdot (-2) \cdot (3)}{r^2} = \frac{-6k}{r^2}$

Taking the ratio of the new force to the initial force:

$\frac{F_2}{F_1} = \frac{\frac{-6k}{r^2}}{\frac{24k}{r^2}} = -\frac{6}{24} = -\frac{1}{4}$

$F_2 = 40 \cdot \left(-\frac{1}{4}\right) = -10$ N

3. **Answer :** A) -q

Let the distance between the two outer charges $4q$ and q be $2x$.
The charge Q is placed at the midpoint, so its distance from both $4q$ and q is x.
For the net force on charge q to be zero, the sum of the forces exerted by $4q$ and Q on q must balance out to zero.

$F_{net_o n_q} = F_{4q} + F_Q = 0$

$\frac{k \cdot (4q) \cdot q}{(2x)^2} + \frac{k \cdot Q \cdot q}{x^2} = 0$

$\frac{4 \cdot k \cdot q^2}{4x^2} + \frac{k \cdot Q \cdot q}{x^2} = 0$

Simplify the first term.

$\frac{k \cdot q^2}{x^2} + \frac{k \cdot Q \cdot q}{x^2} = 0$

Divide the entire equation by $\frac{k \cdot q}{x^2}$.

$q + Q = 0$

$Q = -q$

4. **Answer :** 6.62 cm

For a three-charge system to remain in equilibrium, all three charges must experience zero net force. If the outer charges $+e$ and $+2e$ are strictly fixed, and only q needs to be in equilibrium, we solve for x from $+e$:

$\frac{k \cdot e \cdot q}{x^2} = \frac{k \cdot (2e) \cdot q}{(16 - x)^2}$

$\frac{1}{x^2} = \frac{2}{(16 - x)^2}$

$\frac{1}{x} = \frac{\sqrt{2}}{16 - x}$

$16 - x = \sqrt{2}x \Longrightarrow x\left(1 + \sqrt{2}\right) = 16 \Longrightarrow x = \frac{16}{\sqrt{2} + 1} \approx 6.62$ cm

5. **Answer :** C) $Q/8\epsilon_o$

When a point charge Q is placed exactly at the corner of a cube, that corner is shared by 8 identical adjacent cubes in a three-dimensional lattice.
To form a complete, symmetric, closed Gaussian surface that fully encloses the charge at its center, we need 8 such cubes.
The total electric flux emerging from this large composite cubic surface enclosing the charge Q is $\frac{Q}{\epsilon_o}$.
Since the original single cube represents exactly $\frac{1}{8}$th of this symmetric volume, by symmetry, the flux passing through it is:

$\Phi_{cube} = \frac{1}{8} \cdot \frac{Q}{\epsilon_o} = \frac{Q}{8\epsilon_o}$

6. **Answer :** C) $\frac{q}{2\epsilon_o}$

A cylindrical vessel with one open end can be considered as half of a completely closed, symmetrical structure (like a full closed cylinder of double the length).
When a charge q is placed exactly at the center of the open circular end, the geometrical plane of the opening cuts the total surrounding 3D space exactly in half.

According to Gauss's Law, the total flux radiating outwards in all directions from the charge is $\frac{q}{\epsilon_o}$.

Exactly half of this total flux will travel "downward" into the interior of the vessel and pass through its walls and bottom surface, while the other half escapes into the open space above.

$\Phi_{vessel} = \frac{1}{2} \cdot \Phi_{total} = \frac{q}{2\epsilon_o}$

7. **Answer :** B) ${\epsilon_o}^{-1}$

According to Gauss's law, the total electric flux (Φ_E) coming out of any closed surface enclosing a net charge Q is given by:

$\Phi_E = \frac{Q}{\epsilon_o}$

The question specifies a "positive charge placed in air". This phrasing conventionally refers to a *unit* positive charge ($Q = +1$ C) in electrostatics definitions.

Substituting $Q = 1$ into the equation:

$\Phi_E = \frac{1}{\epsilon_o} = {\epsilon_o}^{-1}$

8. **Answer :** D) $(\phi_2 - \phi_1)\epsilon_o$

According to the convention for electric flux, flux leaving a closed surface is considered positive, and flux entering is considered negative.

Let the inward (entering) flux be Φ_1. The negative contribution is $-\Phi_1$.

Let the outward (leaving) flux be Φ_2. The positive contribution is $+\Phi_2$.

The net electric flux (Φ_{net}) passing through the enclosed surface is the algebraic sum.

$\Phi_{net} = \Phi_2 - \Phi_1$

Using Gauss's Law, the net flux is equal to the total enclosed charge divided by the permittivity of free space.

$\Phi_{net} = \frac{Q_{enclosed}}{\epsilon_o}$

$\Phi_2 - \Phi_1 = \frac{Q_{enclosed}}{\epsilon_o}$

Solving for the enclosed charge $Q_{enclosed}$:

$Q_{enclosed} = (\Phi_2 - \Phi_1) \cdot \epsilon_o$

9. **Answer :** C) -5 V

The electric potential (V) at a specific point is defined as the external work done (W) in bringing a unit positive charge from infinity to that point against the electrostatic force.

$V = \frac{W}{q}$

We are given the work done $W = 50$ J and the charge $q = -10$ C.

$V = \frac{50 \text{ J}}{-10 \text{ C}}$

$V = -5$ V

10. **Answer :** C) 10 V

The electrostatic potential energy (U) of a charge at a specific point is the product of the charge (q) and the electric potential (V) at that point.

$U = q \cdot V$

We are given the potential energy $U = 50$ J and the magnitude of the charge $q = 5$ C.

We can rearrange the formula to solve for the electric potential V.

$V = \frac{U}{q}$

$V = \frac{50}{5}$

$V = 10$ V

11. **Answer :** D) E is directed in the direction of decreasing V.

The relation between the electric field $\vec{E}$ and the electric potential V is mathematically expressed as the negative gradient of the potential.

$\vec{E} = -\nabla V$

In one dimension, this is written as $E = -\frac{dV}{dr}$.

The negative sign indicates that the electric field vector points in the spatial direction where the potential V undergoes the maximum rate of decrease.

12. **Answer :** D) zero

The electric potential V created by a stationary point charge q at a fixed distance r is constant.

$V = \frac{k \cdot q}{r}$

When one sphere is moved around the other at a constant radius of 1 m, it follows a perfectly circular path. This entire path lies on a single equipotential surface.

The change in potential ΔV along any path on an equipotential surface is zero.

The work done W is given by $W = q \cdot \Delta V$.

$W = q \cdot (0) = 0$

13. **Answer :** A, B, C

Let's evaluate the principles of Gauss's Law and electric fields.

Gauss's Law is $\oint \vec{E} \cdot d\vec{A} = \frac{Q_{inside}}{\epsilon_o}$.

A) If Q_1 (inside) changes, the enclosed charge changes, altering the net flux Φ. By superposition, the net electric field $\vec{E}$ at points on the surface also changes. This is True.

B) If Q_2 (outside) changes, the enclosed charge remains identical, so the net flux Φ over the closed surface will not change. However, the external charge contributes to the local electric field $\vec{E}$ at every point on the surface, so $\vec{E}$ will change. This is True.

C) If $Q_1 = 0$ and $Q_2 \neq 0$, the enclosed charge is zero, yielding $\Phi = 0$. However, the external charge Q_2 still projects an electric field across space, so $\vec{E} \neq 0$ on the surface. This is True.

14. **Answer :** A, C

An equipotential surface is defined as a locus of points having the exact same electrostatic potential.

The change in potential dV along any infinitesimally small displacement $d\vec{r}$ exactly strictly tangent to the equipotential surface is zero.

We know $dV = -\vec{E} \cdot d\vec{r} = -E \cdot dr \cdot \cos(\theta)$.

For dV to be zero for any non-zero displacement along the surface, $\cos(\theta)$ must be zero.

$\theta = 90°$

This strictly requires the electric field vector $\vec{E}$ to be perfectly perpendicular (normal) to the equipotential surface at all points (Statement A is True).

Consequently, $\vec{E}$ can have no mathematical component parallel (tangent) to the surface (Statement C is True).

15. Answer : C, D

The net electric field at any specific point in space is determined by the vector sum (superposition) of the existing original electric field and the new electric field generated by the introduced point charge.

$\vec{E}new = \vec{E}original + \vec{E}_{charge}$

The magnitude of this new resultant vector depends fundamentally on the relative directions of the two vectors.

If a positive charge is introduced such that its outward radial field aligns with the original field, the net magnitude increases (Statement C is True).

If a negative charge is introduced such that its inward radial field opposes the original field, the net magnitude decreases (Statement D is True). It entirely depends on placement coordinates.

16. Answer : B, D

The absolute electrostatic potential V at a specific spatial point and the absolute potential energy U of a system rely strictly on the arbitrary choice of a reference state where $V = 0$ (typically at infinity).

However, the mathematical difference between potentials at two distinct points ($\Delta V = V_2 - V_1$) subtracts out this arbitrary reference constant entirely.

$\Delta V = (V_2 - V_{ref}) - (V_1 - V_{ref}) = V_2 - V_1$

Similarly, the change in potential energy ($\Delta U = U_2 - U_1$) is independent of the chosen zero-level reference geometry.

17. Answer : A, C, D

Let's examine the properties of equipotential surfaces.

A) If two equipotential surfaces having distinct, unequal potentials intersected, the points of intersection would paradoxically possess two different potential values simultaneously, which violates physics definitions. They cannot intersect. (True)

C) The electric field magnitude is given by $E \approx -\frac{\Delta V}{\Delta r}$. For a constant potential difference ΔV between two surfaces, the field E is inversely proportional to the spatial separation Δr. The field is maximum where the gap is smallest. (True)

D) Electric lines of force are strictly perpendicular to any equipotential surface they originate from or pass through, preventing work from being done along the surface. (True)

18. Answer : A, B, C, D

A uniform electric field possesses constant magnitude and direction identically at all points in space.

Because $\vec{E}$ is constant and perfectly unidirectional, the surfaces of constant potential (equipotential surfaces) must be flat planes mathematically normal to the field vectors (Statements A and B are True).

The relation is $E = -\frac{\Delta V}{\Delta x}$. Because E is completely constant, for equal increments of potential change ΔV, the physical distance Δx between consecutive planes must be perfectly equal (Statement C is True).

The negative sign in $E = -\frac{dV}{dx}$ rigorously dictates that electrostatic potential naturally decreases as one moves exactly in the direction of the electric field lines (Statement D is True).

19. **Answer :** D) -4 C

To allow the human to overcome their gravitational weight, the upward electrostatic force must equal the downward gravitational force.

$F_{electric} = F_{gravity}$

$q \cdot E = m \cdot g$

The electric field E is given as 150 N/C directed towards the center of the Earth (downwards). For the electric force to act upwards (opposite to the field), the acquired charge q must be negative.

We substitute the given magnitude values using $g \approx 10$ m/s^2 to match the standard simple options.

$|q| \cdot 150 = 60 \cdot 10$

$|q| \cdot 150 = 600$

$|q| = \frac{600}{150} = 4$ C

Considering the sign, the necessary charge is -4 C.

20. **Answer :** D) $1.44 \times 10^7 C$ N

The electrostatic force of repulsion between two identical point charges is given by Coulomb's law.

$F = \frac{k \cdot q_1 \cdot q_2}{r^2}$

We substitute the given values: $k = 9 \times 10^9$ N m^2/C^2, $q_1 = q_2 = 4$ C, and separation $r = 100$ m.

$F = \frac{(9 \times 10^9) \cdot (4) \cdot (4)}{(100)^2}$

$F = \frac{9 \times 10^9 \cdot 16}{10000}$

$F = \frac{144 \times 10^9}{10^4}$

$F = 144 \times 10^5 = 1.44 \times 10^7$ N

21. **Answer :** C) Towards the center of the Earth

This is a direct factual extraction from the provided comprehension passage.

The passage explicitly states: "The electric field near the earth's surface believed to be 150 N/C and directed towards the center of the earth."

22. **Answer :** A) $7.0 \times 10^{-4} C kg^{-1}$

The problem states that the required electrical force on the toner particle must be exactly 10 times its gravitational weight.

$F_{electric} = 10 \cdot F_{gravity}$

$q \cdot E = 10 \cdot m \cdot g$

We are solving for the charge-to-mass ratio, which is $\frac{q}{m}$. We rearrange the equation.

$\frac{q}{m} = \frac{10 \cdot g}{E}$

We substitute $g = 9.8$ m/s^2 and the given electric field magnitude $E = 1.4 \times 10^5$ N/C.

$\frac{q}{m} = \frac{10 \cdot 9.8}{1.4 \times 10^5}$

$\frac{q}{m} = \frac{98}{140000} = \frac{98}{1.4 \times 10^5} = 70 \times 10^{-5}$

$\frac{q}{m} = 7.0 \times 10^{-4}$ C/kg

23. **Answer :** D) 1.15×10^{10}

First, we determine the net mass required to hold one single excess electron based on the calculated charge-to-mass ratio.

The elementary charge is $e = 1.6 \times 10^{-19}$ C.

$$m_{total} = \frac{e}{\left(\frac{q}{m}\right)} = \frac{1.6 \times 10^{-19}}{7.0 \times 10^{-4}} \approx 2.285 \times 10^{-16} \text{ kg}$$

Next, we establish the absolute mass of a single ${}^{12}_{6}C$ carbon atom. One atomic mass unit (u) is approximately 1.66×10^{-27} kg.

$m_{atom} = 12$ u $= 12 \cdot 1.66 \times 10^{-27} \approx 1.992 \times 10^{-26}$ kg

Finally, we calculate the number of carbon atoms (N) by dividing the total required mass by the individual atomic mass.

$$N = \frac{m_{total}}{m_{atom}} = \frac{2.285 \times 10^{-16}}{1.992 \times 10^{-26}}$$

$N \approx 1.147 \times 10^{10}$

Rounding to standard precision yields 1.15×10^{10}.

24. **Answer :** D) ten times its weight

This is explicitly defined in the parameters of the passage provided in the text.

The passage strictly states: "A toner particle is to be attached to the drum with a force that is 10 times the weight of the particle."

25. **Answer :** C) A is false but R is true

The macroscopic law of quantization of charge asserts that all observable electric charge must exist precisely as integral multiples of the elementary charge e.

Reason (R) correctly states that theoretical quarks carry fractional charges such as $\pm\frac{e}{3}$ and $\pm\frac{2e}{3}$.

However, because quarks are subjected to "color confinement," they can never be found freely in isolation; they exclusively bind into hadrons (like protons and neutrons) whose total composite charges are always perfect integers.

Therefore, observable free charges do not violate quantization. Assertion (A) is false.

26. **Answer :** A) Both A and R are true and R is correct explanation of A

An electrically charged body can induce an opposite local charge on a nearby perfectly neutral body (through polarization or electrostatic induction) and subsequently attract it.

Therefore, observing an attractive force between two objects simply proves that at least one object is charged, but does not confirm both are.

Conversely, true electrostatic repulsion can only mechanically occur if both interacting bodies possess a net charge of the exact same sign.

Thus, observing repulsion definitively confirms that both bodies are electrically charged. Reason (R) correctly explains why attraction is ambiguous, thereby proving why repulsion is the only sure test.

27. **Answer :** Match a-P, b-Q, c-R, d-S

The matrix matching resolves based on standard electrostatic definitions.

(a) Quantization of charge implies that any physical amount of net charge must be an exact discrete integer multiple of the elementary charge e. This strictly matches (P) "integral multiples of e".

(b) The Principle of Conservation of charge strictly mandates that the total net charge of a closed, isolated physical system remains invariant over time. This strictly matches (Q) "Net charge of an isolated system remains constant".

(c) Macroscopically, a positive charge on a material body is physically caused by the removal or loss of negative electrons. This strictly matches (R) "Deficiency of electrons".

(d) A negative charge on a material body is physically caused by the addition or gain of extra negative electrons. This strictly matches (S) "Excess of electrons".

28. Answer : Match A-S, B-P, C-R, D-Q

Let us construct the correct mathematical formula for each fundamental electrostatic quantity.

(A) The magnitude of the electrostatic force F between two distinct point charges q_1 and q_2 separated by distance r is $F = \frac{K \cdot q_1 \cdot q_2}{r^2}$. This uniquely matches (S).

(B) The magnitude of the electric field E generated by a single point charge Q at a distance r is $E = \frac{K \cdot Q}{r^2}$. This uniquely matches (P).

(C) The scalar electric potential V created by a single point charge q at distance r is $V = \frac{K \cdot q}{r}$. This uniquely matches (R).

(D) The electrostatic potential energy U stored in a two-point charge system separated by distance r is $U = \frac{K \cdot q_1 \cdot q_2}{r}$. This uniquely matches (Q).

29. Answer : 27

The total electrostatic potential energy (U) of a discrete system of three interacting point charges (q_1, q_2, q_3) placed at the vertices of a triangle is the sum of the potential energies of all unique two-charge combinations.

$$U = \frac{k \cdot q_1 \cdot q_2}{r_{12}} + \frac{k \cdot q_2 \cdot q_3}{r_{23}} + \frac{k \cdot q_3 \cdot q_1}{r_{31}}$$

For an equilateral triangle, all distances are exactly equal: $r = 10$ cm $= 0.1$ m. All charges are identically $q = 10$ muC $= 10^{-5}$ C.

$$U = 3 \cdot \frac{k \cdot q^2}{r}$$

Substitute Coulomb's constant $k = 9 \times 10^9$ N m^2/C^2.

$$U = 3 \cdot \frac{(9 \times 10^9) \cdot (10^{-5})^2}{0.1}$$

$$U = 3 \cdot \frac{(9 \times 10^9) \cdot (10^{-10})}{10^{-1}}$$

$$U = \frac{27 \cdot 10^{-1}}{10^{-1}} = 27 \text{ Joules}$$

30. Answer : 9

The work strictly done by the conservative electric field is physically equal to the negative of the complete change in the electrostatic potential energy of the system.

$W_{field} = -\Delta U = -(U_{final} - U_{initial}) = U_{initial} - U_{final}$

When the infinite separation is eventually achieved, the charges are infinitely far apart, rendering their interaction zero.

$U_{final} = 0$

Thus, $W_{field} = U_{initial}$. We precisely calculate the initial energy for the three charges ($q = 10$ muC $= 10^{-5}$ C) separated by distance $r = 30$ cm $= 0.3$ m.

$$U_{initial} = 3 \cdot \frac{k \cdot q^2}{r}$$

$$U_{initial} = 3 \cdot \frac{(9 \times 10^9) \cdot (10^{-5})^2}{0.3}$$

$U_{initial} = \frac{27 \times 10^{-1}}{0.3} = \frac{2.7}{0.3} = 9$ Joules

31. Answer : -2

The core principle of conservation of electric charge asserts that the absolute net charge of an entirely isolated system must strictly remain constant across any internal process.

Let the initial interacting charges be q_1 and q_2.

$q_1 = +3$ C

$q_2 = -1$ C

The total initial net charge before any contact is the algebraic sum.

$Q_{initial} = q_1 + q_2 = (+3) + (-1) = +2$ C

After physical contact and subsequent separation, let the new terminal charges be q_1' and q_2'. We are provided that one charge becomes $q_1' = +4$ C.

By conservation, the final net charge must perfectly equal the initial net charge.

$Q_{final} = q_1' + q_2' = Q_{initial}$

$(+4 \text{ C}) + q_2' = +2$ C

$q_2' = +2 \text{ C} - 4 \text{ C} = -2$ C

4. Current Electricity

Electric Current and Its Effects

LEVEL-I

1. **Answer :** A) $V_d \propto A^{-1}$

The relationship between current and drift velocity is given by the formula:

$I = n \cdot e \cdot A \cdot V_d$

Since the current I is steady and n, e are constants for a given material, we can rearrange for V_d:

$V_d = \frac{I}{n \cdot e \cdot A}$

$V_d \propto \frac{1}{A}$

$V_d \propto A^{-1}$

2. **Answer :** A) $\frac{ml}{ne^2\tau A}$

The formula for the resistance of a conductor is:

$R = \rho \cdot \frac{l}{A}$

The formula for resistivity in terms of microscopic quantities is:

$\rho = \frac{m}{n \cdot e^2 \cdot \tau}$

Substituting the value of resistivity into the resistance formula gives:

$R = \left(\frac{m}{n \cdot e^2 \cdot \tau}\right) \cdot \frac{l}{A}$

$R = \frac{m \cdot l}{n \cdot e^2 \cdot \tau \cdot A}$

3. **Answer :** C) Drift velocity

For a steady current in a conductor, the current I is constant across any cross-section.

The free electron density n, charge e, and resistivity ρ are material properties and remain constant.

The charge crossing in a given time interval is the current, which is constant.

The drift velocity is inversely proportional to the area of cross-section:

$V_d = \frac{I}{n \cdot e \cdot A}$

Therefore, drift velocity depends on the cross-section.

4. **Answer :** A) $v_d \alpha E$

From Ohm's law in microscopic form, current density J is proportional to the electric field E:

$J = \sigma \cdot E$

We know that current density is also given by:

$J = n \cdot e \cdot v_d$

Equating both expressions for J:

$n \cdot e \cdot v_d = \sigma \cdot E$

$v_d = \left(\frac{\sigma}{n \cdot e}\right) \cdot E$

Since σ, n, e are constant, we get:

$v_d \propto E$

5. **Answer :** A) Independent of area of cross-section

The net quantity of charge crossing any cross-section per second is the definition of electric current.

$I = \frac{q}{t}$

For a steady current, the same amount of charge passes through any cross-section per unit time, regardless of the area.

Therefore, it is independent of the area of cross-section.

6. **Answer :** B) Decrease with the increase of temperature

Relaxation time is the average time between successive collisions of electrons.

When temperature increases, the thermal energy of the positive ions in the lattice increases, causing them to vibrate with greater amplitude.

This increases the frequency of collisions for the drifting electrons.

Since the collisions happen more frequently, the time between collisions decreases.

7. **Answer :** D) Current only

In a steady state, the rate of flow of charge (current) must be the same throughout the conductor.

The area of cross-section is non-uniform, so it changes along the length.

Drift speed is $V_d \propto \frac{1}{A}$, so it changes.

Electric field is $E = \rho \cdot J = \rho \cdot \left(\frac{I}{A}\right)$, so it also changes with area.

Thus, only the current remains constant.

8. **Answer :** C) Does not change

The current through the wire is given by Ohm's law:

$I = \frac{V}{R}$

The resistance of the wire is:

$R = \rho \cdot \frac{l}{A}$

Substituting R into the current equation:

$I = \frac{V \cdot A}{\rho \cdot l}$

The current is also related to drift velocity:

$I = n \cdot e \cdot A \cdot V_d$

Equating the two expressions for current:

$n \cdot e \cdot A \cdot V_d = \frac{V \cdot A}{\rho \cdot l}$

$V_d = \frac{V}{n \cdot e \cdot \rho \cdot l}$

This expression is independent of the area A or diameter d.

9. **Answer :** D) 1500 C

The formula relating charge, current, and time is:

$Q = I \cdot t$

The given current is $I = 5$ A.

The given time is $t = 5$ minutes.

Converting time to seconds:

$t = 5 \cdot 60$

$t = 300$ s

Substituting the values into the formula:

$Q = 5 \cdot 300$

$Q = 1500$ C

10. **Answer :** B) 1.6

The formula for equivalent current of a revolving charge is:

$I = \frac{q}{T}$

The time period T relates to angular frequency ω by:

$T = \frac{2 \cdot \pi}{\omega}$

Substituting T into the current formula:

$I = \frac{q \cdot \omega}{2 \cdot \pi}$

The charge of an electron is $q = 1.6 \times 10^{-19}$ C.

The angular frequency is given as $\omega = 6.28$ rad/s.

Using $\pi \approx 3.14$, we get $2 \cdot \pi \approx 6.28$.

$I = \frac{1.6 \times 10^{-19} \cdot 6.28}{6.28}$

$I = 1.6 \times 10^{-19}$ A

The value in the required unit format is 1.6.

11. **Answer :** B) $E \propto J$

The microscopic form of Ohm's law is:

$J = \sigma \cdot E$

where σ is the conductivity of the material.

Rearranging the formula gives:

$E = \frac{1}{\sigma} \cdot J$

Since conductivity is a constant for a given material at a constant temperature, we can write:

$E = \rho \cdot J$

$E \propto J$

12. **Answer :** A) For metallic conductors at low temperature

Ohm's law states that the current through a conductor is directly proportional to the potential difference across it, provided physical conditions and temperature remain constant.

At high temperatures, the resistance of metallic conductors changes, causing a deviation from Ohm's law.

Electrolytes and diodes are generally non-ohmic devices.

Therefore, Ohm's law is true for metallic conductors at constant, generally lower temperatures where heating effects are negligible.

13. **Answer :** A) Potential

Heat flows from a body at a higher temperature to a body at a lower temperature.

Temperature determines the direction of heat flow.

Similarly, electric charge flows from a point at higher potential to a point at lower potential.

Electric potential determines the direction of flow of positive charge.

Therefore, potential is analogous to temperature.

14. **Answer :** A) t_1

The graph plots current I on the y-axis and potential V on the x-axis.

From Ohm's law, $V = I \cdot R$.

Rearranging gives $I = \left(\frac{1}{R}\right) \cdot V$.

The slope of the $I - V$ graph is equal to $\frac{1}{R}$.

A smaller slope corresponds to a larger resistance R.
For metals, resistance increases with an increase in temperature ($R \propto T$).
The line with the lowest slope is t_1, which means it has the highest resistance.
Therefore, t_1 represents the highest temperature.

15. **Answer :** A) (-) ve pole to (+) ve pole during discharging
Inside a cell, the conventional current flows from the negative terminal to the positive terminal during the discharging process.
This happens as the chemical energy drives the positive charges against the internal electric field.
In the external circuit, current flows from the positive terminal to the negative terminal.
During charging, current is forced into the positive terminal, reversing the internal direction.

16. **Answer :** A) Being charged
The formula for terminal voltage when a cell is discharging is:
$V = E - I \cdot r$
In this case, $V < E$.
When a cell is being charged, current is driven into its positive terminal by an external source.
The formula for terminal voltage during charging is:
$V = E + I \cdot r$
Therefore, the terminal voltage is greater than the emf when the cell is being charged.

17. **Answer :** A) Zero
The terminal voltage of a real cell is $V = E - I \cdot r$.
An ideal cell is a theoretical concept where there is no internal energy loss.
For an ideal cell, the terminal voltage is always equal to its electromotive force (emf), regardless of the current drawn.
This means the voltage drop across the internal resistance must be zero.
Therefore, the internal resistance r of an ideal cell is zero.

18. **Answer :** C) 1.1 Ω
The formula for the current in a circuit with internal resistance is:
$$I = \frac{E}{R + r}$$
The given emf is $E = 50$ V.
The given external resistance is $R = 10$ Ω.
The given current is $I = 4.5$ A.
Substituting the values:
$$4.5 = \frac{50}{10 + r}$$
$$4.5 \cdot (10 + r) = 50$$
$$45 + 4.5 \cdot r = 50$$
$$4.5 \cdot r = 5$$
$$r = \frac{5}{4.5}$$
$$r \approx 1.11\ \Omega$$

19. **Answer :** D) a-h, b-g, c-e, d-f
For a charging cell, the terminal voltage is $V = E + I \cdot r$, so $V > E$ (matches h).

For a discharging cell, the terminal voltage is $V = E - I \cdot r$, so $V < E$ (matches g).
For a short-circuited cell, the external resistance is zero, so terminal voltage $V = 0$ (matches e).
For a cell in an open circuit, no current flows ($I = 0$), so $V = E$ (matches f).
The correct sequence is a-h, b-g, c-e, d-f.

20. **Answer :** B) 7V
The formula for terminal voltage during charging is:
$V = E + I \cdot r$
The given emf is $E = 6$ V.
The given current is $I = 1$ A.
The internal resistance is $r = 1\ \Omega$.
Substituting the values into the formula:
$V = 6 + (1 \cdot 1)$
$V = 6 + 1$
$V = 7$ V

21. **Answer :** C) Doubled
The formula for heat generated by a heater is:
$H = \frac{V^2 \cdot t}{R}$
Let the initial resistance of the coil be R.
When the coil is cut into two equal parts, the length of each part is halved.
Since resistance is directly proportional to length, the new resistance is $R' = \frac{R}{2}$.
The new heat generated is:
$H' = \frac{V^2 \cdot t}{\frac{R}{2}}$
$H' = 2 \cdot \left(\frac{V^2 \cdot t}{R}\right)$
$H' = 2 \cdot H$

22. **Answer :** D) 20 minutes
The formula for heat generated is:
$H = \frac{V^2 \cdot t}{R}$
To boil the same volume of water, the same amount of heat H is required.
Using the first set of conditions:
$H = \frac{(220)^2 \cdot 5}{R}$
Using the second set of conditions:
$H = \frac{(110)^2 \cdot t'}{R}$
Equating both expressions for H:
$\frac{(220)^2 \cdot 5}{R} = \frac{(110)^2 \cdot t'}{R}$
$(220)^2 \cdot 5 = (110)^2 \cdot t'$
$t' = 5 \cdot \left(\frac{220}{110}\right)^2$
$t' = 5 \cdot (2)^2$
$t' = 5 \cdot 4$
$t' = 20$ minutes

23. **Answer :** B) Only 25 watt bulb will fuse

The resistance of a bulb is given by $R = \frac{V_{\text{rated}}^2}{P_{\text{rated}}}$.

For the 25W bulb: $R_{25} = \frac{220^2}{25} = 1936\ \Omega$.

For the 100W bulb: $R_{100} = \frac{220^2}{100} = 484\ \Omega$.

The rated current for the 25W bulb is $I_{25} = \frac{25}{220} \approx 0.114$ A.

The total resistance in series is $R_{\text{total}} = 1936 + 484 = 2420\ \Omega$.

The actual current flowing when connected to 440V is:

$$I_{\text{actual}} = \frac{440}{2420}$$

$$I_{\text{actual}} = \frac{44}{242}$$

$I_{\text{actual}} \approx 0.182$ A

Since I_{actual} (0.182 A) is greater than the rated current for the 25W bulb (0.114 A), it will fuse.

24. **Answer :** D) 25 watt

The formula for power is:

$$P = \frac{V^2}{R}$$

The resistance of the bulb is constant and given by:

$$R = \frac{V_{\text{rated}}^2}{P_{\text{rated}}}$$

When the bulb is operated at a new voltage V_{new}, the power consumed is:

$$P_{\text{new}} = \frac{V_{\text{new}}^2}{R}$$

Substituting R:

$$P_{\text{new}} = P_{\text{rated}} \cdot \left(\frac{V_{\text{new}}}{V_{\text{rated}}}\right)^2$$

$$P_{\text{new}} = 100 \cdot \left(\frac{110}{220}\right)^2$$

$$P_{\text{new}} = 100 \cdot \left(\frac{1}{2}\right)^2$$

$$P_{\text{new}} = 100 \cdot \frac{1}{4}$$

$P_{\text{new}} = 25$ W

25. **Answer :** B) 2

The formula for Joule's heating is:

$H = I^2 \cdot R \cdot t$

The given heat produced is $H = 80$ J.

The given current is $I = 2$ A.

The given time is $t = 10$ s.

Substituting the values into the formula:

$80 = (2)^2 \cdot R \cdot 10$

$80 = 4 \cdot R \cdot 10$

$80 = 40 \cdot R$

$$R = \frac{80}{40}$$

$R = 2\ \Omega$

26. **Answer :** C) 2:1

The formula for power dissipated in a parallel circuit is:

$P = \frac{V^2}{R}$

In a parallel circuit, the voltage V is constant across both resistors.

Therefore, power is inversely proportional to resistance.

$\frac{P_1}{P_2} = \frac{R_2}{R_1}$

The given ratio of resistances is $R_1 : R_2 = 1:2$.

Substituting this ratio:

$\frac{P_1}{P_2} = \frac{2}{1}$

The ratio of powers dissipated is 2:1.

LEVEL-II

1. **Answer :** B) Is opposite to the direction of $\vec{E}$

Electric field lines represent the direction of force on a positive charge.

Electrons carry a negative charge.

The electric force on a charge is given by $\vec{F} = q \cdot \vec{E}$.

For an electron, $q = -e$, so $\vec{F} = -e \cdot \vec{E}$.

Therefore, the force and the resulting motion of free electrons are opposite to the direction of the applied electric field.

2. **Answer :** A) $r \propto \frac{1}{A}$

Note: Assuming "rate of flow of electrons" here refers to the drift velocity concept due to the options provided. The text implies a relation for V_d.

The current I in a conductor is constant.

$I = n \cdot e \cdot A \cdot V_d$

Rearranging for velocity (r):

$V_d = \frac{I}{n \cdot e \cdot A}$

Since I, n, e are constant:

$V_d \propto \frac{1}{A}$

3. **Answer :** D) y/x

The equation for terminal potential difference is:

$V = E - I \cdot r$

From the graph, the y-intercept (when $I = 0$) is X.

So, $X = E$.

The x-intercept (when $V = 0$) is Y.

Setting $V = 0$:

$0 = E - Y \cdot r$

$Y \cdot r = E$

Substitute X for E:

$Y \cdot r = X$

$r = \frac{X}{Y}$

Internal conductance is the reciprocal of internal resistance:

Conductance = $\frac{1}{r}$

Conductance = $\frac{Y}{X}$

4. **Answer :** A) $i = \frac{nE}{R+nr}$

For n cells connected in series, their equivalent electromotive force is:

$E_{eq} = E + E + \cdots$ (n times)

$E_{eq} = n \cdot E$

The equivalent internal resistance of the n cells in series is:

$r_{eq} = r + r + \cdots$ (n times)

$r_{eq} = n \cdot r$

The total resistance of the circuit is the sum of external and internal resistances:

$R_{total} = R + n \cdot r$

By Ohm's law, the current in the circuit is:

$$i = \frac{E_{eq}}{R_{total}}$$

$$i = \frac{n \cdot E}{R + n \cdot r}$$

5. **Answer :** B) The dim bulb

In a house, appliances and bulbs are connected in parallel to ensure they receive the same voltage.

The power dissipated by a bulb in parallel is given by:

$$P = \frac{V^2}{R}$$

Since voltage V is constant, power is inversely proportional to resistance.

A brighter bulb dissipates more power (P is high).

Therefore, a brighter bulb has a lower resistance.

Consequently, the dim bulb dissipates less power and must have a larger resistance.

6. **Answer :** A) The external resistance is infinity

The formula for terminal voltage is:

$V = E - I \cdot r$

The problem states that terminal voltage is equal to emf, so $V = E$.

This means the term $I \cdot r$ must be equal to zero.

This occurs if internal resistance $r = 0$ (ideal cell) or if current $I = 0$.

Current $I = 0$ corresponds to an open circuit, where the external resistance is infinity.

Option A represents the open circuit condition.

7. **Answer :** A) $\frac{E}{2}$

The total resistance of the circuit is $R_{\text{total}} = R + r$.

It is given that external resistance equals internal resistance, $R = r$.

$R_{\text{total}} = r + r = 2 \cdot r$

The current flowing in the circuit is:

$$I = \frac{E}{2 \cdot r}$$

The potential difference across the terminal is the voltage drop across the external resistance:

$V = I \cdot R$

Substitute the expressions for I and R:

$$V = \left(\frac{E}{2 \cdot r}\right) \cdot r$$

$V = \frac{E}{2}$

8. **Answer :** B, C, D

In a steady current state, the current I is constant throughout the entire length of the conductor.

The area of cross-section A is non-uniform and varies along the length.

Current density $J = \frac{I}{A}$. Since A varies, J is not constant.

Electric field $E = \rho \cdot J$. Since J varies, E is not constant.

Drift speed $V_d = \frac{J}{n \cdot e}$. Since J varies, V_d is not constant.

Therefore, B, C, and D are not constant.

9. **Answer :** A, C, D

The formula for drift velocity is:

$V_d = \frac{I}{n \cdot e \cdot A}$

I is the magnitude of the current.

A is the cross-sectional area.

n is the number of free electrons per unit volume.

e is the charge of an electron.

Therefore, drift velocity depends on the cross-section of the wire, number of free electrons, and magnitude of the current.

10. **Answer :** C) Diode

Ohm's law states that $V \propto I$ for linear, ohmic conductors.

A diode is a semiconductor device that allows current to flow easily in one direction but restricts it in the opposite direction.

Its $V - I$ graph is not a straight line passing through the origin.

Therefore, it does not obey Ohm's law and is a non-ohmic resistance.

11. **Answer :** A, B, C

Joule's law of heating is given by the formula:

$H = I^2 \cdot R \cdot t$

Here, I is the strength of the current.

R is the resistance of the conductor.

t is the time for which the current is passed.

Therefore, the heat generated depends on all three quantities.

12. **Answer :** A, B, D

The two cells are connected in parallel, creating a closed loop.

Because their emfs are unequal ($E_1 \neq E_2$), a circulating current will flow through the loop.

The cell with the higher emf will drive the current and discharge, supplying energy.

The cell with the lower emf will be forced to take current in the reverse direction and will charge.

The points A and B are common to both branches, so the potential difference $V_A - V_B$ must be identical for both paths.

Passage - I

13. **Answer :** A) $3.6 \times 10^6 Am^{-2}$

The formula for current density is:

$J = \frac{I}{A}$

The given current is $I = 1.8$ A.

The given area is $A = 0.5\ \text{mm}^2$.
Convert the area to square meters:
$A = 0.5 \times 10^{-6}\ \text{m}^2$
Substitute the values:
$J = \dfrac{1.8}{0.5 \times 10^{-6}}$
$J = 3.6 \times 10^{6}\ \text{A/m}^2$

14. **Answer :** C) $2.55 \times 10^4 A/m^2$
The formula for current is $I = \frac{q}{t}$.
The given charge is $q = 90$ C.
The given time is 1 hour and 15 minutes, which is 75 minutes.
Convert time to seconds:
$t = 75 \cdot 60 = 4500$ s
Calculate current:
$I = \frac{90}{4500} = 0.02$ A
The diameter of the wire is $d = 1\ \text{mm} = 10^{-3}$ m.
The area of the wire is $A = \frac{\pi \cdot d^2}{4}$.
$A = \dfrac{3.14 \cdot (10^{-3})^2}{4}$
$A \approx 0.785 \times 10^{-6}\ \text{m}^2$
Calculate current density $J = \frac{I}{A}$:
$J = \dfrac{0.02}{0.785 \times 10^{-6}}$
$J \approx 2.547 \times 10^{4}\ \text{A/m}^2$

15. **Answer :** A) 4:1
The rods are connected one by one, and the current flowing is the same.
The formula for current density is:
$J = \dfrac{I}{A}$
The area of a cross-section is $A = \pi \cdot r^2$.
Substitute area into the formula:
$J = \dfrac{I}{\pi \cdot r^2}$
Since I is constant, current density is inversely proportional to the square of the radius.
$J \propto \dfrac{1}{r^2}$
The ratio of current densities is:
$\dfrac{J_1}{J_2} = \left(\dfrac{r_2}{r_1}\right)^2$
Substitute the given radii $r_1 = 2$ cm and $r_2 = 4$ cm:
$\dfrac{J_1}{J_2} = \left(\dfrac{4}{2}\right)^2$
$\dfrac{J_1}{J_2} = (2)^2$
$\dfrac{J_1}{J_2} = 4$

Passage - II

16. **Answer :** B) 2.82 KWH
The formula for electrical energy is:
$E = P \cdot t$
Energy for 3 tube lights:
$E_{\text{tubes}} = 3 \cdot 40 \text{ W} \cdot 5 \text{ h}$
$E_{\text{tubes}} = 600 \text{ Wh}$
Energy for 2 fans:
$E_{\text{fans}} = 2 \cdot 80 \text{ W} \cdot 12 \text{ h}$
$E_{\text{fans}} = 1920 \text{ Wh}$
Energy for 1 Television:
$E_{\text{TV}} = 1 \cdot 60 \text{ W} \cdot 5 \text{ h}$
$E_{\text{TV}} = 300 \text{ Wh}$
Total energy consumed per day:
$E_{\text{total}} = 600 + 1920 + 300$
$E_{\text{total}} = 2820 \text{ Wh}$
Convert to Kilowatt-hours:
$E_{\text{total}} = 2.82 \text{ KWH}$

17. **Answer :** C) 84.6
The number of units consumed is equal to the total energy in KWH.
Energy consumed per day = 2.82 KWH.
Total units for 30 days:
Total Units = $2.82 \cdot 30$
Total Units = 84.6

18. **Answer :** A) 253.8
The formula for total cost is:
Cost = Total Units $\cdot$ Rate per Unit
The total units consumed in 30 days is 84.6.
The rate is Rs. 3.00 per KWH.
Substituting the values:
Cost = $84.6 \cdot 3.00$

Cost = 253.8

Assertion and Reason

19. **Answer :** A) Both A and R are true, and R is the correct explanation of A
Ohm's law dictates a linear relationship between voltage and current.
Assertion (A) states Ohm's law is not valid for all materials, which is true.
Reason (R) states materials like semiconductors and electrolytes do not show linear V-I relations, which is also true.
The non-linear behavior of these specific materials is exactly why Ohm's law is not universally valid.

20. **Answer :** A) Both A and R are true, and R is the correct explanation of A
During discharging, the current flows out of the cell.
The formula for terminal voltage is $V = E - I \cdot r$.
Assertion (A) states terminal voltage is less than emf during discharging, which is true because $V < E$.
Reason (R) states there is a potential drop across internal resistance. This is represented by the $I \cdot r$ term.
This potential drop is the mathematical and physical reason why the terminal voltage is less than the emf.

Matrix Matching

21. **Answer :** a-P, b-P, c-R, S, d-Q

Electric charge is a scalar quantity, as it only has magnitude. (P)

Electric current is treated as a scalar quantity in basic circuit theory because it does not obey vector addition laws. (P)

Electric current density ($J = i/A$) (R) has a specific direction (direction of current flow) and is a vector quantity. (S)

The unit for current density is Amperes per square meter (Ampere/m²). (Q)

22. **Answer :** a-P,Q , b-Q, c-R, d-S

Electrical power can be calculated as $P = I^2 \cdot R = \frac{V^2}{R}$

Thermal energy per sec is heat power, given by $\frac{V^2}{R}$. (Q)

When a cell is discharging, the terminal voltage is $V = \epsilon - i \cdot r$. (R)

When a cell is charging, the terminal voltage is $V = \epsilon + i \cdot r$. (S)

This leaves Power to match with $i \cdot R$ (likely implying $i^2 \cdot R$, maybe a typo in options).

Let's assign it to P.

Integer Type

23. **Answer :** 2

The formula for electrical power is:

$P = \frac{V^2}{R}$

Let the initial power be P_1.

When the resistance is reduced to $\frac{R}{2}$, the new power P_2 is:

$P_2 = \frac{V^2}{\frac{R}{2}}$

$P_2 = 2 \cdot \left(\frac{V^2}{R}\right)$

$P_2 = 2 \cdot P_1$

Therefore, the power changes by a factor of 2.

24. **Answer :** 6

The formula for the terminal voltage of a discharging cell is:

$V = E - I \cdot r$

The given electromotive force is $E = 10$ V.

The given internal resistance is $r = 2\ \Omega$.

The given current is $I = 2$ A.

Substituting the values into the formula:

$V = 10 - (2 \cdot 2)$

$V = 10 - 4$

$V = 6$ V

25. **Answer :** 0.8

The formula for electric current is:

$I = \frac{q}{t}$

The total charge q can be expressed as:

$q = n \cdot e$

Substituting q into the current formula:

$I = \frac{n \cdot e}{t}$

The given number of electrons is $n = 5 \times 10^{18}$.
The charge of an electron is $e = 1.6 \times 10^{-19}$ C.
The given time is $t = 1$ s.
Substituting the values:

$$I = \frac{5 \times 10^{18} \cdot 1.6 \times 10^{-19}}{1}$$

$$I = 5 \cdot 1.6 \cdot 10^{18} \cdot 10^{-19}$$

$$I = 8.0 \cdot 10^{-1}$$

$$I = 0.8 \text{ A}$$

Resistance, Resistivity and Electrical Circuits

1. **Answer :** D) Sixteen times

The formula for the resistance of a wire is:

$$R = \rho \cdot \frac{l}{A}$$

When the wire is drawn (stretched), its volume remains constant.

$$V = A \cdot l$$

The new diameter is half the original, so the new radius is $r_2 = \frac{r_1}{2}$.
The new area of cross-section is:

$$A_2 = \pi \cdot \left(\frac{r_1}{2}\right)^2$$

$$A_2 = \frac{A_1}{4}$$

Since volume is constant ($A_1 \cdot l_1 = A_2 \cdot l_2$):

$$l_2 = \frac{A_1 \cdot l_1}{\frac{A_1}{4}}$$

$$l_2 = 4 \cdot l_1$$

The new resistance is:

$$R_2 = \rho \cdot \frac{l_2}{A_2}$$

$$R_2 = \rho \cdot \frac{4 \cdot l_1}{\frac{A_1}{4}}$$

$$R_2 = 16 \cdot R_1$$

2. **Answer :** B) $\frac{R}{4}$

The formula for the resistance of a wire is:

$$R = \rho \cdot \frac{L}{A}$$

For the first wire, the resistance is $R = \rho \cdot \frac{L}{a}$.
For the second wire, the length is the same and the area is $4a$.
The new resistance is:

$$R' = \rho \cdot \frac{L}{4 \cdot a}$$

$$R' = \frac{1}{4} \cdot \left(\rho \cdot \frac{L}{a}\right)$$

$$R' = \frac{R}{4}$$

3. **Answer :** A) 12 Ω

The formula for resistance in terms of volume and length is:

$R = \rho \cdot \frac{l^2}{V}$

The length is increased by 10%, so the new length is $l' = 1.1 \cdot l$.

The new resistance is:

$R' = \rho \cdot \frac{(1.1 \cdot l)^2}{V}$

$R' = 1.21 \cdot R$

Substituting the initial resistance $R = 10$:

$R' = 1.21 \cdot 10$

$R' = 12.1\Omega$

Note: The closest integer approximation provided in the options is 12 Ω.

4. **Answer :** A) $\frac{R}{n^4}$

The formula for resistance is:

$R = \rho \cdot \frac{l}{A}$

The new radius is $r' = n \cdot r$.

The new area of cross-section is:

$A' = \pi \cdot (n \cdot r)^2$

$A' = n^2 \cdot A$

Since the volume remains constant ($A \cdot l = A' \cdot l'$):

$l' = \frac{A \cdot l}{n^2 \cdot A}$

$l' = \frac{l}{n^2}$

The new resistance is:

$R' = \rho \cdot \frac{\frac{l}{n^2}}{n^2 \cdot A}$

$R' = \frac{R}{n^4}$

5. **Answer :** A) 300%

The formula for resistance based on constant volume is:

$R = \rho \cdot \frac{l^2}{V}$

The length is increased by 100%, meaning it is doubled ($l' = 2 \cdot l$).

The new resistance is:

$R' = \rho \cdot \frac{(2 \cdot l)^2}{V}$

$R' = 4 \cdot R$

The change in resistance is:

$\Delta R = 4 \cdot R - R$

$\Delta R = 3 \cdot R$

The percentage change is:

Percentage Change $= \frac{3 \cdot R}{R} \cdot 100$

Percentage Change = 300%

6. **Answer :** D) $R < R_1$

For two resistors in parallel,

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

Since $R_1 > R_2$, the equivalent resistance is always less than the smallest resistance in the parallel combination.

Therefore,

$R < R_2 < R_1$

Therefore, it must also be true that $R < R_1$.

7. **Answer :** B) n^2x

The equivalent resistance x of n equal resistors in parallel is:

$$x = \frac{R}{n}$$

Solving for R:

$R = n \cdot x$

When connected in series, the equivalent resistance is:

$R_s = n \cdot R$

Substituting the value of R:

$R_s = n \cdot (n \cdot x)$

$R_s = n^2 \cdot x$

8. **Answer :** C) 3 Ω

The point D divides the bottom 4 ohm resistor into two equal halves of 2 ohm each.

There are two parallel paths from point A to point D.

Resistance of the first path (A to B to D) is:

$R_{path1} = 4 + 2$

$R_{path1} = 6\Omega$

Resistance of the second path (A to C to D) is:

$R_{path2} = 4 + 2$

$R_{path2} = 6\Omega$

The equivalent resistance between A and D is:

$$\frac{1}{R_{AD}} = \frac{1}{6} + \frac{1}{6}$$

$$\frac{1}{R_{AD}} = \frac{2}{6}$$

$R_{AD} = 3\Omega$

9. **Answer :** D) 48 V

The formula for potential difference between two points is based on Kirchhoff's Voltage Law:

$V_A - \Sigma(I \cdot R) + \Sigma E = V_B$

The total resistance in the branch is:

$R_{total} = 6 + 9 + 5$

$R_{total} = 20\Omega$

The net electromotive force is (assuming standard polarities from left to right as drawn):

$E_{net} = -4 - 12$

$E_{net} = -16\,V$

Using the voltage drop equation from A to B with a current of 2 A:

$V_A - I \cdot R_{total} + E_{net} = V_B$

$V_A - V_B = I \cdot R_{total} - E_{net}$

$V_A - V_B = 2 \cdot 20 - (-16)$
$V_A - V_B = 40 + 16$
$V_A - V_B = 56\,V$

10. **Answer :** C) 3.0 V

Step 1: Left combination (between A and the middle junction)
There are three 6 Ω resistors in parallel.
$R_L = \left(\frac{1}{6} + \frac{1}{6} + \frac{1}{6}\right)^{-1} = 2\,\Omega$
Step 2: Right combination (between the middle junction and Q)
There are two branches:
Top branch: $6\Omega + 6\Omega = 12\Omega$
Bottom branch: 6Ω
These are in parallel:
$R_R = \frac{12 \times 6}{12 + 6} = \frac{72}{18} = 4\,\Omega$
Step 3: Total equivalent resistance
$R_{\text{eq}} = 2 + 4 = 6\,\Omega$
Step 4: Potential difference $V_P - V_Q$
The current through the circuit is 0.5 A.
$V_P - V_Q = IR = (0.5)(6) = 3.0$ V
$V = 3.0V$

11. **Answer :** C) 2Ω

For an infinite ladder, the equivalent resistance R_{eq} remains the same if one repeating unit is added.
$R_{eq} = R_1 + \frac{R_2 \cdot R_{eq}}{R_2 + R_{eq}}$
Substitute the given values ($R_1 = 1$ and $R_2 = 2$):
$R_{eq} = 1 + \frac{2 \cdot R_{eq}}{2 + R_{eq}}$
Multiply both sides by $(2 + R_{eq})$:
$R_{eq} \cdot (2 + R_{eq}) = 1 \cdot (2 + R_{eq}) + 2 \cdot R_{eq}$
$2 \cdot R_{eq} + R_{eq}^2 = 2 + R_{eq} + 2 \cdot R_{eq}$
$R_{eq}^2 - R_{eq} - 2 = 0$
Factorizing the quadratic equation:
$(R_{eq} - 2) \cdot (R_{eq} + 1) = 0$
Since resistance cannot be negative:
$R_{eq} = 2\Omega$

12. **Answer :** C) n^2

The maximum resistance occurs when all n resistors are in series.
$R_{max} = n \cdot R$
The minimum resistance occurs when all n resistors are in parallel.
$R_{min} = \frac{R}{n}$
The ratio of the maximum to the minimum resistance is:
Ratio $= \frac{R_{max}}{R_{min}}$

$$\text{Ratio} = \frac{n \cdot R}{\frac{R}{n}}$$

$$\text{Ratio} = n^2$$

13. **Answer :** D) $\frac{8}{3}\Omega$

When connected in an equilateral triangle, two resistors are in series, and this combination is in parallel with the third resistor.

The resistance of the series part is:

$R_s = 4 + 4$

$R_s = 8\Omega$

The equivalent resistance R_{eq} between any two corners is:

$$R_{eq} = \frac{R_s \cdot 4}{R_s + 4}$$

$$R_{eq} = \frac{8 \cdot 4}{8 + 4}$$

$$R_{eq} = \frac{32}{12}$$

$$R_{eq} = \frac{8}{3}\Omega$$

14. **Answer :** C) 5 Ω

Left resistor = 2 Ω

Right resistor = 2 Ω

These are in series with the middle network.

The middle section has two 2 Ω resistors connected in parallel.

Their equivalent resistance is

$$R = \frac{2 \times 2}{2 + 2} = \frac{4}{4} = 1\ \Omega$$

Since all three parts are in series,

$R_{\text{eq}} = 2 + 1 + 2 = 5\ \Omega$

15. **Answer :** D) $2\frac{2}{3}$ ohm

First, observe the three resistors connected in parallel between terminal A and the middle junction. Since each resistor has a value of 2 Ω, we can calculate their equivalent resistance (R_p) using the formula for parallel resistors:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{R_p} = \frac{3}{2}$$

$$R_p = \frac{2}{3}\ \Omega$$

Next, this parallel combination (R_p) is in series with the fourth resistor located before terminal B, which is also 2 Ω. To find the total equivalent resistance (R_{AB}), we add the series values together:

$R_{AB} = R_p + 2\ \Omega$

$$R_{AB} = \frac{2}{3} + 2$$

To add these, convert the whole number to a fraction with a common denominator:

$R_{AB} = \frac{2}{3} + \frac{6}{3}$

$R_{AB} = \frac{8}{3}\ \Omega$

Convert the improper fraction $\frac{8}{3}$ into a mixed fraction:

$\frac{8}{3} = 2\frac{2}{3}\ \Omega$

16. **Answer :** B) $\frac{p_1+p_2}{2}$

The formula for the equivalent resistance of two wires in series is:

$R_{eq} = R_1 + R_2$

Expressing resistance in terms of resistivity, length, and area (since dimensions are the same, L and A are constant):

$\rho_{eq} \cdot \frac{2 \cdot L}{A} = \rho_1 \cdot \frac{L}{A} + \rho_2 \cdot \frac{L}{A}$

Dividing the entire equation by $\frac{L}{A}$:

$2 \cdot \rho_{eq} = \rho_1 + \rho_2$

$\rho_{eq} = \frac{\rho_1 + \rho_2}{2}$

17. **Answer :** A) $\frac{14}{3}\Omega$

The circuit represents a Wheatstone bridge. We first check for balance by comparing the ratio of arms.

Ratio 1 $= \frac{3}{6}$

Ratio 1 $= \frac{1}{2}$

Ratio 2 $= \frac{4}{8}$

Ratio 2 $= \frac{1}{2}$

Since the ratios are equal, the bridge is balanced, and no current flows through the central 7 ohm resistor.

The equivalent resistance is the parallel combination of the top series branch and the bottom series branch.

$R_{top} = 3 + 4$

$R_{top} = 7\Omega$

$R_{bottom} = 6 + 8$

$R_{bottom} = 14\Omega$

$R_{eq} = \frac{R_{top} \cdot R_{bottom}}{R_{top} + R_{bottom}}$

$R_{eq} = \frac{7 \cdot 14}{7 + 14}$

$R_{eq} = \frac{98}{21}$

$R_{eq} = \frac{14}{3}\Omega$

18. **Answer :** A) $\frac{7}{3}$ A from a to b through e

The circuit contains two voltage sources in a single loop. We apply Kirchhoff's Voltage Law.

The 10V and 4V batteries are connected such that their voltages aid each other in the loop.

$E_{net} = 10 + 4$

$E_{net} = 14V$

The total resistance in the circuit is the sum of the series resistors:

$R_{total} = 1 + 2 + 3$

$R_{total} = 6\Omega$

The current in the circuit is:

$$I = \frac{E_{net}}{R_{total}}$$

$$I = \frac{14}{6}$$

$$I = \frac{7}{3}A$$

The current direction is driven by the net positive potential, moving from the positive terminal of the 10V cell (a) towards (b) through the central line (e).

19. **Answer :** B) $\frac{(E-V)R}{V}$

The formula for the terminal potential difference V of a cell discharging is:

$V = E - I \cdot r$

Rearranging for the internal resistance r:

$I \cdot r = E - V$

$$r = \frac{E - V}{I}$$

By Ohm's Law for the external circuit, the current is:

$$I = \frac{V}{R}$$

Substitute this expression for I into the equation for r:

$$r = \frac{E - V}{\frac{V}{R}}$$

$$r = \frac{(E - V) \cdot R}{V}$$

20. **Answer :** C) Energy

Kirchhoff's law of meshes (or Kirchhoff's Voltage Law) states that the sum of all voltages around a closed loop is zero.

This principle is a direct consequence of the law of conservation of energy.

It ensures that the work done to move a charge around a closed loop is zero.

21. **Answer :** A) Zero

According to Kirchhoff's Voltage Law, the algebraic sum of the potential changes in a closed loop is zero.

This includes the products of current and resistance (voltage drops) and the electromotive forces (e.m.f).

$\Sigma E - \Sigma(I \cdot R) = 0$

Therefore, the net sum is zero.

22. **Answer :** A) $R = \frac{r}{2}$

The maximum power theorem states that maximum energy is delivered to an external resistor when its resistance equals the equivalent internal resistance of the source.

The two cells of electromotive force E and internal resistance r are connected in parallel.
The equivalent internal resistance r_{eq} of two identical resistors in parallel is:
$r_{eq} = \frac{r}{2}$
For maximum energy transfer to the external resistor R:
$R = r_{eq}$
$R = \frac{r}{2}$

23. **Answer :** A) 1.7 amp
According to Kirchhoff's Current Law (KCL), the sum of currents entering a junction equals the sum of currents leaving the junction.
At the first (left) junction, two currents of 2 A enter.
Current entering middle wire = 2 + 2
$I_{\text{middle}} = 4$ A
At the second (right) junction, this 4 A current splits into three branches.
$I_{\text{middle}} = I_1 + I_2 + i$
Substituting the given values for the leaving currents:
$4 = 1 + 1.3 + i$
$4 = 2.3 + i$
$i = 4 - 2.3$
$i = 1.7$ A

24. **Answer :** A) 0.1
The circuit consists of three parallel branches connected across the same potential difference.
The resistances of the branches are:
Branch 1: 60 Ω
Branch 2: 15 Ω + 5 Ω = 20 Ω
Branch 3: 10 Ω
The equivalent resistance R_{eq} of these parallel branches is given by:
$\frac{1}{R_{eq}} = \frac{1}{60} + \frac{1}{20} + \frac{1}{10}$
$\frac{1}{R_{eq}} = \frac{1+3+6}{60}$
$\frac{1}{R_{eq}} = \frac{10}{60}$
$R_{eq} = 6\ \Omega$
The total current entering the parallel combination is 1 A.
The potential difference V across the parallel branches is:
$V = I_{total} \cdot R_{eq}$
$V = 1 \cdot 6$
$V = 6$ V
The current i flows through the top 60 Ω branch. By Ohm's Law:
$i = \frac{V}{R_{top}}$
$i = \frac{6}{60}$
$i = 0.1$ A

25. **Answer :** C) Network is not balanced

For a standard Wheatstone bridge with arms P, Q, R, and S, the balance condition is:

$\frac{P}{Q} = \frac{R}{S}$

This can be rearranged as $P \cdot S = R \cdot Q$.

If the resistances in arms Q and S are interchanged, the new condition for balance would require:

$\frac{P}{S} = \frac{R}{Q}$

This implies $P \cdot Q = R \cdot S$, which generally contradicts the original balance condition $P \cdot S = R \cdot Q$ unless $Q = S$ or $P = R$.

Therefore, for arbitrary resistance values, interchanging Q and S disrupts the equality, and the network is no longer balanced.

26. **Answer :** B) The bridge still balances

The balance condition for a Wheatstone bridge with arms R_1, R_2, R_3, and R_4 is:

$\frac{R_1}{R_2} = \frac{R_3}{R_4}$

This equation can be algebraically rearranged by cross-multiplying and dividing:

$\frac{R_1}{R_3} = \frac{R_2}{R_4}$

When the galvanometer and the battery are interchanged, the nodes that were previously equipotential become the input nodes, and the input nodes become the equipotential nodes.

The new effective resistance ratios of the arms perfectly match the rearranged equation above.

Thus, the condition for zero deflection in the galvanometer remains satisfied, and the bridge remains balanced.

LEVEL – II

1. **Answer :** C) 1:2

The formula for the resistance of a wire is:

$R = \rho \cdot \frac{l}{A}$

The volume of a wire is given by:

$V = A \cdot l$

Rearranging for area A:

$A = \frac{V}{l}$

Substituting this into the resistance formula gives:

$R = \rho \cdot \frac{l}{\frac{V}{l}}$

$R = \rho \cdot \frac{l^2}{V}$

Since both wires are made of the same material, their resistivity ρ is the same.

The ratio of their resistances is:

$\frac{R_1}{R_2} = \left(\frac{l_1}{l_2}\right)^2 \cdot \left(\frac{V_2}{V_1}\right)$

Given that the ratio of lengths is $\frac{l_1}{l_2} = \frac{1}{2}$ and the ratio of volumes is $\frac{V_1}{V_2} = \frac{1}{2}$:

$\frac{R_1}{R_2} = \left(\frac{1}{2}\right)^2 \cdot \left(\frac{2}{1}\right)$

$\frac{R_1}{R_2} = \frac{1}{4} \cdot 2$

$\frac{R_1}{R_2} = \frac{1}{2}$

The ratio of their resistances is 1:2.

2. **Answer :** C) 80 Ω

The formula for the resistance of a wire is:

$R = \rho \cdot \frac{l}{A}$

For a cylindrical wire, the cross-sectional area is $A = \pi \cdot r^2$, so:

$R = \rho \cdot \frac{l}{\pi \cdot r^2}$

The original resistance is given as $R = 10\,\Omega$.

For the new wire, the length is twice the original ($l' = 2l$) and the radius is half the original ($r' = \frac{r}{2}$).

The new area is:

$A' = \pi \cdot \left(\frac{r}{2}\right)^2$

$A' = \frac{\pi \cdot r^2}{4}$

$A' = \frac{A}{4}$

The new resistance R' is:

$R' = \rho \cdot \frac{l'}{A'}$

$R' = \rho \cdot \frac{2l}{\frac{A}{4}}$

$R' = 8 \cdot \left(\rho \cdot \frac{l}{A}\right)$

$R' = 8 \cdot R$

$R' = 8 \cdot 10$

$R' = 80\,\Omega$

3. **Answer :** 0.16 A

Let the potential at the central junction of the star connection be x.

Applying Kirchhoff's Current Law (KCL) at the central junction, the sum of currents leaving the node is zero:

$\frac{x-10}{10} + \frac{x-7}{30} + \frac{x-5}{30} = 0$

To eliminate the denominators, multiply the entire equation by 30:

$3 \cdot (x-10) + 1 \cdot (x-7) + 1 \cdot (x-5) = 0$

$3x - 30 + x - 7 + x - 5 = 0$

$5x - 42 = 0$

$5x = 42$

$x = 8.4$ V

The current I through the 10 Ω resistor is driven by the potential difference across it:

$$I = \frac{10 - x}{10}$$
$$I = \frac{10 - 8.4}{10}$$
$$I = \frac{1.6}{10}$$
$$I = 0.16 \text{ A}$$

4. **Answer :** C) 280 m

The formula relating resistance, resistivity, length, and area is:

$$R = \rho \cdot \frac{l}{A}$$

The cross-sectional area A of the wire is:

$$A = \frac{\pi \cdot d^2}{4}$$

Given the diameter $d = 1.4$ mm $= 1.4 \times 10^{-3}$ m:

$$A = \frac{\frac{22}{7} \cdot (1.4 \times 10^{-3})^2}{4}$$
$$A = \frac{\frac{22}{7} \cdot 1.96 \times 10^{-6}}{4}$$
$$A = \frac{22 \cdot 0.28 \times 10^{-6}}{4}$$
$$A = 22 \cdot 0.07 \times 10^{-6}$$
$$A = 1.54 \times 10^{-6} \text{ m}^2$$

Rearranging the resistance formula to solve for length l:

$$l = \frac{R \cdot A}{\rho}$$

Substituting the given values ($R = 4\ \Omega$, $\rho = 2.2 \times 10^{-8}\ \Omega$m):

$$l = \frac{4 \cdot 1.54 \times 10^{-6}}{2.2 \times 10^{-8}}$$
$$l = \frac{6.16 \times 10^{-6}}{2.2 \times 10^{-8}}$$
$$l = 2.8 \times 10^2$$
$$l = 280 \text{ m}$$

5. **Answer :** A) 10 Ω

Let's label the unmarked junctions to make it easier to trace:
Let the top junction be **Node A**.
Let the bottom-left junction be **Node B**.
Now, let's map out where each resistor is connected relative to the terminals **P** and **Q**:
Between P and Node A: **10 Ω**
Between Node A and Q: **5 Ω**
Between P and Node B: **20 Ω**
Between Node B and Q: **10 Ω**
Between Node A and Node B: **15 Ω**

This specific configuration forms a standard **Wheatstone bridge**, where the 15 Ω resistor acts as the central cross-connection (galvanometer arm) between the two parallel branches.

Check for a Balanced Bridge

A Wheatstone bridge is "balanced" if the ratio of the resistances in the upper arm equals the ratio in the lower arm. When measuring between P and Q, we check the ratios of the adjacent resistors:

Upper arm ratio: $\frac{R_{PA}}{R_{AQ}} = \frac{10}{5} = 2$

Lower arm ratio: $\frac{R_{PB}}{R_{BQ}} = \frac{20}{10} = 2$

Because both ratios are equal (2 = 2), the bridge is perfectly balanced.

Simplify the Circuit

In a balanced Wheatstone bridge, the electrical potential at Node A is equal to the potential at Node B. Because there is no voltage difference between these two points, zero current will flow through the 15 Ω resistor.

Therefore, we can safely remove the 15 Ω resistor from our calculations, simplifying the circuit into two simple parallel branches:

Upper branch (Node A): The 10 Ω and 5 Ω resistors are now in series.

$R_{upper} = 10 + 5 = 15\ \Omega$

Lower branch (Node B): The 20 Ω and 10 Ω resistors are now in series.

$R_{lower} = 20 + 10 = 30\ \Omega$

Calculate the Effective Resistance

Finally, we calculate the equivalent resistance (R_{eq}) of these two parallel branches connected between P and Q:

$$\frac{1}{R_{eq}} = \frac{1}{R_{upper}} + \frac{1}{R_{lower}}$$

$$\frac{1}{R_{eq}} = \frac{1}{15} + \frac{1}{30}$$

$$\frac{1}{R_{eq}} = \frac{2}{30} + \frac{1}{30} = \frac{3}{30}$$

$$R_{eq} = \frac{30}{3} = 10\ \Omega$$

6. **Answer :** A) $\frac{3}{8}A, \frac{1}{8}A$

The circuit consists of four parallel branches connected across two main nodes. Let the right node be at potential V and the left node be at 0V.

Applying KCL at the right node (sum of currents leaving = 0):

$$\frac{V}{4} + \frac{V-2}{4} + \frac{V}{4} + \frac{V}{4} = 0$$

Multiplying the entire equation by 4:

$V + V - 2 + V + V = 0$

$4V - 2 = 0$

$4V = 2$

$V = 0.5$ V

The current from the battery (in the second branch) flows leftward, driven by the 2V source against the 0.5V node potential:

$$I_{battery} = \frac{2 - V}{4}$$

$$I_{battery} = \frac{2 - 0.5}{4}$$

$$I_{battery} = \frac{1.5}{4}$$

$$I_{battery} = \frac{3}{8}\ \text{A}$$

The ammeter is in the third branch, measuring the current flowing through it:

$$I_{ammeter} = \frac{V - 0}{4}$$

$$I_{ammeter} = \frac{0.5}{4}$$

$$I_{ammeter} = \frac{1}{8}\ \text{A}$$

7. **Answer :** B) $8r/7$

The circuit is a highly symmetric bridge structure. We apply a voltage V between nodes A and B and find the total current I entering A.

Due to the symmetry of the network, the central node O sits at a potential of $\frac{V}{2}$ relative to B.

Applying Kirchhoff's laws and simplifying the delta-star transformations for the upper symmetric portion reveals the current distribution.

Let $V_A = 1\text{V}$ and $V_B = 0\text{V}$.

The potentials at the upper intermediate nodes can be found via nodal analysis, yielding $V_{M1} = \frac{5}{8}\text{V}$.

The total current leaving node A through its two connected resistors (r to M1 and r to O) is:

$$I_A = \frac{V_A - V_{M1}}{r} + \frac{V_A - V_O}{r}$$

$$I_A = \frac{1 - \frac{5}{8}}{r} + \frac{1 - \frac{1}{2}}{r}$$

$$I_A = \frac{\frac{3}{8}}{r} + \frac{\frac{4}{8}}{r}$$

$$I_A = \frac{7}{8r}$$

The equivalent resistance R_{eq} is the ratio of total voltage to total current:

$$R_{eq} = \frac{V_A}{I_A}$$

$$R_{eq} = \frac{1}{\frac{7}{8r}}$$

$$R_{eq} = \frac{8r}{7}$$

8. **Answer :** D) 2Ω

The circuit consists of outer edge resistors of 3 Ω and internal radial spokes of 6 Ω connected to node A.

We can simplify the circuit by starting from the node farthest from B and folding it recursively, as the components form parallel and series combinations.

The path A-F-E has resistance 3 + 3 = 6 Ω.

This is in parallel with the spoke AE (6 Ω):

$R_{AE_eq} = \frac{6 \cdot 6}{6+6} = 3\ \Omega$

Now, node E is connected to A via this equivalent 3 Ω resistor, and to D via the outer 3 Ω resistor.

The path A-E-D has resistance $3 + 3 = 6\ \Omega$.

This is in parallel with the spoke AD (6 Ω):

$R_{AD_eq} = \frac{6 \cdot 6}{6+6} = 3\ \Omega$

Similarly, the path A-D-C has resistance $3 + 3 = 6\ \Omega$, which is in parallel with the spoke AC (6 Ω):

$R_{AC_eq} = \frac{6 \cdot 6}{6+6} = 3\ \Omega$

Finally, the path A-C-B has resistance $3 + 3 = 6\ \Omega$.

This entire branch is in parallel with the direct outer edge resistor AB (3 Ω).

The equivalent resistance between A and B is:

$R_{eq} = \frac{6 \cdot 3}{6+3}$

$R_{eq} = \frac{18}{9}$

$R_{eq} = 2\ \Omega$

9. **Answer :** B) $\frac{n+1}{2}$

For n wires connected in series, the equivalent resistance is the sum of their individual resistances:

$R_{eq} = R_1 + R_2 + R_3 + \cdots + R_n$

Using the formula $R = \rho \cdot \frac{L}{A}$, and knowing they have the same dimensions (L and A):

$\rho_{eq} \cdot \frac{n \cdot L}{A} = \rho_1 \cdot \frac{L}{A} + \rho_2 \cdot \frac{L}{A} + \cdots + \rho_n \cdot \frac{L}{A}$

Dividing the entire equation by $\frac{L}{A}$:

$n \cdot \rho_{eq} = 1 + 2 + 3 + \cdots + n$

The sum of the first n natural numbers is $\frac{n(n+1)}{2}$:

$n \cdot \rho_{eq} = \frac{n(n+1)}{2}$

Dividing by n gives the equivalent resistivity:

$\rho_{eq} = \frac{n+1}{2}$

10. **Answer :** A, B, C, D

The resistance R of a conductor is given by the formula:

$R = \rho \cdot \frac{L}{A}$

Where L is the length of the conductor and A is its cross-sectional area.

The resistivity ρ depends on the nature of the material and the temperature of the conductor.

Therefore, resistance depends on all four factors: length, area of cross-section, temperature, and nature of the material.

11. **Answer :** A, B, D

Let the potential at the central junction D be x.

Applying Kirchhoff's Current Law (KCL) at node D:

$\frac{x-70}{10}+\frac{x-0}{20}+\frac{x-10}{30}=0$

Multiplying the equation by 60 to clear the denominators:

$6\cdot(x-70)+3\cdot x+2\cdot(x-10)=0$

$6x-420+3x+2x-20=0$

$11x-440=0$

$11x=440$

$x=40$ V

So, the potential at D is 40 V (Option A is correct).

The currents in each section are:

$I_{AD}=\frac{70-40}{10}=\frac{30}{10}=3$ A

$I_{DB}=\frac{40-0}{20}=\frac{40}{20}=2$ A

$I_{DC}=\frac{40-10}{30}=\frac{30}{30}=1$ A

The ratio of currents is $I_{AD}:I_{DB}:I_{DC}=3:2:1$ (Option B is correct).

The total power drawn by the network is the sum of power dissipated in each resistor ($P=I^2\cdot R$):

$P_{total}=(3^2\cdot 10)+(2^2\cdot 20)+(1^2\cdot 30)$

$P_{total}=(9\cdot 10)+(4\cdot 20)+(1\cdot 30)$

$P_{total}=90+80+30$

$P_{total}=200$ W

(Option D is correct).

12. **Answer :** A, B, C, D

The circuit is a balanced Wheatstone bridge framework.

Solving nodal equations for the network reveals the potentials at nodes P, Q, S, and T.

Let the source voltage be 12 V applied across the input terminals.

Due to the symmetry and specific resistance ratios of the branches:

$V_P=8$ V and $V_Q=8$ V

$V_S=4$ V and $V_T=4$ V

Since $V_P=V_Q$, the potential difference across PQ is zero, hence the current through PQ is zero (Option A is correct).

The total equivalent resistance of the balanced bridge is:

$R_{eq}=\frac{(2+2)\cdot(4+4)}{(2+2)+(4+4)}$

$R_{eq}=\frac{4\cdot 8}{4+8}=\frac{32}{12}=\frac{8}{3}$ Omega

Total current $I_1=\frac{12}{4}=3$ A (Option B is correct).

The potential at S (4 V) is less than the potential at Q (8 V) (Option C is correct).

The current I_2 through the branch PS is:

$I_2=\frac{V_P-V_S}{2}=\frac{8-4}{2}=2$ A

(Option D is correct).

13. **Answer :** C

The total resistance of the wire is R. For a regular n-sided polygon, the resistance of each side is $\frac{R}{n}$.

The equivalent resistance between two corners separated by k sides involves two parallel paths: one with k sides and the other with $(n-k)$ sides.

The resistance of path 1 is $R_1 = k \cdot \frac{R}{n}$.

The resistance of path 2 is $R_2 = (n-k) \cdot \frac{R}{n}$.

The equivalent resistance R_{eq} is:

$$R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$R_{eq} = \frac{\left(k \cdot \frac{R}{n}\right) \cdot \left((n-k) \cdot \frac{R}{n}\right)}{R}$$

$$R_{eq} = \frac{k(n-k)}{n^2} \cdot R$$

The minimum value of R_{eq} occurs when $k = 1$ (adjacent corners):

$$R_{min} = \frac{1 \cdot (n-1)}{n^2} \cdot R$$

$$R_{min} = R\left(\frac{n-1}{n^2}\right)$$

14. Answer : B

Let the top common node (A-C-F) be at potential V and the bottom common node (B-D-G) be at 0 V.

By Kirchhoff's Current Law at node D, the sum of entering currents equals leaving currents.

From the diagram, I_1 flows upwards through the 12 V battery branch.

$V_C - V_D = V - 0 = V$

$V = 12 - 4I_1$

For the branch with the 4 V battery, the current I_2 flows from G to D, meaning it flows downwards in the branch:

$V = 4 + 6I_2$

Equating the two expressions for V and using the KCL relationship $I_1 = 0.5 + I_2$ gives the solution where:

$I_1 = 1.1$ A

This satisfies the nodal voltage requirement perfectly:

$V = 12 - 4(1.1) = 12 - 4.4 = 7.6$ V

$V = 4 + 6(0.6) = 4 + 3.6 = 7.6$ V

(where $I_2 = 0.6$ A).

15. Answer : A) $\frac{R}{n}$

The uniform wire of total resistance R is bent into a regular n-sided polygon.

Since it is a regular polygon, all n sides are equal in length.

Because resistance is directly proportional to length, the total resistance R is divided equally among the n sides.

Therefore, the resistance of each individual side is:

$$r = \frac{R}{n}$$

16. Answer : D) $\frac{R}{4}$

When calculating the resistance between opposite corners of an even n-sided polygon, the circuit divides into two equal parallel paths.
Each path contains exactly half of the polygon's sides, which is $\frac{n}{2}$ sides.
The resistance of one such path is:

$$R_{path} = \frac{n}{2} \cdot \frac{R}{n}$$

$$R_{path} = \frac{R}{2}$$

The equivalent resistance R_{eq} of these two paths in parallel is:

$$R_{eq} = \frac{\frac{R}{2} \cdot \frac{R}{2}}{\frac{R}{2} + \frac{R}{2}}$$

$$R_{eq} = \frac{\frac{R^2}{4}}{R}$$

$$R_{eq} = \frac{R}{4}$$

17. **Answer :** B) $R(n-1)/n^2$

For adjacent corners, the circuit consists of two parallel paths: one path is a single side, and the other path comprises the remaining $(n-1)$ sides.
Resistance of the first path (one side):

$$R_1 = \frac{R}{n}$$

Resistance of the second path ($(n-1)$ sides):

$$R_2 = (n-1) \cdot \frac{R}{n}$$

The equivalent resistance R_{eq} is:

$$R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$R_{eq} = \frac{\left(\frac{R}{n}\right) \cdot \left(\frac{(n-1)R}{n}\right)}{R}$$

$$R_{eq} = \frac{(n-1)R^2}{n^2 \cdot R}$$

$$R_{eq} = \frac{n-1}{n^2} \cdot R$$

18. **Answer :** 2.75 V

Top Branch:
E_1 = 2 V
R_1 = 2 Ω
Middle Branch: (The two resistors are in series)
E_2 = 3 V
R_2 = 1 Ω + 3 Ω = 4 Ω
Bottom Branch:
E_3 = 4 V
R_3 = 4 Ω
Equivalent EMF Calculation
We will use Millman's Theorem again to find the equivalent voltage (E_{eq}) across the parallel branches:

$$E_{eq} = \frac{\sum \frac{E_i}{R_i}}{\sum \frac{1}{R_i}}$$

Substitute the corrected values into the formula:

$$E_{eq} = \frac{\frac{2}{2} + \frac{3}{4} + \frac{4}{4}}{\frac{1}{2} + \frac{1}{4} + \frac{1}{4}}$$

Calculate the numerator (sum of currents):

Numerator = 1 + 0.75 + 1 = 2.75 A

Calculate the denominator (equivalent conductance):

Denominator = 0.5 + 0.25 + 0.25 = 1 Ω^{-1}

Divide the numerator by the denominator:

$$E_{eq} = \frac{2.75}{1}$$

The correct equivalent emf between the terminals A and B is **2.75 V**.

19. **Answer :** 1 Ω

Top Branch (R_1): 2 Ω

Middle Branch (R_2): The 1 Ω and 3 Ω resistors are in series, so we add them together: 1 + 3 = 4 Ω

Bottom Branch (R_3): 4 Ω

Calculate the equivalent parallel resistance:

Since these three branches are connected in parallel between terminals A and B, we use the parallel resistance formula:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Substitute the values we found:

$$\frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{4} + \frac{1}{4}$$

Find a common denominator (which is 4) to add the fractions:

$$\frac{1}{R_{eq}} = \frac{2}{4} + \frac{1}{4} + \frac{1}{4}$$

$$\frac{1}{R_{eq}} = \frac{4}{4} = 1$$

Invert the result to find R_{eq}:

$$R_{eq} = 1$$

20. **Answer :** 0.75 V

Top Branch: Positive terminal faces A.

$E_1 = 2$ V

$R_1 = 2\ \Omega$

Middle Branch: Positive terminal faces A.

$E_2 = 3$ V

$R_2 = 1\ \Omega + 3\ \Omega = 4\ \Omega$

Bottom Branch (Reversed): Negative terminal now faces A.

$E_3 = -4$ V

$R_3 = 4\ \Omega$

Equivalent EMF Calculation

Using Millman's Theorem:

$$E_{eq} = \frac{\sum \frac{E_i}{R_i}}{\sum \frac{1}{R_i}}$$

Substitute the updated values into the formula:

$$E_{eq} = \frac{\frac{2}{2} + \frac{3}{4} + \frac{-4}{4}}{\frac{1}{2} + \frac{1}{4} + \frac{1}{4}}$$

Calculate the numerator:

Numerator $= 1 + 0.75 - 1 = 0.75$ A

Calculate the denominator (equivalent conductance remains the same):

Denominator $= 0.5 + 0.25 + 0.25 = 1\ \Omega^{-1}$

Divide the numerator by the denominator:

$$E_{eq} = \frac{0.75}{1} = 0.75 \text{ V}$$

21. **Answer :** B) Both Assertion and Reason are true, but Reason is not the correct explanation

Assertion (A) states that in a parallel combination, the potential difference is the same across each resistor. This is true because all resistors are connected directly between the same two nodes.

Reason (R) states that current divides among different branches. This is also a true statement describing the behavior of parallel circuits.

However, the current dividing (Reason) is a consequence of the parallel arrangement and individual resistances, not the fundamental physical reason *why* the potential difference is the same. The potential difference is the same purely due to the topological connection to common equipotential nodes. Therefore, R does not correctly explain A.

22. **Answer :** A) Both Assertion and Reason are true, and Reason is the correct explanation

Assertion (A) states that in a balanced Wheatstone bridge, no current flows through the galvanometer. This is the definition of a balanced bridge and is true.

Reason (R) states that the potential difference across the galvanometer is zero. This is also true.

According to Ohm's Law ($I = \frac{V}{R}$), if the potential difference (V) across a component is zero, the current (I) flowing through it must be zero. Therefore, the zero potential difference perfectly explains why no current flows.

23. **Answer :** a-Q, b-R, S, c-Q, d-P

The reciprocal of resistivity is conductivity. Its unit is $\text{ohm}^{-1}\text{m}^{-1}$ (Q).

Resistivity (ρ)($\rho = \frac{R\,A}{l}$) (R) in terms of microscopic quantities is given by $\rho = \frac{m}{ne^2\tau}$ (S).

Conductivity is the reciprocal of resistivity; it also shares the unit $\text{ohm}^{-1}\text{m}^{-1}$ (Q) (or matching to its respective formula if explicitly provided).

Drift speed (v_d) is related to current by $I = neAv_d$, so $v_d = \frac{i}{neA}$ (P).

24. **Answer :**

Let the potential at node **F** be the reference, so $V_F = 0$ V.

The battery between A and F is 4 V with the positive terminal at F (longer line). So, $V_F - V_A = 4 \Longrightarrow V_A = -4$ V.

The battery between F and E is 4 V with the positive terminal at E. So, $V_E - V_F = 4 \Longrightarrow V_E = 4$ V.

The entire top wire (B-C-D) acts as a single node. Let its potential be V.
The 6 V battery in the middle branch has its positive terminal facing up (towards C).
The unknown battery ϵ has its positive terminal facing up (towards D).
Now, let's write the expressions for the currents leaving the top node (B-C-D):
Current through 2Ω (from B to A):

$$I_{BA} = \frac{V - V_A}{2} = \frac{V - (-4)}{2} = \frac{V + 4}{2}$$

Current through 4Ω (from C to F):
Moving from C to F, we pass the resistor and then go against the 6 V battery.

$$I_{CF} = \frac{V - 6 - V_F}{4} = \frac{V - 6}{4}$$

Current through R (from D to E):
Moving from D to E, we pass resistor R and go against battery ϵ.

$$I_{DE} = \frac{V - \epsilon - V_E}{R} = \frac{V - \epsilon - 4}{R}$$

Applying Kirchhoff's Current Law (sum of currents leaving node = 0):

$$\frac{V + 4}{2} + \frac{V - 6}{4} + \frac{V - \epsilon - 4}{R} = 0$$

Multiplying the entire equation by 4 to clear standard denominators:

$$2(V + 4) + (V - 6) + \frac{4(V - \epsilon - 4)}{R} = 0$$

$$3V + 2 + \frac{4V - 4\epsilon - 16}{R} = 0$$

Solving for V:

$$V(3R + 4) + 2R - 4\epsilon - 16 = 0 \Longrightarrow V = \frac{4\epsilon + 16 - 2R}{3R + 4}$$

Now we can find the specific currents in terms of ϵ and R:
Current through 4Ω (I_{CF}):

$$I_{CF} = \frac{V - 6}{4} = \frac{1}{4}\left(\frac{4\epsilon + 16 - 2R}{3R + 4} - 6\right) = \frac{1}{4}\left(\frac{4\epsilon - 20R - 8}{3R + 4}\right) = \frac{\epsilon - 5R - 2}{3R + 4}$$

Current through 2Ω (I_{BA}):

$$I_{BA} = \frac{V + 4}{2} = \frac{1}{2}\left(\frac{4\epsilon + 16 - 2R}{3R + 4} + 4\right) = \frac{1}{2}\left(\frac{4\epsilon + 10R + 32}{3R + 4}\right) = \frac{2\epsilon + 5R + 16}{3R + 4}$$

25. Evaluating Column I Statements
(Note: In Column II, options P, Q, and R are identical: "Possible if $\epsilon = 6V$". We will evaluate against $\epsilon = 6V$ and general values)

a) Current passing through 4Ω resistance can be zero
For $I_{CF} = 0$, we need the numerator to be zero: $\epsilon - 5R - 2 = 0 \Longrightarrow R = \frac{\epsilon - 2}{5}$.
Since R must be a positive, non-zero resistance, this is only possible if $\epsilon > 2$ V.
If $\epsilon = 6$ V *(Options P, Q, R)*: $R = \frac{6-2}{5} = 0.8\Omega$. This is a valid finite resistance, so it is possible.

b) Current passing through 4Ω resistance can be from F to C direction
This means I_{CF} is negative, so $\epsilon - 5R - 2 < 0 \Longrightarrow R > \frac{\epsilon - 2}{5}$.
Since R is a variable that can be arbitrarily large, we can always choose an R that satisfies this for *any* given ϵ.
If $\epsilon = 6$ V *(Options P, Q, R)*: We just need $R > 0.8\Omega$, which is easily possible.
For any value (Option S): As proven, this is also possible for any value of ϵ.

c) Current passing through 4Ω resistance can be from C to F direction

This means I_{CF} is positive, so $\epsilon - 5R - 2 > 0 \Longrightarrow R < \frac{\epsilon-2}{5}$.

Since R must be > 0, we need $0 < R < \frac{\epsilon-2}{5}$, which requires $\epsilon > 2$ V.

If $\epsilon = 6$ V *(Options P, Q, R)*: We need $0 < R < 0.8\Omega$, which is a valid range. It is possible.

For any value (Option S): If $\epsilon \leq 2$ V, this is impossible. So S does not apply.

d) Current passing through 2Ω resistance will be from B to A direction

Look at the derived current: $I_{BA} = \frac{2\epsilon+5R+16}{3R+4}$.

Since R is positive and ϵ is a non-negative magnitude, the numerator $(2\epsilon + 5R + 16)$ is always positive, and the denominator $(3R + 4)$ is always positive. Therefore, I_{BA} is always greater than zero.

The current *will always* flow from B to A, making this possible (and certain) for any value of ϵ from zero to infinite (Option S).

26. Answer : a → P , b → Q, c → R, d → S

We are given that all four bulbs consume the same power, which we will denote as P. We are also given the resistance of bulb 1, $R_1 = 36\ \Omega$.

Finding Resistance of Bulb 3 (a)

Bulbs 2 and 3 are in series, meaning the exact same current flows through them. Since they consume the same power ($P = I^2R$), their resistances must be perfectly equal: $R_2 = R_3$.

Let the voltage across the parallel combination (Bulb 1 and the series pair 2 & 3) be V_p.

Power of Bulb 1: $P = \frac{V_p^2}{R_1} = \frac{V_p^2}{36}$

The current through the top branch is $I_{23} = \frac{V_p}{R_2+R_3} = \frac{V_p}{2R_3}$.

Power of Bulb 3: $P = I_{23}^2 R_3 = \left(\frac{V_p}{2R_3}\right)^2 R_3 = \frac{V_p^2}{4R_3^2} \cdot R_3 = \frac{V_p^2}{4R_3}$

Equating the two power expressions: $\frac{V_p^2}{36} = \frac{V_p^2}{4R_3}$

$36 = 4R_3 \Longrightarrow R_3 = 9\ \Omega$.

Match: (a) → (P)

Finding Resistance of Bulb 4 (b)

Let the total current from the battery be I_{total}. This total current must flow through Bulb 4.

I_{total} splits into I_1 (through Bulb 1) and I_{23} (through Bulbs 2 & 3).

From $P = I^2R$, we can find the individual currents in terms of P:

$$I_1 = \sqrt{\frac{P}{R_1}} = \sqrt{\frac{P}{36}} = \frac{\sqrt{P}}{6}$$

$$I_{23} = \sqrt{\frac{P}{R_3}} = \sqrt{\frac{P}{9}} = \frac{\sqrt{P}}{3} = \frac{2\sqrt{P}}{6}$$

Total current $I_{total} = I_1 + I_{23} = \frac{\sqrt{P}}{6} + \frac{2\sqrt{P}}{6} = \frac{3\sqrt{P}}{6} = \frac{\sqrt{P}}{2}$.

Power of Bulb 4: $P = I_{total}^2 R_4 = \left(\frac{\sqrt{P}}{2}\right)^2 R_4 = \frac{P}{4} \cdot R_4$.

Dividing both sides by P: $1 = \frac{R_4}{4} \Longrightarrow R_4 = 4\ \Omega$.

Match: (b) → (Q)

Finding Equivalent Resistance (d)

The circuit consists of R_4 in series with the parallel combination of R_1 and $(R_2 + R_3)$.
Resistance of the parallel part (R_p):

$$R_p = \frac{R_1 \cdot (R_2 + R_3)}{R_1 + (R_2 + R_3)}$$

$R_p = \frac{36 \cdot (9+9)}{36+9+9} = \frac{36 \cdot 18}{54} = \frac{36 \cdot 1}{3} = 12\ \Omega.$

Total Equivalent Resistance (R_{eq}) = $R_4 + R_p = 4 + 12 = 16\ \Omega.$

Match: (d) → (S)

Finding Voltage Output (c)

We are given the condition that the power of each bulb is 4 W.

Total power consumed by all 4 bulbs together = 4×4 W = 16 W.

Total power can also be calculated for the whole circuit as $P_{total} = \frac{V^2}{R_{eq}}$, where V is the battery's voltage output.

$$16 = \frac{V^2}{16}$$

$V^2 = 16 \times 16 = 256$

$V = 16$ V.

Match: (c) → (R)

27. **Answer :** 10

When a wire is stretched to 3 times its original length ($l' = 3l$), its volume remains constant.

The new resistance is proportional to the square of the length change factor:

$R_{new} = n^2 \cdot R_{old}$

$R_{new} = 3^2 \cdot 10$

$R_{new} = 9 \cdot 10 = 90\ \Omega$

The wire is then cut into 3 equal parts. The resistance of each part is:

$$R_{part} = \frac{90}{3} = 30\ \Omega$$

These 3 parts are connected in parallel. The equivalent resistance is:

$$R_{eq} = \frac{30}{3}$$

$R_{eq} = 10\ \Omega$

28. **Answer :** 21

Let the original length be l and the original resistance be R.

When the wire is stretched by 10%, the new length is:

$l' = l + 0.1l = 1.1l$

Since volume is constant during stretching, the new resistance R' is proportional to $(l')^2$:

$R' = R \cdot (1.1)^2$

$R' = 1.21 \cdot R$

The increase in resistance is:

$\Delta R = 1.21R - R = 0.21R$

The percentage increase is:

Percentage Increase = $0.21 \cdot 100$

Percentage Increase = 21%

29. **Answer :** 4

A wire of resistance 16 Ω is cut into n equal parts.

Because resistance is directly proportional to length, the resistance of each part is:

$R_{part} = \frac{16}{n}$

These n parts are connected in parallel. The formula for n equal resistors in parallel is:

$R_{eq} = \frac{R_{part}}{n}$

Substitute the given equivalent resistance ($R_{eq} = 1\ \Omega$) and the expression for R_{part}:

$1 = \frac{\frac{16}{n}}{n}$

$1 = \frac{16}{n^2}$

$n^2 = 16$

$n = 4$

Previous Year Questions

1. **Answer :** C) $5/4\ \Omega$

Step 1: Calculate the series equivalent of the resistors between Q, R, and S.

$R_{QRS} = 2 + 2 = 4\Omega$

Step 2: Calculate the parallel equivalent of R_{QRS} and the diagonal resistor connecting Q and S directly.

$R_{QS(eq)} = \frac{4 \times 2}{4 + 2} = \frac{8}{6} = \frac{4}{3}\Omega$

Step 3: Calculate the series equivalent of the branch starting from P to Q, and then Q to S.

$R_{PQS(eq)} = 2 + \frac{4}{3} = \frac{10}{3}\Omega$

Step 4: Calculate the final equivalent resistance by putting the upper branch $R_{PQS(eq)}$ in parallel with the bottom resistor connecting P and S.

$R_{eq} = \frac{\frac{10}{3} \times 2}{\frac{10}{3} + 2} = \frac{\frac{20}{3}}{\frac{16}{3}} = \frac{20}{16} = \frac{5}{4}\Omega$

2. **Answer :** C) (i), (ii) and (iv) only

The potential difference V across a cell with emf E and internal resistance r is $V = E - ir$.

Graph (i) plots V versus i, which is a straight line with a negative slope, matching the equation.

Thermal power in the external circuit is $P = i^2R = \frac{E^2R}{(R+r)^2}$.

Graph (ii) plots P versus R. This curve starts at zero, reaches a peak maximum when $R = r$ (Maximum Power Transfer), and then decreases, matching the graph exactly.

The ratio $K = \frac{\text{Total Power Generated}}{\text{Thermal Power in cell}} = \frac{E \cdot i}{i^2 \cdot r} = \frac{E}{i \cdot r}$.

Using $ir = E - V$, we get $K = \frac{E}{E-V}$.

As $V \to 0$, $K \to 1$. As $V \to E$, $K \to \infty$. This asymptotic behavior matches the standard graphs provided.

3. **Answer :** C) -1A

The three cells are connected in parallel. We calculate the equivalent emf and internal resistance.

$$E_{eq} = \frac{\frac{3}{1}+\frac{2}{1}+\frac{1}{1}}{\frac{1}{1}+\frac{1}{1}+\frac{1}{1}}$$

$$E_{eq} = \frac{3+2+1}{3}$$

$$E_{eq} = \frac{6}{3} = 2 \text{ V}$$

The potential difference V_{PQ} across the parallel combination is 2 V (with P positive relative to Q).

For branch 3, the current I_3 is determined by the potential difference driving it against its own emf:

$$I_3 = \frac{V_{PQ} - E_3}{r_3}$$

$$I_3 = \frac{2-1}{1}$$

$I_3 = 1$ A

This 1A current flows from the higher potential node P to the lower potential node Q.

The question asks for the current from Q to P, which is in the opposite direction.

$I_{Q\to P} = -1$ A

4. **Answer :** B) T1<T2

Step 1: Establish the mathematical relationship between the graph's slope and electrical resistance.

Slope $m = \frac{I}{V} = \frac{1}{R}$

Step 2: Compare the slopes visually provided in the graph to find the relationship between the resistances.

$m_1 > m_2 \Longrightarrow \frac{1}{R_1} > \frac{1}{R_2} \Longrightarrow R_1 < R_2$

Step 3: Relate the resistance to temperature. For metallic conductors, resistance is directly proportional to temperature.

$R \propto T \Longrightarrow T_1 < T_2$

5. **Answer :** 3.33 V

Step 1: Analyze the circuit diagram topology. The diagram shows five resistors connected between nodes X and Y. There is a top branch containing one resistor. There is a middle branch containing two resistors. A diagonal resistor connects the node after the top resistor to the node between the two middle resistors. Finally, there is a bottom branch containing one resistor. Assuming the node after the top resistor is connected directly to Y as drawn, the top resistor is in parallel with the whole structure. The diagonal resistor and the rightmost middle resistor r are in parallel. This parallel pair is in series with the leftmost middle resistor. Finally, the bottom resistor is in parallel with everything.

Step 2: Calculate the equivalent resistance R_{eq} assuming all five resistors have the same resistance R. The middle branch has a resistance of R_{mid}.

$$R_{mid} = R + \left(\frac{R \times R}{R+R}\right) = R + 0.5R = 1.5R$$

Step 3: The total equivalent resistance is the parallel combination of the top branch, middle branch, and bottom branch.

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{1.5R} + \frac{1}{R} = \frac{2}{R} + \frac{2}{3R} = \frac{8}{3R}$$

$$R_{eq} = \frac{3R}{8}$$

Step 4: Use the given equivalent resistance to find R.

$$\frac{3R}{8} = 2 \Longrightarrow R = \frac{16}{3}\Omega$$

Step 5: Calculate the potential difference across resistor r. The entire middle branch is connected across the 10 V source. The voltage divides between the series resistor R and the parallel pair that includes r.

$$V_r = 10 \times \left(\frac{0.5R}{R + 0.5R}\right) = 10 \times \left(\frac{0.5}{1.5}\right) = \frac{10}{3}\ \text{V} \approx 3.33\ \text{V}$$

6. **Answer :** C) 82.9%

Step 1: Identify the given parameters for the transmission line. The power delivered P_{out} is 20000 W, the voltage V is 220 V, and the line resistance R is 0.5 Ω.

Step 2: Calculate the current I flowing through the transmission line.

$$I = \frac{P_{out}}{V} = \frac{20000}{220} = \frac{1000}{11}\ \text{A}$$

Step 3: Calculate the power lost as heat in the line.

$$P_{loss} = I^2R = \left(\frac{1000}{11}\right)^2 \times 0.5 = \frac{1000000}{121} \times 0.5 \approx 4132.2\ \text{W}$$

Step 4: Calculate the total power input P_{in}.

$$P_{in} = P_{out} + P_{loss} = 20000 + 4132.2 = 24132.2\ \text{W}$$

Step 5: Calculate the efficiency of transmission.

$$\text{Efficiency} = \left(\frac{P_{out}}{P_{in}}\right) \times 100\% = \left(\frac{20000}{24132.2}\right) \times 100\% \approx 82.9\%$$

7. **Answer :** B) 7.5 A

Step 1: Determine the resistance of the individual segments of the circuit. The circular wire has a total resistance of 4 Ω, which is divided into 4 equal quarter-circle arcs of 1 Ω each. The two perpendicular diameters have a resistance of 2 Ω each, meaning their individual radii from the center O are 1 Ω each.

Step 2: Analyze the circuit symmetry and assign node potentials. The 5 V battery is connected between points P and R. Let V_R be 0 V and V_P be 5 V. Due to the symmetry of the circuit across the PR axis, nodes Q and S must be at the same potential, so $V_Q = V_S$.

Step 3: Apply Kirchhoff's Current Law (KCL) at node Q and node O. Let $V_Q = V_1$ and $V_O = V_2$.

At node Q:

$$\frac{V_1 - 5}{1} + \frac{V_1 - V_2}{1} + \frac{V_1 - 0}{1} = 0 \Longrightarrow 3V_1 - V_2 = 5$$

At node O:

$$\frac{V_2 - 5}{1} + \frac{V_2 - V_1}{1} + \frac{V_2 - 0}{1} + \frac{V_2 - V_1}{1} = 0 \Longrightarrow -2V_1 + 4V_2 = 5$$

Step 4: Solve the system of equations for the node voltages. From the first equation, $V_2 = 3V_1 - 5$. Substitute this into the second equation:

$$-2V_1 + 4(3V_1 - 5) = 5$$

$$-2V_1 + 12V_1 - 20 = 5 \Longrightarrow 10V_1 = 25 \Longrightarrow V_1 = 2.5\ \text{V}$$

Then, substitute back to find V_2:

$V_2 = 3(2.5) - 5 = 7.5 - 5 = 2.5$ V

Step 5: Calculate the total current supplied by the battery. The total current I enters at node P and splits into three parallel paths: PQ, PO, and PS.

$$I = \frac{5 - V_1}{1} + \frac{5 - V_2}{1} + \frac{5 - V_1}{1}$$

$I = (5 - 2.5) + (5 - 2.5) + (5 - 2.5) = 2.5 + 2.5 + 2.5 = 7.5$ A

8. **Answer :** D)

Step 1: Determine the total equivalent resistance of the circuit. The circuit consists of a resistor R in series with a parallel combination of R and r.

$$R_{total} = R + \left(\frac{R \cdot r}{R + r}\right) = \frac{R(R + r) + Rr}{R + r} = \frac{R^2 + 2Rr}{R + r}$$

Step 2: Calculate the total current flowing from the voltage source V.

$$I_{total} = \frac{V}{R_{total}} = \frac{V(R + r)}{R^2 + 2Rr}$$

Step 3: Determine the current I_r flowing specifically through the variable resistor r using the current divider rule.

$$I_r = I_{total} \times \left(\frac{R}{R + r}\right) = \left(\frac{V(R + r)}{R(R + 2r)}\right) \times \left(\frac{R}{R + r}\right) = \frac{V}{R + 2r}$$

Step 4: Express the rate of heat generation (power) in r.

$$P_r = I_r^2 \cdot r = \left(\frac{V}{R + 2r}\right)^2 \cdot r = \frac{V^2 \cdot r}{(R + 2r)^2}$$

Step 5: Substitute $r = fR$ into the power equation.

$$P_r = \frac{V^2(fR)}{(R + 2fR)^2} = \frac{V^2 fR}{\big(R(1 + 2f)\big)^2} = \frac{V^2 fR}{R^2(1 + 2f)^2} = \frac{V^2 f}{R(2f + 1)^2}$$

9. **Answer :** C) 9 : 1

Step 1: Define the initial power dissipation P_1. Let the initial resistance of the heater wire be R and the supply voltage be $V = 230$ V.

$$P_1 = \frac{V^2}{R}$$

Step 2: Determine the resistance of the wire pieces when cut. When the wire is cut into three equal pieces, the resistance of each individual piece becomes $\frac{R}{3}$.

Step 3: Calculate the equivalent resistance when the three pieces are connected in parallel.

$$\frac{1}{R_p} = \frac{1}{R/3} + \frac{1}{R/3} + \frac{1}{R/3} = \frac{3}{R} + \frac{3}{R} + \frac{3}{R} = \frac{9}{R}$$

$$R_p = \frac{R}{9}$$

Step 4: Calculate the new power dissipation P_2 and find the ratio.

$$P_2 = \frac{V^2}{R_p} = \frac{V^2}{R/9} = 9\left(\frac{V^2}{R}\right) = 9P_1$$

Ratio $P_2 : P_1 = 9P_1 : P_1 = 9 : 1$

10. **Answer :** C) 1.6 V

Step 1: Analyze the circuit structure to determine the voltage across each branch. The circuit features a 4 V battery connected across two parallel branches. Because they are in parallel with the ideal voltage source, the full 4 V is applied across the entire bottom branch, which consists of a 2 Ω and a 3 Ω resistor in series.

Step 2: Calculate the potential difference across the 2 Ω resistor using the voltage divider rule. The total resistance of the bottom branch is $2 + 3 = 5\Omega$. The voltage divides proportionally across the series resistors.

$$V_{2\Omega} = V_{total} \times \left(\frac{R_2}{R_2 + R_3}\right) = 4 \times \left(\frac{2}{2+3}\right) = 4 \times \left(\frac{2}{5}\right) = \frac{8}{5} = 1.6 \text{ V}$$

5. Magnetic Effects of Electric Current

Magnetic Fields, Forces, and Effects of Electric Current

LEVEL I

1. **Answer :** B) It repels a known magnet

The conclusive test for magnetism is repulsion.

When a piece of metal attracts a known magnet, it could simply be a magnetic material (like iron) rather than an actual magnet.

However, if it repels a known magnet, it must possess like poles, confirming it is a permanent magnet itself.

2. **Answer :** D) Magnetic moment

Magnetic pole strength is a scalar quantity.

Magnetic permeability is a scalar property of a medium.

Magnetic moment is a vector quantity that points from the South Pole to the North Pole of a magnet.

3. **Answer :** B) $2M/\pi$

Let the initial length of the wire be L and its pole strength be m.

$M = m \times L$

When the wire is bent into a semicircular arc of radius r, the length L forms the circumference of the semicircle.

$L = \pi \times r$

$r = \frac{L}{\pi}$

The new effective length (distance between the two poles) is the diameter of the semicircle.

$L' = 2 \times r$

$L' = \frac{2 \times L}{\pi}$

The new magnetic moment M' is the product of pole strength and the new effective length.

$M' = m \times L'$

$M' = m \times \frac{2 \times L}{\pi}$

$M' = \frac{2 \times (m \times L)}{\pi}$

$M' = \frac{2 \times M}{\pi}$

4. **Answer :** D) More for shorter magnet

The magnetic moment is given by the formula:

$M = m \times l$

Rearranging for pole strength m:

$m = \frac{M}{l}$

Since the magnetic moments M are equal for both magnets, the pole strength is inversely proportional to the magnetic length.

$m \propto \frac{1}{l}$

Therefore, the shorter magnet (smaller l) will have a higher pole strength m.

5. **Answer :** C) 1:2

Let the lengths of the two magnets be l_1 and l_2, and their pole strengths be m_1 and m_2.

$\frac{l_1}{l_2} = \frac{2}{3}$

$\frac{m_1}{m_2} = \frac{3}{4}$

The formula for magnetic moment is:

$M = m \times l$

The ratio of their magnetic moments is:

$\frac{M_1}{M_2} = \frac{m_1 \times l_1}{m_2 \times l_2}$

$\frac{M_1}{M_2} = \left(\frac{m_1}{m_2}\right) \times \left(\frac{l_1}{l_2}\right)$

$\frac{M_1}{M_2} = \left(\frac{3}{4}\right) \times \left(\frac{2}{3}\right)$

$\frac{M_1}{M_2} = \frac{6}{12}$

$\frac{M_1}{M_2} = \frac{1}{2}$

6. **Answer :** C) 25 A-m

The formula for the magnetic moment of a bar magnet is:

$M = m \times 2l$

Here, the magnetic moment $M = 5$ A-m^2 and the distance between poles $2l = 20$ cm.

$2l = 0.2$ m

Substituting the given values into the formula:

$5 = m \times 0.2$

$m = \frac{5}{0.2}$

$m = 25$ A-m

7. **Answer :** D) One-fourth

The Coulomb force of attraction or repulsion between two magnetic poles is inversely proportional to the square of the distance between them.

$F \propto \frac{1}{d^2}$

When the distance is doubled, the new distance is $2d$.

$F' \propto \frac{1}{(2d)^2}$

$F' \propto \frac{1}{4d^2}$

$F' = \frac{1}{4} \times \left(\frac{1}{d^2}\right)$

$F' = \frac{F}{4}$

8. **Answer :** C) Remains unchanged

The force between two magnetic poles is given by Coulomb's Law for magnetism:

$F = \frac{\mu_0}{4\pi} \times \frac{m_1 \times m_2}{d^2}$

Let the new pole strengths be ${m_1}' = 2m_1$ and ${m_2}' = 2m_2$, and the new distance be $d' = 2d$.

$F' = \frac{\mu_0}{4\pi} \times \frac{{m_1}' \times {m_2}'}{(d')^2}$

$F' = \frac{\mu_0}{4\pi} \times \frac{(2m_1) \times (2m_2)}{(2d)^2}$

$F' = \frac{\mu_0}{4\pi} \times \frac{4 \times m_1 \times m_2}{4 \times d^2}$

$F' = \frac{\mu_0}{4\pi} \times \frac{m_1 \times m_2}{d^2}$

$F' = F$

9. **Answer :** $2\ A - m$

The force between a unit north pole (m_1 = 1 A-m) and another magnetic pole ($m_2 = m$) is given by:

$F = \frac{\mu_0}{4\pi} \times \frac{m_1 \times m_2}{r^2}$

The value of $\frac{\mu_0}{4\pi}$ in SI units is 10^{-7} T-m/A.

$2 \times 10^{-7} = 10^{-7} \times \frac{1 \times m}{(1)^2}$

$2 = m$

$m = 2$ A-m

10. **Answer :** B) Are closed curves

Magnetic lines of force always form continuous, closed loops.

They emerge from the North Pole and enter the South Pole externally, and then travel from the South Pole to the North Pole internally.

11. **Answer :** B) Run continuously through the bar and outside the bar

In a bar magnet, magnetic lines of induction do not start or end at the poles like electric field lines.

They form closed continuous loops, traveling from the North Pole to the South Pole outside the magnet, and from the South Pole to the North Pole inside the magnet.

12. **Answer :** D) $\frac{1}{d^3}$

For a short bar magnet, the magnetic induction B at a point on the axial or equatorial line is inversely proportional to the cube of the distance d from the center of the magnet.

$B \propto \frac{1}{d^3}$

13. **Answer :** D) Both B and C

The magnetic force acting on a moving charge is given by the cross product:

$\vec{F}_m = q(\vec{V} \times \vec{B})$

By the definition of the cross product, the resulting force vector $\vec{F}_m$ is always perpendicular to the plane containing the velocity vector $\vec{V}$ and the magnetic field vector $\vec{B}$.

Thus, the angle between $\vec{F}_m$ and $\vec{V}$ is 90°, and the angle between $\vec{F}_m$ and $\vec{B}$ is 90°.

14. **Answer :** C) Helix

When a charged particle enters a uniform magnetic field at an oblique angle (other than 0°, 90°, or 180°), its velocity can be resolved into two components.

The component parallel to the magnetic field causes linear motion, while the perpendicular component causes circular motion.
The superposition of these two motions results in a helical path.

15. **Answer :** C) Both electric and magnetic fields
A stationary electric charge produces only an electric field.
However, an electric charge in motion constitutes an electric current, which generates a magnetic field around it in addition to its inherent electric field.

16. **Answer :** C) $\propto m$
When a charged particle enters perpendicularly into a uniform magnetic field, the magnetic force provides the necessary centripetal force for circular motion.
$q \times v \times B = \frac{m \times v^2}{r}$
Rearranging this formula to solve for the radius r:
$r = \frac{m \times v}{q \times B}$
For a constant velocity v, charge q, and magnetic field B, the radius r is directly proportional to the mass m of the particle.
$r \propto m$

17. **Answer :** C) Vertically upward
The direction of the magnetic force on a moving positive charge is given by Fleming's Left-Hand Rule or the right-hand cross-product rule.
The velocity vector $\vec{V}$ is directed towards the South.
The magnetic field vector $\vec{B}$ is directed towards the East.
Using $\vec{F} = q(\vec{V} \times \vec{B})$, the cross product of South and East points vertically upwards.

18. **Answer :** A) Straight line
The magnetic force on a moving charge is given by:
$F = q \times v \times B \times \sin(\theta)$
Since the proton's velocity is parallel to the magnetic field, the angle $\theta = 0°$.
$\sin(0°) = 0$
$F = 0$
Because no magnetic force acts on the proton, it will not experience any deflection and will continue to travel in a straight line.

19. **Answer :** B) Trajectory of proton is less curved
The radius of the circular path of a charged particle in a magnetic field is given by:
$r = \frac{\sqrt{2 \times m \times K}}{q \times B}$
Here, K is the kinetic energy, q is the charge, and B is the magnetic field.
Since both the electron and proton have the same kinetic energy K, the same magnitude of charge q, and are in the same magnetic field B, the radius r is directly proportional to the square root of the mass m.
$r \propto \sqrt{m}$
Because the mass of a proton is much greater than the mass of an electron, its radius of curvature will be much larger.
A larger radius of curvature means the trajectory is less curved.

20. **Answer :** C) Momentum of the particle
The formula for the radius of curvature r of a charged particle moving perpendicularly in a magnetic field is:

$r = \frac{m \times v}{q \times B}$

The product of mass m and velocity v is the momentum p of the particle.

$r = \frac{p}{q \times B}$

Thus, the radius of curvature is directly proportional to the momentum of the particle.

21. **Answer :** B) $2.4 \times 10^{-11} N$

Velocity is 2×10^8 m/s.

The magnetic force is given by the formula:

$F = q \times v \times B \times \sin(\theta)$

Given values:

$q = 1.6 \times 10^{-19}$ C

$B = 1.5$ T

$\theta = 30°$

Assuming $v = 2 \times 10^8$ m/s

$F = 1.6 \times 10^{-19} \times 2 \times 10^8 \times 1.5 \times \sin(30°)$

$\sin(30°) = 0.5$

$F = 1.6 \times 10^{-19} \times 2 \times 10^8 \times 1.5 \times 0.5$

$F = 1.6 \times 10^{-19} \times 1.5 \times 10^8$

$F = 2.4 \times 10^{-11}$ N

22. **Answer :** D) $8 \times 10^{-12} N$

First, convert the kinetic energy from MeV to Joules.

$K = 2$ MeV

$K = 2 \times 10^6$ eV

$K = 2 \times 10^6 \times 1.6 \times 10^{-19}$ J

$K = 3.2 \times 10^{-13}$ J

The formula for kinetic energy is:

$K = \frac{1}{2} \times m \times v^2$

$v = \sqrt{\frac{2 \times K}{m}}$

Using the mass of a proton $m = 1.67 \times 10^{-27}$ kg:

$v = \sqrt{\frac{2 \times 3.2 \times 10^{-13}}{1.67 \times 10^{-27}}}$

$v = \sqrt{3.83 \times 10^{14}}$

$v = 1.95 \times 10^7$ m/s

The magnetic force is given by:

$F = q \times v \times B \times \sin(90°)$

$F = 1.6 \times 10^{-19} \times 1.95 \times 10^7 \times 2.5 \times 1$

$F = 7.8 \times 10^{-12}$ N

Rounding to the nearest given option, we get 8×10^{-12} N.

23. **Answer :** $1.6 \times 10^{-13} \tilde{K} N$

The force is given by the Lorentz force formula:

$\vec{F} = q(\vec{V} \times \vec{B})$

Given vectors:

$\vec{V} = 5 \times 10^5 \hat{\imath}$ m/s
$\vec{B} = 3\hat{\imath} + 2\hat{\jmath}$ T
$q = 1.6 \times 10^{-19}$ C
Substitute the values into the cross product:
$\vec{F} = 1.6 \times 10^{-19} \times [(5 \times 10^5 \hat{\imath}) \times (3\hat{\imath} + 2\hat{\jmath})]$
Expand the cross product:
$\vec{F} = 1.6 \times 10^{-19} \times \left[\left(15 \times 10^5 \times (\hat{\imath} \times \hat{\imath})\right) + \left(10 \times 10^5 \times (\hat{\imath} \times \hat{\jmath})\right)\right]$
Since $\hat{\imath} \times \hat{\imath} = 0$ and $\hat{\imath} \times \hat{\jmath} = \hat{k}$:
$\vec{F} = 1.6 \times 10^{-19} \times \left[0 + 10 \times 10^5 \hat{k}\right]$
$\vec{F} = 1.6 \times 10^{-19} \times 10^6 \hat{k}$
$\vec{F} = 1.6 \times 10^{-13} \hat{k}$ N

24. **Answer :** 4 cm, 25 milli sec
Given values:
$q = 100\mu\text{C} = 100 \times 10^{-6}$ C $= 10^{-4}$ C
$m = 10$ mg $= 10 \times 10^{-6}$ kg $= 10^{-5}$ kg
$B = 25$ T
$v = 10$ m/s
The formula for the radius of the circular path is:
$$r = \frac{m \times v}{q \times B}$$
$$r = \frac{10^{-5} \times 10}{10^{-4} \times 25}$$
$$r = \frac{10^{-4}}{25 \times 10^{-4}}$$
$$r = \frac{1}{25} \text{ m}$$
$r = 0.04$ m
$r = 4$ cm
$$T = \frac{2\pi m}{qB}$$
Substitute the values into the period equation.
$$T = \frac{2\pi \times 10^{-5}}{10^{-4} \times 25}$$
$$T = \frac{2\pi \times 10^{-1}}{25}$$
$$T = \frac{0.2\pi}{25}$$
$T = 0.008\pi$ s
$T = 0.025$ s $= 25$ milli seconds

25. **Answer :** B) Oersted
Hans Christian Oersted experimentally discovered that an electric current passing through a wire produces a magnetic field, deflecting a nearby compass needle.

26. **Answer :** D) A stationary charge
A stationary charge only generates a constant electric field around it.
It does not generate a magnetic field, which requires the charge to be in motion.

27. **Answer :** C) $\frac{\pi}{2}$
The force on a current-carrying conductor in a uniform magnetic field is given by:

$F = I \times L \times B \times \sin(\theta)$
The force F is maximum when the term $\sin(\theta)$ is maximum.
The maximum value of $\sin(\theta)$ is 1, which occurs when $\theta = 90°$.
$\theta = \frac{\pi}{2}$ radians

28. **Answer :** A) Zero
The force on a current-carrying conductor is:
$F = I \times L \times B \times \sin(\theta)$
If the conductor is placed parallel to the magnetic field, the angle $\theta = 0°$.
$\sin(0°) = 0$
$F = I \times L \times B \times 0$
$F = 0$

29. **Answer :** C) Vertically upwards
Apply Fleming's Left-Hand Rule or the right-hand vector cross product rule.
The magnetic field ($\vec{B}$) is pointing towards the North.
The current direction ($\vec{L}$) is pointing towards the East.
Using $\vec{F} = I(\vec{L} \times \vec{B})$, the cross product of East and North points vertically upwards.

30. **Answer :** C) It will move inwards to plane of paper
According to the provided diagram, the magnetic field $\vec{B}$ points from left to right across the page.
The current I in wire W flows vertically upwards.
Using Fleming's Left-Hand Rule: Point your index finger to the right (magnetic field) and your middle finger upwards (current). Your thumb will point directly into the plane of the paper.
Thus, the force acting on the wire is directed inwards.

31. **Answer :** C) Away from you, into the page
The magnetic field $\vec{B}$ flows from the North pole to the South pole.
Since the North pole is to your left and the South pole is to your right, the magnetic field points from left to right.
The current flows vertically upwards.
Using Fleming's Left-Hand Rule, with the index finger pointing right (field) and the middle finger pointing up (current), the thumb points into the page (away from you).

32. **Answer :** A) 30°
The formula for the magnetic force on a current-carrying wire is:
$F = I \times L \times B \times \sin(\theta)$
Given values:
$F = 15$ N
$I = 10$ A
$L = 1.5$ m
$B = 2$ T
Substitute the values into the formula:
$15 = 10 \times 1.5 \times 2 \times \sin(\theta)$
$15 = 30 \times \sin(\theta)$
$\sin(\theta) = \frac{15}{30}$
$\sin(\theta) = 0.5$
$\theta = \arcsin(0.5)$

$\theta = 30°$

33. **Answer :** D) Zero

The torque τ acting on a current-carrying loop in a magnetic field is given by:

$\tau = M \times B \times \sin(\theta)$

Where M is the magnetic moment and θ is the angle between the area vector of the loop and the magnetic field.

Because the plane of the loop is perpendicular to the magnetic field, its area vector is parallel to the magnetic field.

Therefore, $\theta = 0°$.

$\tau = M \times B \times \sin(0°)$

$\tau = 0$

34. **Answer :** C) 1

The magnetic force on a straight current-carrying wire is $F = I \times L \times B \times \sin(\theta)$.

For wire ab, it is placed perpendicularly to the magnetic lines of force. Its length is l and the angle is 90°.

$F_{ab} = I \times l \times B \times \sin(90°)$

$F_{ab} = I \times l \times B$

For wire bc, we can view its length in terms of a right-angled triangle abc where $\angle abc = 45°$.

The hypotenuse bc has a length of $\frac{l}{\cos(45°)} = l\sqrt{2}$.

The angle between wire bc and the magnetic field B is 45°.

$F_{bc} = I \times (bc) \times B \times \sin(45°)$

$F_{bc} = I \times \left(l\sqrt{2}\right) \times B \times \left(\frac{1}{\sqrt{2}}\right)$

$F_{bc} = I \times l \times B$

The ratio of the force on bc to the force on ab is:

$\frac{F_{bc}}{F_{ab}} = \frac{I \times l \times B}{I \times l \times B}$

$\frac{F_{bc}}{F_{ab}} = 1$

LEVEL II

1. **Answer :** B) 1.25 A-m²

The formula for the magnetic moment of a bar magnet is given by the product of its pole strength and its magnetic length:

$M = m \times 2l$

Given the pole strength is 25 A-m.

The length of the magnet is given as 5 cm. We must convert this distance to standard SI units (meters).

$2l = 5$ cm $= 0.05$ m

Substitute these values into the magnetic moment formula:

$M = 25 \times 0.05$

$M = 1.25$

The magnetic moment is 1.25 A-m².

2. **Answer :** B) or C) 4 A-m and 16 A-m

Let the pole strength of the weaker pole be m.

The strength of the stronger pole is $4m$.

The distance between them is 10 cm, which is 0.1 m.
The force of repulsion between two magnetic poles is given by Coulomb's Law for magnetism:
$F = \frac{\mu_0}{4\pi} \times \frac{m_1 \times m_2}{d^2}$
The force is given as 64 mg.wt. We must convert this into Newtons.
Mass in kg is 64×10^{-6} kg.
$F = 64 \times 10^{-6} \times 10$
$F = 6.4 \times 10^{-4}$ N
Substitute the values into Coulomb's law equation, using 10^{-7} for the magnetic constant:
$6.4 \times 10^{-4} = 10^{-7} \times \frac{m \times 4m}{(0.1)^2}$
$6.4 \times 10^{-4} = 10^{-7} \times \frac{4m^2}{0.01}$
$6.4 \times 10^{-4} = 10^{-7} \times 400 \times m^2$
$6.4 \times 10^{-4} = 4 \times 10^{-5} \times m^2$
$m^2 = \frac{6.4 \times 10^{-4}}{4 \times 10^{-5}}$
$m^2 = 16$
$m = 4$
The weaker pole is 4 A-m.
The stronger pole is $4 \times 4 = 16$ A-m.

3. **Answer :** B) F
Let the two initial poles be placed at vertices A and B of an equilateral triangle.
Since they attract each other, one must be a North pole (+m) and the other a South pole (-m).
The magnitude of the force F between them at distance a is:
$F = \frac{\mu_0}{4\pi} \times \frac{m^2}{a^2}$
A third North pole (+m) is placed at vertex C.
The force exerted on C by the North pole at A is a repulsive force of magnitude F directed outward along the line AC.
The force exerted on C by the South pole at B is an attractive force of magnitude F directed inward along the line CB.
The angle between these two force vectors is 120°.
The resultant force is calculated using the vector addition formula:
$F_{res} = \sqrt{F_1^2 + F_2^2 + 2 \times F_1 \times F_2 \times \cos(\theta)}$
$F_{res} = \sqrt{F^2 + F^2 + 2 \times F \times F \times \cos(120°)}$
$\cos(120°) = -0.5$
$F_{res} = \sqrt{2F^2 - F^2}$
$F_{res} = \sqrt{F^2}$
$F_{res} = F$

4. **Answer :** C) 9:1
The force between two magnetic poles is inversely proportional to the square of the distance between them.

$F \propto \frac{1}{d^2}$

Let the initial force be F_1 at a distance d_1 of 10 cm.

Let the final force be F_2 at a new distance d_2 of 30 cm.

The ratio of the forces can be written as:

$\frac{F_1}{F_2} = \left(\frac{d_2}{d_1}\right)^2$

Substitute the distance values:

$\frac{F_1}{F_2} = \left(\frac{30}{10}\right)^2$

$\frac{F_1}{F_2} = (3)^2$

$\frac{F_1}{F_2} = \frac{9}{1}$

The ratio of the forces is 9:1.

5. **Answer :** C) $\frac{\mu_0}{4\pi}\frac{\sqrt{3}m^2}{a^2}$

Since all three poles are isolated North poles of strength m, they will all repel each other.

The North pole at the third vertex experiences two repulsive forces from the other two North poles at the base vertices.

The magnitude of each individual repulsive force is:

$F = \frac{\mu_0}{4\pi} \times \frac{m^2}{a^2}$

Because the triangle is equilateral, the angle between these two outward repulsive force vectors is 60°.

The resultant force is calculated using vector addition:

$F_{res} = \sqrt{F^2 + F^2 + 2 \times F \times F \times \cos(60°)}$

$\cos(60°) = 0.5$

$F_{res} = \sqrt{2F^2 + F^2}$

$F_{res} = \sqrt{3F^2}$

$F_{res} = F\sqrt{3}$

Substitute the expression for F back into the result:

$F_{res} = \frac{\mu_0}{4\pi} \times \frac{\sqrt{3}m^2}{a^2}$

6. **Answer :** A) The lines of force may intersect

This statement is wrong. Magnetic lines of force never intersect each other. If they were to intersect, it would mean there are two different directions of the magnetic field at a single point, which is physically impossible.

7. **Answer :** B) $\frac{r}{2}$

The radius of a charged particle moving perpendicular to a magnetic field is given by:

$r = \frac{m \times v}{q \times B}$

For a proton, the mass is m_p and the charge is e. Its radius r_p is:

$r_p = \frac{m_p \times v}{e \times B}$

For a doubly ionized helium ion (alpha particle), the mass is $4m_p$ and the charge is $2e$. Its radius r is:

$r = \frac{4m_p \times v}{2e \times B}$

$r = 2 \times \left(\frac{m_p \times v}{e \times B}\right)$

Substitute the proton's radius expression into this equation:

$r = 2 \times r_p$

Solving for the proton's radius r_p:

$r_p = \frac{r}{2}$

8. **Answer :** C) 1.6×10^{-13}

the speed is be

The magnetic force vector is calculated using the cross product of velocity and magnetic field:

$\vec{F} = q(\vec{v} \times \vec{B})$

The velocity vector is along the positive x-direction:

$\vec{v}\ 2 \times 10^5\ \text{m/s} = (2 \times 10^5)\hat{\imath}$

The magnetic field vector is:

$\vec{B} = \hat{\imath} + 4\hat{\jmath} - 3\hat{k}$

Calculate the cross product $\vec{v} \times \vec{B}$:

$\vec{v} \times \vec{B} = (2 \times 10^5)\hat{\imath} \times (\hat{\imath} + 4\hat{\jmath} - 3\hat{k})$

Since $\hat{\imath} \times \hat{\imath} = 0$, $\hat{\imath} \times \hat{\jmath} = \hat{k}$, and $\hat{\imath} \times \hat{k} = -\hat{\jmath}$:

$\vec{v} \times \vec{B} = 2 \times 10^5 \times (4\hat{k} + 3\hat{\jmath})$

Find the magnitude of this cross product vector:

$|\vec{v} \times \vec{B}| = 2 \times 10^5 \times \sqrt{3^2 + 4^2}$

$|\vec{v} \times \vec{B}| = 2 \times 10^5 \times \sqrt{25}$

$|\vec{v} \times \vec{B}| = 2 \times 10^5 \times 5$

$|\vec{v} \times \vec{B}| = 10^6$

Now, multiply by the magnitude of the electron's charge (1.6×10^{-19} C) to find the force:

$F = 1.6 \times 10^{-19} \times 10^6$

$F = 1.6 \times 10^{-13}$ N

If the speed is 2×10^3 m/s

$F = 1.6 \times 10^{-15}$ N

9. **Answer :** B) There are lines of force and they are circular lines encircling the wire

According to Oersted's discovery and the Right-Hand Grip Rule, a straight current-carrying wire generates a magnetic field consisting of concentric circular lines of force that completely encircle the wire in a plane perpendicular to it.

10. **Answer :** A, B, C, D

All options given are correct statements regarding the forces acting on a charged particle: A moving charged particle entering a magnetic field from the outside may only complete a partial arc and exit, rather than a full circle. If it traces a helical path, the central axis of that helix aligns parallel to the uniform magnetic field lines. Because magnetic force is always perpendicular to the instantaneous velocity, no work is done, meaning the power associated with the magnetic force is always zero. If both uniform electric and magnetic fields exist, the electric field will

constantly do work on the particle, altering its speed and preventing it from maintaining a perfect circular path.

11. **Answer :** A, D

When a charged particle is fired at an oblique angle θ (where $0° < \theta < 90°$) to a uniform magnetic field, its velocity has a constant parallel component ($v\cos\theta$) and a rotating perpendicular component ($v\sin\theta$). Because the perpendicular component of velocity never becomes zero, the total velocity vector is never purely parallel to the x-axis. Thus, statement A is correct. Because the parallel component of velocity ($v\cos\theta$) remains constant and non-zero, the particle never moves purely perpendicular to the x-axis. Thus, statement D is correct.

12. **Answer :** A) The magnetic field may be perpendicular to the direction of the velocity

To keep the particle moving perfectly in a horizontal direction, the downward force of gravity must be exactly canceled by an upward magnetic force. For a magnetic force to be generated in the vertical direction while the particle moves horizontally, the magnetic field must be applied horizontally and perpendicular to the velocity vector. (If the field were parallel to the velocity, the magnetic force would be zero, making option B incorrect. Calculations show option C's value is incorrect.)

13. **Answer :** A) $3.2 \times 10^{-15} N$

The magnetic force acting on a moving charge is:

$F = q \times v \times B \times \sin(\theta)$

For a doubly ionized helium ion, the charge is $2e$.

$q = 2 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-19}$ C

The velocity is 10 km/s, which must be converted to m/s.

$v = 10 \times 10^3 = 10^4$ m/s

The magnetic field B is 1.0 T, and the angle is 90° so $\sin(90°) = 1$.

$F = 3.2 \times 10^{-19} \times 10^4 \times 1.0 \times 1$

$F = 3.2 \times 10^{-15}$ N

14. **Answer :** B) 2.1×10^{-4} m

The formula for the radius of a charged particle in a magnetic field is:

$$r = \frac{m \times v}{q \times B}$$

The mass of a helium ion is roughly 4 atomic mass units (amu).

$m = 4 \times 1.67 \times 10^{-27} = 6.68 \times 10^{-27}$ kg

Substitute the values into the formula:

$$r = \frac{6.68 \times 10^{-27} \times 10^4}{3.2 \times 10^{-19} \times 1.0}$$

$$r = \frac{6.68 \times 10^{-23}}{3.2 \times 10^{-19}}$$

$r \approx 2.08 \times 10^{-4}$ m

Rounding to match the closest option format gives 2.1×10^{-4}.

15. **Answer :** D) 1.32×10^{-7}s

The time period for one complete circular revolution is given by:

$$T = \frac{2\pi \times m}{q \times B}$$

Substitute the known values:

$$T = \frac{2 \times 3.1415 \times 6.68 \times 10^{-27}}{3.2 \times 10^{-19} \times 1.0}$$

$$T = \frac{41.97 \times 10^{-27}}{3.2 \times 10^{-19}}$$

$T \approx 13.1 \times 10^{-8}$ s

$T = 1.31 \times 10^{-7}$ s

16. Answer : A) $\frac{mv \sin\theta}{Bq}$

When a particle is projected at an angle θ, only the perpendicular component of its velocity ($v \sin\theta$) contributes to the circular motion that forms the radius of the helix. The standard radius equation $r = \frac{mv}{qB}$ is modified using this perpendicular velocity component:

$r = \frac{m \times (v \sin\theta)}{B \times q}$

17. Answer : C) $\frac{2\pi m}{Bq}$

The time required for one revolution (the period) depends only on the mass, charge, and magnetic field strength. It is completely independent of the velocity and projection angle.

$T = \frac{2\pi \times m}{B \times q}$

18. Answer : B) $\frac{2\pi mv \cos\theta}{Bq}$

The pitch is the linear distance traveled along the magnetic field axis during one complete revolution.

It is calculated by multiplying the parallel component of velocity ($v \cos\theta$) by the time period T.

$Pitch = (v \cos\theta) \times T$

Substitute the formula for the time period:

$Pitch = (v \cos\theta) \times \left(\frac{2\pi \times m}{B \times q}\right)$

$Pitch = \frac{2\pi \times m \times v \cos\theta}{B \times q}$

19. Answer : a - P, R, S; b - P, Q; c - S; d - P

For a): If there is only a magnetic field, a charged particle can travel in a straight line (if parallel), a circle (if perpendicular), or a helix (if at an oblique angle). For b): If there is only an electric field, the particle will travel in a straight line (if fired parallel to the field) or a parabola (if fired at an angle). For c): If both fields are parallel and velocity is perpendicular to both, the magnetic field causes circular motion while the electric field continuously accelerates the particle along the axis, creating a helical path with increasing (non-uniform) pitch. For d): In crossed electric and magnetic fields, a particle can travel in a perfectly straight line without deflection if its velocity allows the electric and magnetic forces to perfectly balance each other.

20. Answer : a - Q, S; b - R; c - Q; d - P, R

For a): Radius $r = \frac{mv}{qB}$. It is directly proportional to the perpendicular component of velocity. Because velocity is proportional to the square root of kinetic energy, radius is also directly proportional to the square root of K.E. For b): Time period $T = \frac{2\pi m}{qB}$. It is independent of the particle's velocity. For c): Centripetal force $F = qvB$. It is directly proportional to the component of velocity perpendicular to the magnetic field. For d): Angular frequency $\omega = \frac{qB}{m}$. It is directly proportional to the specific charge (q/m) and is completely independent of velocity.

21. **Answer :** 2

To move horizontally, the net vertical force acting on the particle must be zero.
Three vertical forces are in play: gravity (downward), electric force (upward), and magnetic force.
First, calculate the downward gravitational force:
$F_g = m \times g$
$F_g = (2 \times 10^{-5}) \times 9.8$
$F_g = 19.6 \times 10^{-5}$ N
Next, calculate the upward electric force:
$F_e = q \times E$
$F_e = (10^{-6}) \times 200$
$F_e = 2 \times 10^{-4}$ N
$F_e = 20 \times 10^{-5}$ N
Since the upward electric force (20×10^{-5} N) is slightly larger than the downward gravitational force (19.6×10^{-5} N), the magnetic force must push downward to perfectly balance the forces.
The required magnetic force is the difference between these two forces:
$F_m = F_e - F_g$
$F_m = (20 \times 10^{-5}) - (19.6 \times 10^{-5})$
$F_m = 0.4 \times 10^{-5}$ N
The formula for the magnetic force on a moving charge is:
$F_m = q \times v \times B$
Substitute the known values into the equation to find the velocity:
$0.4 \times 10^{-5} = (10^{-6}) \times v \times (2.0)$
$0.4 \times 10^{-5} = 2 \times 10^{-6} \times v$
Divide both sides by 2×10^{-6}:
$$v = \frac{0.4 \times 10^{-5}}{2 \times 10^{-6}}$$
$$v = \frac{4 \times 10^{-6}}{2 \times 10^{-6}}$$
$v = 2$

Torque on Dipole, Faraday's Law, and Applications

LEVEL I

1. **Answer :** D) All the above

When a bar magnet is freely suspended in a uniform magnetic field, the net translatory force on it is zero.
This is because the North pole and the South pole experience equal and opposite magnetic forces.
However, these parallel forces act along different lines of action, forming a couple that creates a torque, causing only rotatory motion.
The torque is defined by the cross product:
$\vec{\tau} = \vec{M} \times \vec{B}$

Because the torque vector is perpendicular to the plane containing the magnetic moment and the magnetic field, its direction aligns vertically along the suspension wire.
The magnitude of this torque is given by:
$\tau = M \times B \times \sin(\theta)$
This torque reaches its maximum value when the magnet is normal (perpendicular) to the magnetic field, where $\theta = 90°$.
Since all the provided statements accurately describe the physical behavior, all options are correct.

2. **Answer :** A) 0°
The torque acting on a freely suspended magnet in a uniform magnetic field is given by the formula:
$\tau = M \times B \times \sin(\theta)$
To find the rate of change of torque with respect to the deflection angle θ, we take the derivative of the torque function:
$\frac{d\tau}{d\theta} = \frac{d}{d\theta}(M \times B \times \sin(\theta))$
$\frac{d\tau}{d\theta} = M \times B \times \cos(\theta)$
This rate of change is maximum when the cosine function reaches its maximum possible value.
The maximum value of $\cos(\theta)$ is 1, which occurs when $\theta = 0°$.

3. **Answer :** $320\ Nm$
Length of the magnet $L = 10$ cm $= 0.1$ m
Pole strength $m = 20\ \text{A} \cdot \text{m}$
Deflection angle $\theta = 30°$
Earth's horizontal magnetic field component $B = 320\ \text{Wb/m}^2$
Calculate the magnetic dipole moment M of the bar magnet. The magnetic moment is defined as the product of its pole strength and its magnetic length.
$M = m \times L$
$M = 20 \times 0.1 = 2\ \text{A} \cdot \text{m}^2$
Set up the formula for the moment of the deflecting couple (torque τ) acting on a magnetic dipole in a uniform magnetic field.
$\tau = MB \sin(\theta)$
Substitute the calculated magnetic moment and the given values into the torque equation to find the final magnitude.
$\tau = 2 \times 320 \times \sin(30°)$
$\tau = 640 \times 0.5$
$\tau = 320\ \text{N} \cdot \text{m}$

4. **Answer :** B) $7.2 \times 10^{-4} Nm$
The formula for the torque acting on a magnetized needle in a uniform magnetic field is:
$\tau = M \times B \times \sin(\theta)$
The given values are $M = 4.8 \times 10^{-2}$ J/T, $B = 3 \times 10^{-2}$ T, and the angle $\theta = 30°$.
Substitute these values into the formula:
$\tau = (4.8 \times 10^{-2}) \times (3 \times 10^{-2}) \times \sin(30°)$
Since $\sin(30°) = 0.5$:
$\tau = 14.4 \times 10^{-4} \times 0.5$

$\tau = 7.2 \times 10^{-4}$ N-m

5. **Answer :** $150\hat{K}N - m$

The magnetic moment vector is aligned along the x-axis:

$\vec{M} = 50\hat{\imath}$ A-m^2

The magnetic field vector is given as:

$\vec{B} = 0.5\hat{\imath} + 3.0\hat{\jmath}$ T

The torque vector is the cross product of the magnetic moment and the magnetic field:

$\vec{\tau} = \vec{M} \times \vec{B}$

$\vec{\tau} = (50\hat{\imath}) \times (0.5\hat{\imath} + 3.0\hat{\jmath})$

Distribute the cross product:

$\vec{\tau} = (50 \times 0.5)(\hat{\imath} \times \hat{\imath}) + (50 \times 3.0)(\hat{\imath} \times \hat{\jmath})$

Since the cross product of parallel vectors is zero ($\hat{\imath} \times \hat{\imath} = 0$) and $\hat{\imath} \times \hat{\jmath} = \hat{k}$:

$\vec{\tau} = 0 + 150\hat{k}$

$\vec{\tau} = 150\hat{k}$ N-m

6. **Answer :** A) 60°

The torque acting on a bar magnet in a uniform magnetic field is:

$\tau = M \times B \times \sin(\theta)$

The magnet experiences its maximum torque when it is perpendicular to the field ($\theta_1 = 90°$).

$\tau_{max} = M \times B \times \sin(90°)$

$\tau_{max} = M \times B$

We need to find the new angle θ_2 where the torque is exactly half of the maximum torque.

$\tau_{new} = \frac{1}{2} \times \tau_{max}$

Substitute the expressions for the torques:

$M \times B \times \sin(\theta_2) = \frac{1}{2} \times (M \times B)$

Divide both sides by $M \times B$:

$\sin(\theta_2) = 0.5$

The angle whose sine is 0.5 is 30°. Therefore, the new position is at 30° to the magnetic field.

The question asks for the angle through which it must be rotated from the maximum torque position (90°).

Rotation Angle = $90° - 30°$

Rotation Angle = 60°

7. **Answer :** B) $3 \times 10^{-3}N - m$

First, convert the dimensions of the rectangular coil into area in standard SI units (square meters).

Area $A = 10$ cm $\times$ 15 cm

$A = 150$ cm^2

$A = 150 \times 10^{-4}$ m^2

Calculate the magnetic moment M of the coil:

$M = N \times I \times A$

Substitute the given values: number of turns $N = 100$, current $I = 2$ A.

$M = 100 \times 2 \times (150 \times 10^{-4})$

$M = 200 \times 150 \times 10^{-4}$

$M = 30000 \times 10^{-4}$

$M = 3$ A-m^2

The torque on the coil is calculated using the angle θ between the normal to the coil and the magnetic field, which is given as 30°.

$\tau = M \times B \times \sin(\theta)$

Substitute the values, including $B = 2 \times 10^{-3}$ Wb/m^2:

$\tau = 3 \times (2 \times 10^{-3}) \times \sin(30°)$

Since $\sin(30°) = 0.5$:

$\tau = 6 \times 10^{-3} \times 0.5$

$\tau = 3 \times 10^{-3}$ N-m

8. **Answer :** D) 0.66 J and 0 N-m

The initial position of the magnet is along the direction of the uniform magnetic field, meaning the initial angle $\theta_1 = 0°$.

The final position is opposite to the field direction, meaning the final angle $\theta_2 = 180°$.

The work done in rotating a magnet in a magnetic field is:

$W = -M \times B \times (\cos(\theta_2) - \cos(\theta_1))$

Substitute the given values $M = 1.5$ J/T and $B = 0.22$ T:

$W = -1.5 \times 0.22 \times (\cos(180°) - \cos(0°))$

Since $\cos(180°) = -1$ and $\cos(0°) = 1$:

$W = -0.33 \times (-1 - 1)$

$W = -0.33 \times (-2)$

$W = 0.66$ J

The torque required to keep the magnet in this final position ($\theta = 180°$) is:

$\tau = M \times B \times \sin(180°)$

Since $\sin(180°) = 0$:

$\tau = 0$ N-m

9. **Answer :** B) -5J

The potential energy of a bar magnet in a uniform magnetic field is given by the formula:

$U = -M \times B \times \cos(\theta)$

Substitute the provided values: $M = 5$ A-m^2, $B = 2$ T, and the angle $\theta = 60°$.

$U = -5 \times 2 \times \cos(60°)$

Since the cosine of 60° is 0.5:

$U = -10 \times 0.5$

$U = -5$ J

10. **Answer :** C) Weber

The standard SI unit of magnetic flux is the Weber.

11. **Answer :** B) 1 weber

According to the provided diagram description, the angle θ between the normal to the area ($\vec{n}$) and the magnetic field ($\vec{B}$) is 60°.

The magnetic flux Φ passing through the coil is calculated by the dot product of the magnetic field and the area vector.

$\Phi = B \times A \times \cos(\theta)$

Substitute the given values: $A = 0.5$ m^2 and $B = 4.0$ Wb/m^2.

$\Phi = 4.0 \times 0.5 \times \cos(60°)$

Since $\cos(60°) = 0.5$:

$\Phi = 2.0 \times 0.5$

Φ = 1 weber

12. **Answer :** C) Magnetic shell
A circular coil carrying an electric current generates a magnetic field that mimics a very short bar magnet.
This configuration is physically and mathematically modeled as a magnetic dipole or a magnetic shell.

13. **Answer :** B) North Pole
According to the Right-Hand Grip Rule (or the clock face rule), if you look at the face of a circular coil and the current appears to flow in an anti-clockwise direction, that face behaves as a North Pole.

14. **Answer :** C) B being non-uniform in space
According to Faraday's Law of Induction, an electromotive force (emf) is induced only when there is a change in magnetic flux over time.
Changes in the magnetic field magnitude with time (A), the radius of the loop with time (B), or the angle of the loop with time (D) all result in a change in magnetic flux.
However, if a stationary loop is placed in a magnetic field that is non-uniform in space but constant in time, the magnetic flux passing through the loop does not change, meaning no emf is induced.

15. **Answer :** D) Is useful in deciding about the direction of an induced emf
Lenz's Law states that the direction of an induced electromotive force (and consequently the induced current) will always be such that it opposes the change in magnetic flux that produced it.
Therefore, its primary utility is determining the direction of the induced emf.

16. **Answer :** A) A force acting on the strip to oppose the motion
When a non-magnetic metallic strip is moved out from between the poles of a magnet, the magnetic flux passing through the strip changes over time.
This changing flux induces eddy currents within the body of the metallic strip.
According to Lenz's Law, these eddy currents will interact with the magnetic field to produce a magnetic force that opposes the motion that created them, thus making it harder to pull the strip away.

17. **Answer :** D) The current will be in clockwise as the electron passes by the loop
As the electron moves along the line AB, it constitutes a current flowing in the opposite direction (from B to A).
This effective current generates a magnetic field. Assuming the loop is positioned above the wire, the right-hand rule dictates that the magnetic field points outwards through the loop.
As the electron approaches, the outward magnetic flux through the loop increases. By Lenz's law, the loop induces a current to oppose this increase by generating an inward magnetic field, resulting in a clockwise current.
As the electron moves away, the outward flux decreases. The loop opposes this decrease by generating an outward magnetic field, resulting in an anti-clockwise current.

18. **Answer :** D) 5
First, calculate the area of one square loop.
$A = 10 \text{ cm} \times 10 \text{ cm} = 0.1 \text{ m} \times 0.1 \text{ m}$
$A = 0.01 \text{ m}^2$

The induced electromotive force (emf) according to Faraday's law for a coil of N turns is:

$E = N \times A \times \frac{dB}{dt}$

Substitute the given values: $N = 500$, $A = 0.01\ \text{m}^2$, and the rate of change of magnetic field $\frac{dB}{dt} = 1\ \text{T/s}$.

$E = 500 \times 0.01 \times 1$

$E = 5\ \text{V}$

19. **Answer :** A) 0.1s

Area is 100 cm^2 (0.01 m^2)

The average induced electromotive force (emf) is calculated using Faraday's Law:

$E = \frac{N \times A \times B}{t}$

Substitute the assumed area and the given values: $E = 0.1\ \text{V}$, $N = 50$, and $B = 2 \times 10^{-2}\ \text{T}$.

$0.1 = \frac{50 \times 0.01 \times (2 \times 10^{-2})}{t}$

Simplify the numerator:

$0.1 = \frac{0.5 \times (2 \times 10^{-2})}{t}$

$0.1 = \frac{1 \times 10^{-2}}{t}$

$0.1 = \frac{0.01}{t}$

Solve for time t:

$t = \frac{0.01}{0.1}$

$t = 0.1\ \text{s}$

20. **Answer :** BLu

Note: The standard formula for motional emf is $E = B \times L \times u$. Option B in the source text is printed as just "BL", omitting the velocity term "u", but it represents the correct intended formula among the choices.

When a conducting rod of length L falls with a velocity u perpendicular to a uniform horizontal magnetic field B, it sweeps through the magnetic flux.

The potential difference (motional emf) generated between its two ends is derived directly from the Lorentz force on the charges within the conductor.

The expression for this induced motional emf is the product of the magnetic field strength, the length of the rod, and its velocity.

$E = B \times u \times L$

21. **Answer :** A) Zero

An insulating rod does not possess free charge carriers (electrons) that are free to move throughout the material.

Even though the Lorentz force still acts on the bound charges within the moving atoms, no macroscopic separation of charge can occur at the ends of the rod.

Without the accumulation of free charges at the ends, no potential difference (emf) can be established across the rod.

22. **Answer :** B) 16 V

The magnitude of the induced electromotive force (emf) is equal to the rate of change of the magnetic flux with respect to time.

$E = \frac{d\phi}{dt}$

Given the time-dependent magnetic flux equation $\phi = 3t^2 + 4t + 9$.

Take the derivative of the flux with respect to time t:

$\frac{d\phi}{dt} = \frac{d}{dt}(3t^2) + \frac{d}{dt}(4t) + \frac{d}{dt}(9)$

$E = 6t + 4$

To find the induced emf exactly at $t = 2$ s, substitute the value of time into the derivative.

$E = 6(2) + 4$

$E = 12 + 4$

$E = 16$ V

23. **Answer :** C) 0.02 Wb

First, calculate the cross-sectional area of the circular disc.

$A = \pi \times r^2$

Given the radius $r = 0.2$ m.

$A = \pi \times (0.2)^2$

$A = 0.04\pi$ m^2

The magnetic flux Φ is the dot product of the magnetic field vector and the area vector. The angle between the axis (normal) and the magnetic field is $\theta = 60°$.

$\Phi = B \times A \times \cos(\theta)$

Substitute the given values, including $B = \frac{1}{\pi}$ Wb/m^2.

$\Phi = \left(\frac{1}{\pi}\right) \times (0.04\pi) \times \cos(60°)$

The π terms cancel out, and $\cos(60°) = 0.5$.

$\Phi = 0.04 \times 0.5$

$\Phi = 0.02$ Wb

24. **Answer :** B) Magnetic flux through the loop is non-zero and induced emf is zero

Because the conducting square loop is placed inside a uniform magnetic field, a definite amount of magnetic field lines pass through its area.

Therefore, the magnetic flux $\Phi = B \times A$ through the loop is constant and non-zero.

As the loop translates within its own plane at a constant velocity, it remains entirely within the uniform field, meaning the area exposed to the field does not change.

Since the flux is constant, the rate of change of magnetic flux ($\frac{d\Phi}{dt}$) is zero.

According to Faraday's law, an induced emf only occurs when flux changes over time. Thus, the induced emf is zero.

25. **Answer :** B) 2.5 A

To keep the movable conductor moving at a constant speed, the net force acting on it must be exactly zero.

This means the external mechanical force applied to the left must perfectly balance the opposing magnetic force generated by the current interacting with the magnetic field.

$F_{applied} = F_{magnetic}$

The formula for the magnetic force on a straight current-carrying conductor is:

$F_{magnetic} = I \times L \times B$

Substitute the known values: $F_{applied} = 0.5$ N, $L = 0.5$ m, and $B = 0.4$ T.

$0.5 = I \times 0.5 \times 0.4$

$0.5 = I \times 0.2$

Solve for the current I:

$I = \frac{0.5}{0.2}$

$I = 2.5$ A

LEVEL-II

1. **Answer :** A) $2 \times 10^{-4} N$

The formula for the torque on a magnet in a uniform magnetic field is:

$\tau = M \times B \times \sin(\theta)$

The magnetic moment M can be written as the product of pole strength m and the magnetic length $2l$.

$M = m \times 2l$

Substitute this into the torque equation:

$\tau = (m \times 2l) \times B \times \sin(\theta)$

The magnetic force acting on a single pole is defined as $F = m \times B$. Substituting this into the equation yields:

$\tau = F \times 2l \times \sin(\theta)$

Now, substitute the provided values: torque $\tau = 2 \times 10^{-5}$ N-m, length $2l = 20$ cm = 0.20 m, and $\theta = 30°$.

$2 \times 10^{-5} = F \times 0.20 \times \sin(30°)$

Since $\sin(30°) = 0.5$:

$2 \times 10^{-5} = F \times 0.20 \times 0.5$

$2 \times 10^{-5} = F \times 0.10$

Solve for the force F:

$F = \frac{2 \times 10^{-5}}{0.10}$

$F = 2 \times 10^{-4}$ N

2. **Answer :** C) $8 \times 10^{-5} J$

The work done in rotating a magnet in a uniform magnetic field from an initial angle θ_1 to a final angle θ_2 is given by:

$W = M \times B \times (\cos(\theta_1) - \cos(\theta_2))$

The magnet is initially parallel to the field, so $\theta_1 = 0°$. It is rotated through 60°, making $\theta_2 = 60°$.

The work done for this first rotation is given as 8×10^{-5} J.

$W_1 = M \times B \times (\cos(0°) - \cos(60°))$

$8 \times 10^{-5} = M \times B \times (1 - 0.5)$

$8 \times 10^{-5} = M \times B \times 0.5$

Isolate the term $M \times B$:

$M \times B = \frac{8 \times 10^{-5}}{0.5}$

$M \times B = 16 \times 10^{-5}$ J

Now, we calculate the work done in rotating it through another 30°, meaning it rotates from its current position of 60° to a new position of 90°.

$W_2 = M \times B \times (\cos(60°) - \cos(90°))$

Substitute the previously found value for $M \times B$:

$W_2 = 16 \times 10^{-5} \times (0.5 - 0)$

$W_2 = 16 \times 10^{-5} \times 0.5$

$W_2 = 8 \times 10^{-5}$ J

3. **Answer :** B) 1.2 mJ

The kinetic energy gained by the magnet is equal to the work done by the magnetic field as the magnet rotates from its initial position to its final position.

The work done (and thus the kinetic energy K) in rotating a magnetic dipole in a uniform magnetic field is given by the difference in potential energy:

$K = U_i - U_f$

The formula for the potential energy of a magnetic dipole is:

$U = -M \times B \times \cos(\theta)$

The initial position is east-west, which is perpendicular to the north-south magnetic field.

$\theta_i = 90°$

The final position is north-south, which is aligned with the magnetic field.

$\theta_f = 0°$

Substitute these angles into the kinetic energy equation:

$K = (-M \times B \times \cos(90°)) - (-M \times B \times \cos(0°))$

Since $\cos(90°) = 0$ and $\cos(0°) = 1$:

$K = 0 - (-M \times B \times 1)$

$K = M \times B$

Substitute the given values for magnetic moment $M = 40$ A-m^2 and magnetic field $B = 30\mu\text{T} = 30 \times 10^{-6}$ T:

$K = 40 \times (30 \times 10^{-6})$

$K = 1200 \times 10^{-6}$ J

$K = 1.2 \times 10^{-3}$ J

$K = 1.2$ mJ

4. **Answer :** 0.2 A

According to Faraday's law of electromagnetic induction, the induced electromotive force (e) is the negative rate of change of magnetic flux:

$$e = -\frac{d\phi}{dt}$$

The given equation for magnetic flux is:

$\phi = 6t^2 - 5t + 1$

Take the derivative of the flux with respect to time t:

$$\frac{d\phi}{dt} = \frac{d}{dt}(6t^2 - 5t + 1)$$

$$\frac{d\phi}{dt} = 12t - 5$$

Substitute $t = 0.25$ s into the derivative:

$$\frac{d\phi}{dt} = 12(0.25) - 5$$

$$\frac{d\phi}{dt} = 3 - 5$$

$$\frac{d\phi}{dt} = -2 \text{ Wb/s}$$

Calculate the induced emf:

$e = -(-2)$

$e = 2$ V

By Ohm's law, the induced current i is the emf divided by the resistance R:

$$i = \frac{e}{R}$$

Substitute $e = 2$ V and $R = 10\Omega$:

$i = \frac{2}{10}$

$i = 0.2$ A

5. **Answer :** C) Rate of change of magnetic flux

According to Faraday's Law of Electromagnetic Induction, the magnitude of the total induced electromotive force (emf) in a circuit is directly proportional to the time rate of change of the magnetic flux linking the circuit.

6. **Answer :** D) Zero

The conducting square loop is moving completely within a uniform magnetic field that is constant in both time and space.

Because the field is uniform everywhere and the area of the loop is constant, the total magnetic flux passing through the loop remains constant as it moves.

Since the rate of change of magnetic flux ($\frac{d\phi}{dt}$) is zero, Faraday's law dictates that the induced emf is zero, resulting in zero induced current.

7. **Answer :** C) Change in the deflection will be observed when magnet changes in its direction

When the magnet is pushed into the coil, the magnetic flux linked with the coil increases, inducing a current in one direction.

When the magnet is pulled out of the coil, the magnetic flux decreases, inducing a current in the opposite direction according to Lenz's Law.

Therefore, the galvanometer will show deflections in opposite directions when the magnet's direction of motion changes.

8. **Answer :** D) Respectively anticlockwise and clockwise

When switch S is closed, the clockwise current I_p in loop P increases from zero to a maximum value, causing the magnetic flux linked with loop Q to increase.

According to Lenz's law, loop Q will induce a current I_{Q1} to oppose this increase, which means it will flow in the opposite direction (anticlockwise).

When switch S is opened, the current I_p in loop P decreases, causing the magnetic flux linked with loop Q to decrease.

Loop Q will induce a current I_{Q2} to oppose this decrease, which means it will flow in the same direction as the original current to support the collapsing field (clockwise).

9. **Answer :** B, D

The rate of change of the uniform magnetic field is given as $\frac{dB}{dt} = \alpha$.

The induced electromotive force (e) in the conducting ring of radius a is equal to the area of the ring multiplied by the rate of change of the magnetic field:

$e = A \times \frac{dB}{dt}$

$e = (\pi \times a^2) \times \alpha$

This confirms option B is correct.

The induced emf is also equal to the line integral of the induced non-conservative electric field E around the ring:

$e = \oint E \cdot dl$

Because of the circular symmetry, the electric field magnitude E is constant along the circumference ($2\pi \times a$):

$e = E \times (2\pi \times a)$

Equate the two expressions for the induced emf:

$E \times (2\pi \times a) = \pi \times a^2 \times \alpha$

Solve for E:

$$E = \frac{\pi \times a^2 \times \alpha}{2\pi \times a}$$

$$E = \frac{a \times \alpha}{2}$$

This confirms option D is correct.

10. **Answer :** A, B, C

The magnetic flux is given by the equation:

$\phi = 4t^n + 6$

The magnitude of the induced emf $|e|$ is the rate of change of flux:

$$|e| = \left|\frac{d\phi}{dt}\right|$$

$$|e| = \frac{d}{dt}(4t^n + 6)$$

$$|e| = 4n \times t^{n-1}$$

Analyze for different values of n:

If $0 < n < 1$, the exponent $(n - 1)$ is negative. This means $|e|$ is inversely proportional to a positive power of t, so $|e|$ decreases with time. This makes statement A correct.

If $n = 1$, the exponent is zero. $|e| = 4(1)t^0 = 4$, which is a constant value independent of time. This makes statement B correct.

If $n > 1$, the exponent $(n - 1)$ is positive. This means $|e|$ is directly proportional to a positive power of t, so $|e|$ increases with time. This makes statement C correct.

11. **Answer :** A, D

As the bar magnet moves towards loop B, the North pole approaches it. According to Lenz's law, loop B will induce a current to repel the approaching North pole by creating a North pole on the side facing the magnet (anticlockwise direction when viewed from the magnet).

As the bar magnet moves away from loop A, the South pole recedes from it. Loop A will induce a current to attract the receding South pole by creating a North pole on the side facing the magnet (anticlockwise direction when viewed from the magnet).

Since the induced currents in both loops circulate in the same direction relative to a common axis, they act as parallel currents.

Parallel currents attract each other, meaning the loops will attract each other.

12. **Answer :** A, C

The bar magnet moves towards both loops, meaning the North pole approaches both A and B.

Both loops will induce a current in the same direction (anticlockwise) to oppose the increasing magnetic flux.

Because loop A is closer to the moving magnet, it experiences a stronger magnetic field and a higher rate of change of flux compared to loop B.

Therefore, the induced current in loop A is greater than the induced current in loop B, making statement C correct.

Since the currents in both loops are in the same direction, they will attract each other, making statement A correct.

13. **Answer :** A, B, D

When the rod is pulled with a constant velocity, the external mechanical power supplied by the person is completely converted into electrical power, which is then dissipated as thermal heat in the resistor R. Thus, statement A is correct.
If the applied external force is suddenly doubled, it becomes greater than the opposing magnetic drag force. The net force will accelerate the rod, increasing its kinetic energy and velocity until a new, higher constant velocity is reached where forces balance again. Thus, statement B is correct.

The power required to maintain a constant velocity v_0 is $P = \frac{B^2 \times l^2 \times v_0^2}{R}$. If the resistance R is doubled, the power P becomes half, making statement D correct.

14. Answer : B) $250\mu V$

The total induced emf in the moving bar is the result of both the changing magnetic field and the changing area due to the rod's motion.
The area of the rectangular loop at any instant is:
$A = l \times x$
The magnetic flux is:
$\Phi = B \times A = B \times (l \times x)$
The total rate of change of flux is found using the product rule:
$$\frac{d\Phi}{dt} = \frac{dB}{dt} \times (l \times x) + B \times \frac{d(l \times x)}{dt}$$
$$\frac{d\Phi}{dt} = (l \times x) \times \frac{dB}{dt} + (B \times l) \times \frac{dx}{dt}$$
Given values at $t = 0$: $l = 5$ cm $= 0.05$ m, $x = 5$ cm $= 0.05$ m, $B = 0.1$ T, and $\frac{dB}{dt} = 0.2$ T/s.
According to the provided diagram, the velocity vector v points to the left, meaning the length x is decreasing.
$$\frac{dx}{dt} = -v = -5 \text{ cm/s} = -0.05 \text{ m/s}$$
Substitute all values into the rate of change of flux equation:
$$\frac{d\Phi}{dt} = (0.05 \times 0.05) \times 0.2 + (0.1 \times 0.05) \times (-0.05)$$
$$\frac{d\Phi}{dt} = (0.0025) \times 0.2 - (0.005) \times 0.05$$
$$\frac{d\Phi}{dt} = 0.0005 - 0.00025$$
$$\frac{d\Phi}{dt} = 0.00025 \text{ V}$$
Convert the voltage to microvolts (μV):
$e = 250 \times 10^{-6}$ V
$e = 250\mu$V

15. Answer : A) 2.5 A

The induced current in the circuit is calculated using Ohm's law:
$$I = \frac{e}{R}$$
From the previous question, the induced emf is $e = 250 \times 10^{-6}$ V.
The resistance of the circuit is given as $R = 10^{-4}\Omega$.
$$I = \frac{250 \times 10^{-6}}{10^{-4}}$$
$$I = 250 \times 10^{-2}$$
$$I = 2.5 \text{ A}$$

16. **Answer :** A) 0.16 A, clockwise

The magnetic field is time-dependent and given by $B = 2t$. Its rate of change is:

$\frac{dB}{dt} = 2$ T/s

At $t = 0$, the rod is stationary. The area of the loop is:

$A = l_2 \times x$

Given dimensions: $l_2 = 20$ cm $= 0.2$ m, and the initial length of the loop $x = l_1 = 40$ cm $= 0.4$ m.

$A = 0.2 \times 0.4 = 0.08$ m^2

The induced emf e is:

$e = A \times \frac{dB}{dt}$

$e = 0.08 \times 2 = 0.16$ V

The total length of the conducting rail forming the loop is $x + l_2 + x = 0.4 + 0.2 + 0.4 = 1.0$ m.

Since the resistance of the wire is $1\Omega m^{-1}$, the total resistance R is:

$R = 1.0$ m $\times 1\Omega m^{-1} = 1.0\Omega$

The induced current is:

$I = \frac{e}{R} = \frac{0.16}{1.0} = 0.16$ A

The magnetic field is pointing out of the page and increasing. To oppose this, the induced current will create a magnetic field pointing into the page, which requires a clockwise current.

17. **Answer :** B) 0.08 V

The rod moves to the left at a constant speed of $v = 5$ cm/s $= 0.05$ m/s.

At $t = 2$ s, the distance moved by the rod is:

$d = v \times t = 0.05 \times 2 = 0.1$ m

The new length of the loop x is:

$x = 0.4$ m $- 0.1$ m $= 0.3$ m

The area of the loop at $t = 2$ s is:

$A = 0.3 \times 0.2 = 0.06$ m^2

The magnetic field at $t = 2$ s is:

$B = 2t = 2(2) = 4$ T

The total rate of change of flux is:

$\frac{d\Phi}{dt} = A \times \frac{dB}{dt} + B \times \frac{dA}{dt}$

The rate of change of area is:

$\frac{dA}{dt} = l_2 \times \frac{dx}{dt} = 0.2 \times (-0.05) = -0.01$ m^2/s

Substitute all values:

$\frac{d\Phi}{dt} = (0.06 \times 2) + (4 \times -0.01)$

$\frac{d\Phi}{dt} = 0.12 - 0.04$

$\frac{d\Phi}{dt} = 0.08$ V

18. **Answer :** C) 0.08 N

To keep the rod moving at a constant speed, the external applied force must be equal in magnitude to the opposing magnetic force.

$F = I \times l_2 \times B$

First, determine the induced current I at $t = 2$ s. The induced emf is 0.08 V.
The total length of the wire loop at $t = 2$ s is:
$L = x + l_2 + x = 0.3 + 0.2 + 0.3 = 0.8$ m
The resistance is $R = 0.8\Omega$.
$I = \frac{e}{R} = \frac{0.08}{0.8} = 0.1$ A
Now, calculate the magnetic force using $B = 4$ T and $l_2 = 0.2$ m:
$F = 0.1 \times 0.2 \times 4$
$F = 0.08$ N

19. **Answer :** A) Both Assertion and Reason are true, and Reason is correct explanation
The torque acting on a current-carrying coil is given by the equation:
$\tau = M \times B \times \sin(\theta)$
Here, θ is the angle between the area vector (magnetic moment M) and the magnetic field B.
When the plane of the coil is perpendicular to the magnetic field, its area vector is parallel to the field, making $\theta = 0°$ (or 180°).
Because $\sin(0°) = 0$, the torque becomes zero. Therefore, both the assertion and reason are true, and the reason correctly explains the assertion.

20. **Answer :** A) Both Assertion and Reason are true, and Reason is correct explanation
The potential energy of a magnetic dipole in a magnetic field is defined by the formula:
$U = -M \times B \times \cos(\theta)$
When the dipole is aligned with the magnetic field, the angle $\theta = 0°$.
$U = -M \times B \times \cos(0°)$
$U = -M \times B \times (1)$
$U = -MB$
This negative value represents the minimum possible potential energy, confirming the assertion is true and is directly explained by the provided reason.

21. **Answer :** a - Q, b - P, c - R, d - S
According to the Right-Hand Grip Rule, an upward current in the straight wire creates a magnetic field that is directed into the plane of the page at the location of the square loop.
If the current is increased, the inward magnetic flux through the loop increases. By Lenz's law, the loop will induce a current to create an outward magnetic field to oppose this change. An outward field is produced by an anticlockwise current.

22. If the current is decreased, the inward magnetic flux decreases. The loop induces a current to create an inward magnetic field to support it, resulting in a clockwise current.
If the loop is moved away, the inward magnetic flux decreases, inducing a clockwise current. The left arm of the loop (closest to the wire) will have an upward current, parallel to the straight wire. Parallel currents attract. Since the attractive force on the closer left arm is stronger than the repulsive force on the farther right arm, the net force is attractive.
If the loop is moved toward the wire, the inward flux increases, inducing an anticlockwise current. The downward current in the left arm is antiparallel to the straight wire, causing repulsion. The net force is repulsive.

23. **Answer :** a - R, b - Q, c - P, d - S

Flux of the magnetic field (Φ) is measured in Webers (Wb). Since $V = I \times R$, the unit Volt-second can be written as (Ampere × Ohm × second), which simplifies to Coulomb-ohm. Thus, a matches with R.

The rate of change of flux ($\frac{d\Phi}{dt}$) is equal to the induced electromotive force, which is measured in Volts. Thus, b matches with Q.

Magnetic field strength (B) is measured in Teslas. Thus, c matches with P.

Potential energy is measured in Joules. The unit Ampere-weber represents (Current × Flux), and $\int I\, d\Phi$ yields units of Energy (Joules). Thus, d matches with S.

24. **Answer :** 200

According to Faraday's Law, the average induced electromotive force (emf) is given by:

$$e = -N \times \frac{\Delta\Phi}{\Delta t}$$

Where N is the number of turns, $\Delta\Phi$ is the change in flux, and Δt is the time interval.

Given values: $N = 100$, Initial flux $\Phi_1 = 5$ Wb, Final flux $\Phi_2 = 1$ Wb, and $\Delta t = 2$ s.

Change in flux $\Delta\Phi = \Phi_2 - \Phi_1$:

$\Delta\Phi = 1 - 5 = -4$ Wb

Substitute these values into the equation:

$$e = -100 \times \frac{-4}{2}$$

$$e = -100 \times (-2)$$

$e = 200$ V

25. **Answer :** 0.5

The work done in rotating a current-carrying coil in a magnetic field is given by:

$W = M \times B \times (\cos(\theta_1) - \cos(\theta_2))$

The magnetic moment M of a coil is:

$M = N \times I \times A$

Substitute $N = 50$, $A = 0.02$ m^2, and assume $I = 1$ A:

$M = 50 \times 1 \times 0.02$

$M = 1$ A-m^2

The coil rotates from $\theta_1 = 0°$ to $\theta_2 = 90°$ in the magnetic field $B = 0.5$ T:

$W = 1 \times 0.5 \times (\cos(0°) - \cos(90°))$

Since $\cos(0°) = 1$ and $\cos(90°) = 0$:

$W = 0.5 \times (1 - 0)$

$W = 0.5$ J

26. **Answer :** 5

The motional electromotive force (emf) induced in a straight rod moving perpendicularly through a magnetic field is:

$e = B \times l \times v$

Substitute the given values: $B = 1.5$ T, length $l = 2$ m, and velocity $v = 5$ m/s.

$e = 1.5 \times 2 \times 5$

$e = 15$ V

The induced current I is calculated using Ohm's Law:

$$I = \frac{e}{R}$$

Given the circuit resistance $R = 3\Omega$:

$I = \frac{15}{3}$

$I = 5$ A

Previous Year Questions

1. **Answer :** B) (ii) and (iii) only

By carefully tracing the wire winding from the provided image, the current directions on the front and back portions of the plastic rod can be determined.

In the left loop, the current goes UP on the front segment (I) and DOWN on the back segment (II).

In the right loop, the winding direction reverses, so the current goes DOWN on the front segment (III) and UP on the back segment (IV).

According to the laws of magnetic force, wire segments carrying currents in opposite directions repel each other.

Part I (UP) and Part III (DOWN) are in opposite directions, so they repel. This makes statement (iii) true.

Part II (DOWN) and Part IV (UP) are in opposite directions, so they repel. This makes statement (ii) true.

Therefore, only statements (ii) and (iii) are correct.

2. **Answer :** B) (i) T, (ii) F, (iii) T, (iv) F

True. Inserting a soft iron core into a current-carrying solenoid vastly increases the magnetic field strength because soft iron has high magnetic permeability.

False. According to the Right-Hand Grip Rule, if you point your thumb vertically upwards along the direction of the current, your fingers will curl in an anticlockwise direction when viewed from above.

True. Reversing the direction of the current only reverses the polarity (direction) of the magnetic field; the magnitude (strength) remains unchanged.

False. Soft iron is the most suitable material for the core of an electromagnet, not steel, because it easily loses its magnetism when the current is switched off (low retentivity).

3. **Answer :** C) II only

The magnetic field generated by a solenoid of finite length carrying an electric current is geometrically and mathematically equivalent to the magnetic field produced by a classic bar magnet.

Both produce magnetic field lines that emerge from one end (acting as a North pole) and enter the other end (acting as a South pole), forming continuous loops.

An infinitely long wire produces concentric circular fields, and a simple loop acts only as a very short dipole, failing to replicate the extended physical field of a full bar magnet.

4. **Answer :** C) 3 cm mark

Let the first conductor ($I_1 = 1$ A) be at position $x_1 = 2$ cm.

Let the second conductor ($I_2 = 3$ A) be at position $x_2 = 6$ cm.

The total distance between the two conductors is $d = 6 - 2 = 4$ cm.

Since the currents flow in the same direction, the zero magnetic field (null point) will lie exactly between them where their individual magnetic fields are equal in magnitude and opposite in direction.

Let the null point be at a distance r from the first conductor (1 A). The distance from the second conductor will be $(4 - r)$.

Equate the magnitudes of the magnetic fields:

$$\frac{\mu_0 \times I_1}{2\pi \times r} = \frac{\mu_0 \times I_2}{2\pi \times (4 - r)}$$

Cancel the common terms (μ_0 and 2π):

$$\frac{1}{r} = \frac{3}{4 - r}$$

Cross-multiply to solve for r:

$4 - r = 3 \times r$

$4 = 4 \times r$

$r = 1$ cm

The null point is 1 cm to the right of the first conductor.

Position on the scale = 2 cm + 1 cm = 3 cm.

5. **Answer :** C)

The magnitude of the magnetic field B produced by a long, straight, current-carrying conductor at a perpendicular distance r is given by Biot-Savart's Law (or Ampere's Law):

$$B = \frac{\mu_0 \times I}{2\pi \times r}$$

This equation shows that the magnetic field strength B is inversely proportional to the distance r ($B \propto \frac{1}{r}$).

Mathematically, a graph of an inversely proportional relationship is a rectangular hyperbola that curves downwards and asymptotes towards the axes without touching them, which matches graph C.

6. **Answer :** C) Both (i) and (ii) are correct

Statement (i) is correct because the magnetic field generated by the current in the straight wire exerts a torque on the compass needle, causing a noticeable deflection (as famously discovered by Oersted).

Statement (ii) is also correct because the magnetic field lines around a straight current-carrying wire form concentric circles. A compass needle always aligns itself tangentially to the magnetic field line passing through its location.

7. **Answer :** D)

According to a standard principle of electromagnetism, if an electric current enters and leaves a uniform, closed polygonal wire frame (like a square) at any two nodes (junctions), the magnetic field at the exact geometric center is always perfectly zero due to the symmetrical cancellation of the fields from the divided current paths.

This principle applies directly to the closed loops shown in configurations A, B, and C, meaning the central field in those cases is zero.

However, configuration D depicts a single continuous wire bent into a square shape where the current does not split into parallel paths, but rather the entire 5 A current flows sequentially through all four sides before exiting.

Because the current travels consistently in a single loop around the center, the magnetic fields generated by each of the four sides point in the same direction and add constructively, resulting in a non-zero net magnetic field at the center.

8. **Answer :** A) The upper face of the loop

According to the Right-Hand Grip Rule (or the Clock Face Rule), the magnetic polarity of a current-carrying loop is determined by the direction of the current flow.

When looking at a face of the loop, if the current circulates in an anticlockwise direction, that face acts as a magnetic North pole (field lines emerge from it). Since the current appears anticlockwise when observed from above, the upper face of the loop constitutes the North pole of the equivalent magnet.